

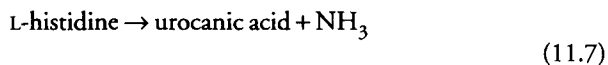
A limited number of commercially-important enzyme conversions, such as glucose isomerisation and starch hydrolysis, are treated as reversible reactions. In these systems, the reaction mixture at equilibrium contains significant amounts of reactants as well as products. However, for many reactions $\Delta G_{\text{rxn}}^\circ$ is negative and large in magnitude. As a result, K is also very large, the reaction favours the products rather than the reactants, and the reaction is regarded as *irreversible*. Most enzyme and cell reactions fall into this category. For example, the equilibrium constant for sucrose hydrolysis by invertase is about 10^4 ; for fermentation of glucose to ethanol and carbon dioxide, K is about 10^{30} . The equilibrium ratio of products to reactants is so overwhelmingly large for these reactions that they are considered to proceed to completion, i.e. the reaction stops only when the concentration of one of the reactants falls to zero. Equilibrium thermodynamics has therefore only limited application to enzyme and cell reactions. Moreover, the thermodynamic principles outlined in this section apply only to closed systems; true thermodynamic equilibrium does not exist in living cells which exchange matter with their surroundings. Metabolic processes in cells are in a dynamic state; products formed are constantly removed or broken down so that reactions are driven forward. Most reactions in biological systems proceed to completion in a finite period of time at a finite rate.

If we know that complete conversion will eventually take place, the most useful reaction parameter to know is the rate at which the transformation proceeds. Another important characteristic, especially for systems in which many different reactions take place at the same time, is the proportion of reactant that is converted to the desired products. These properties of reactions are discussed in the remainder of this chapter.

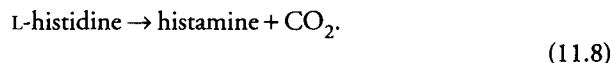
11.1.2 Reaction Yield

The extent to which reactants are converted to products is expressed as the reaction *yield*. Generally speaking, yield is the amount of product formed or accumulated per amount of reactant provided or consumed. Unfortunately, there is no strict definition of yield; several different yield parameters are applicable in different situations. The terms used to express yield in this text do not necessarily have universal acceptance and are defined here for our convenience. Be prepared for other books to use different definitions.

Consider the simple enzyme reaction:



catalysed by histidase. According to the reaction stoichiometry, 1 gmol urocanic acid is produced for each gmol L-histidine consumed; the yield of urocanic acid from histidine is therefore 1 gmol gmol⁻¹. However, let us assume that the histidase used in this reaction is contaminated with another enzyme, histidine decarboxylase. Histidine decarboxylase catalyses the following reaction:



If both enzymes are active, some L-histidine will react with histidase according to Eq. (11.7), while some will be decarboxylated according to Eq. (11.8). After addition of the enzymes to the substrate, analysis of the reaction mixture shows that 1 gmol urocanic acid and 1 gmol histamine are produced for every 2 gmol histidine consumed. The *observed* or *apparent yield* of urocanic acid from L-histidine is $1 \text{ gmol} / 2 \text{ gmol} = 0.5 \text{ gmol gmol}^{-1}$. The observed yield of 0.5 gmol gmol⁻¹ is different from the *stoichiometric, true* or *theoretical yield* of 1 gmol gmol⁻¹ calculated from reaction stoichiometry because the reactant was channelled in two separate reaction pathways. An analogous situation arises if product rather than reactant is consumed in other reactions; the observed yield of product would be lower than the theoretical yield. *When reactants or products are involved in additional reactions, the observed yield may be different from the theoretical yield.*

The above analysis leads to two useful definitions of yield for reaction systems:

$$\left(\begin{array}{c} \text{true, stoichiometric or} \\ \text{theoretical yield} \end{array} \right) = \frac{\left(\begin{array}{c} \text{total mass or moles of} \\ \text{product formed} \end{array} \right)}{\left(\begin{array}{c} \text{mass or moles of reactant used} \\ \text{to form that particular product} \end{array} \right)} \quad (11.9)$$

and

$$\left(\begin{array}{c} \text{observed or} \\ \text{apparent yield} \end{array} \right) = \frac{\left(\begin{array}{c} \text{mass or moles of product present} \end{array} \right)}{\left(\begin{array}{c} \text{total mass or moles of reactant} \\ \text{consumed} \end{array} \right)} \quad (11.10)$$

There is a third type of yield applicable in certain situations. For reactions with incomplete conversion of reactant, it may be of interest to specify the amount of product formed per amount of reactant *provided to the reaction* rather than actually consumed. For example, consider the isomerisation reaction catalysed by glucose isomerase:

glucose $\rightleftharpoons$ fructose.

(11.11)

The reaction is carried out in a closed reactor with pure enzyme. At equilibrium the sugar mixture contains 55 mol% glucose and 45 mol% fructose. The *theoretical yield* of fructose from glucose is 1 gmol gmol⁻¹ because, from

stoichiometry, formation of 1 gmol fructose requires 1 gmol glucose. The *observed yield* would also be 1 gmol gmol⁻¹ if the reaction occurs in isolation. However if the reaction is started with glucose present only, the equilibrium yield of fructose per gmol glucose added to the reactor is 0.45 gmol gmol⁻¹. This type of yield for incomplete reactions may be denoted *gross yield*.

Example 11.2 Incomplete enzyme reaction

An enzyme catalyses the reaction:



At equilibrium, the reaction mixture contains 63 wt% A.

- What is the equilibrium constant?
- If the reaction starts with A only, what is the equilibrium yield of B from A?

Solution:

- From stoichiometry the molecular weights of A and B must be equal: therefore wt% = mol%. From Eq. (11.2):

$$K = \frac{C_{Be}}{C_{Ae}}.$$

Using a basis of 1 gmol l⁻¹, C_{Ae} is 0.63 gmol l⁻¹ and C_{Be} is 0.37 gmol l⁻¹. The value of K therefore is $0.37/0.63 = 0.59$.

- From stoichiometry, the true yield of B from A is 1 gmol gmol⁻¹. However the gross yield is $0.37/1.0 = 0.37$ gmol gmol⁻¹.

11.1.3 Reaction Rate

Consider the general irreversible reaction:



The rate of this reaction can be represented by the rate of conversion of compound A; let us use the symbol R_A to denote the *rate of reaction with respect to A*. R_A has units of, for example kg s⁻¹.

How do we measure reaction rates? For a general reaction system, rate of reaction is related to rate of change of mass in the system by the unsteady-state mass-balance equation derived in Chapter 6:

$$\frac{dM}{dt} = \hat{M}_i - \hat{M}_o + R_G - R_C. \quad (6.5)$$

In Eq (6.5), M is mass, t is time, $\hat{M}_i$ is mass flow rate into the

system, $\hat{M}_o$ is mass flow rate out of the system, R_G is mass rate of generation by reaction and R_C is mass rate of consumption by reaction. Let us apply Eq. (6.5) to compound A, assuming that the reaction of Eq. (11.12) is the only reaction taking place that involves A. Rate of consumption R_C is equal to R_A , and $R_G = 0$. The mass-balance equation becomes:

$$\frac{dM_A}{dt} = \hat{M}_{Ai} - \hat{M}_{Ao} - R_A. \quad (11.13)$$

Therefore, rate of reaction R_A can be determined if we measure the rate of change in mass of A, dM_A/dt , and the rates of flow of A in and out of the system, $\hat{M}_{Ai}$ and $\hat{M}_{Ao}$. In a closed system where $\hat{M}_{Ai} = \hat{M}_{Ao} = 0$, Eq. (11.13) becomes:

$$R_A = -\frac{dM_A}{dt} \quad (11.14)$$

and reaction rate is measured simply by monitoring the change in mass of A in the system. Most measurements of reaction rate are carried out in closed systems so that the data can be analysed according to Eq. (11.14). $\frac{dM_A}{dt}$ is negative when A is consumed by reaction; therefore the minus sign in Eq. (11.14) is necessary to make R_A a positive quantity. Rate of reaction is sometimes called *reaction velocity*. Reaction velocity can also be measured in terms of components B, Y or Z. In a closed system:

$$R_B = \frac{-dM_B}{dt} \quad R_Y = \frac{dM_Y}{dt} \quad R_Z = \frac{dM_Z}{dt} \quad (11.15)$$

where M_B , M_Y and M_Z are masses of B, Y and Z, respectively. When reporting reaction rate, the reactant being monitored should be specified. Because R_Y and R_Z are based on product accumulation, these reaction rates are called *production rates* or *productivity*.

Eqs (11.14) and (11.15) define the rate of reaction in a closed system. However, reaction rate can be expressed using different measurement bases. In bioprocess engineering there are three distinct ways of expressing reaction rate which can be applied in different situations.

- (i) *Total rate*. Total reaction rate is defined in Eqs (11.14) and (11.15) and is expressed as either mass or moles per unit time. Total rate is useful for specifying the output of a particular reactor or manufacturing plant. Production rates for factories are often expressed as total rates; for example: 'The production rate is 100 000 tonnes per year'. If additional reactors are built so that the reaction volume in the plant is increased, then clearly the total reaction rate would increase. Similarly, if the amount of cells or enzyme used in each reactor were also increased, then the total production rate would be improved even further.
- (ii) *Volumetric rate*. Because the total mass of reactant converted in a reaction mixture depends on the size of the system, it is often convenient to specify reaction rate as the rate per unit volume. Units of volumetric rate are, e.g. $\text{kg m}^{-3} \text{s}^{-1}$. Rate of reaction expressed on a volumetric basis is used to account for differences in volume between reaction systems. Therefore, if the reaction mixture in a closed system has volume V :

$$r_A = \frac{R_A}{V} = \frac{-1}{V} \frac{dM_A}{dt} \quad (11.16)$$

where r_A is the volumetric rate of reaction with respect to A. When V is constant, Eq. (11.16) can be written:

$$r_A = \frac{-dC_A}{dt} \quad (11.17)$$

where C_A is the concentration of A in units of, e.g. kg m^{-3} . Volumetric rates are particularly useful for comparing the performance of reactors of different size. A common objective in optimising reaction processes is to maximise volumetric productivity so that the desired total production rate can be achieved with reactors of minimum size and therefore minimum cost.

- (iii) *Specific rate*. Biological reactions involve enzyme and cell catalysts. Because the total rate of conversion depends on the amount of catalyst present, it is sometimes useful to specify reaction rate as the rate per quantity of enzyme or cells involved in the reaction. In a closed system, specific reaction rate can be measured as follows:

$$r_A = -\left(\frac{1}{X} \text{ or } \frac{1}{E}\right) \frac{dM_A}{dt} \quad (11.18)$$

where r_A is the specific rate of reaction with respect to A, X is the quantity of cells, E is the quantity of enzyme and $\frac{dM_A}{dt}$ is the rate of change of mass of A in the system. As quantity of cells is usually expressed as mass, units of specific rate for a cell-catalysed reaction would be, e.g. $\text{kg (kg cells)}^{-1} \text{s}^{-1}$ or simply s^{-1} . On the other hand, the mass of a particular enzyme added to a reaction is rarely known; most commercial enzyme preparations contain several components in unknown and variable proportions depending on the batch obtained from the manufacturer. To overcome these difficulties, enzyme quantity is often expressed as *units of activity* measured under specified conditions. One unit of enzyme is usually taken as the amount which catalyses conversion of 1 μmol substrate per minute at the optimal temperature, pH and substrate concentration. Therefore, if E in Eq. (11.18) is expressed as units of enzyme activity, the specific rate of reaction under process conditions could be reported as, e.g. $\text{kg (unit enzyme)}^{-1} \text{s}^{-1}$. In a closed system where the volume of reaction mixture remains constant, an alternative expression for specific reaction rate is:

$$r_A = -\left(\frac{1}{x} \text{ or } \frac{1}{e}\right) \frac{dC_A}{dt} \quad (11.19)$$

where x is cell concentration and e is enzyme concentration.

Volumetric and total rates are not a direct reflection of catalyst performance; this is represented by the specific rate. Specific rates are employed when comparing different cells or enzymes. Specific rate is the rate achieved per unit catalyst and, under usual circumstances, is not dependent on the size of the system or the amount of catalyst present. Some care is necessary when interpreting results for reaction rate. For example, if two fermentations are carried out with different cell lines and the volumetric rate of reaction is greater in the first fermentation than in the second, you should not jump to the conclusion that the cell line in the first experiment is 'better', or capable of greater metabolic activity. It could be that the faster volumetric rate is due to the first fermenter being operated at a higher cell density than the second, leading to measurement of a more rapid rate per unit volume. Different strains of organism should be compared in terms of specific reaction rates.

Total, volumetric and specific productivities are inter-related concepts in process design. For example, high total productivity could be achieved with a catalyst of low specific activity if the reactor is loaded with a high catalyst concentration. If this is not possible, the volumetric productivity will be relatively low and a larger reactor is required to achieve the desired total productivity. In this book, the symbol R_A will be used to denote total reaction rate with respect to component A; r_A represents either volumetric or specific rate.

11.1.4 Reaction Kinetics

As reactions proceed, the concentrations of reactants decrease. In general, rate of reaction depends on reactant concentration so that the specific rate of conversion decreases simultaneously. Reaction rate also varies with temperature; most reactions speed up considerably as the temperature rises. *Reaction kinetics* refers to the relationship between rate of reaction and conditions which affect reaction velocity, such as reactant concentration and temperature. These relationships are conveniently described using *kinetic expressions* or *kinetic equations*.

Consider again the general irreversible reaction of Eq. (11.12). Often but not always, the volumetric rate of this reaction can be expressed as a function of reactant concentrations using the following mathematical form:

$$r_A = k C_A^a C_B^b \quad (11.20)$$

where k is the *rate constant* or *rate coefficient* for the reaction. By definition, the rate constant is independent of the concentration of reacting species but is dependent on other variables that

influence reaction rate, such as temperature. When the kinetic equation has the form of Eq. (11.20), the reaction is said to be of *order a* with respect to component A and *order b* with respect to B. The order of the overall reaction is $(a + b)$. It is not usually possible to predict the order of reactions from stoichiometry. The mechanism of single reactions and the functional form of the kinetic expression must be determined by experiment. The dimensions and units of k depend on the order of the reaction.

11.1.5 Effect of Temperature on Reaction Rate

Temperature has a significant kinetic effect on reactions. Variation of the rate constant k with temperature is described by the *Arrhenius equation*:

$$k = A e^{-E/RT} \quad (11.21)$$

where k is the rate constant, A is the *Arrhenius constant* or *frequency factor*, E is the *activation energy* for the reaction, R is the ideal gas constant, and T is absolute temperature. Values of R are listed in Table 2.5 (p. 20). According to the Arrhenius equation, as T increases, k increases exponentially. Taking the natural logarithm of both sides of Eq. (11.21):

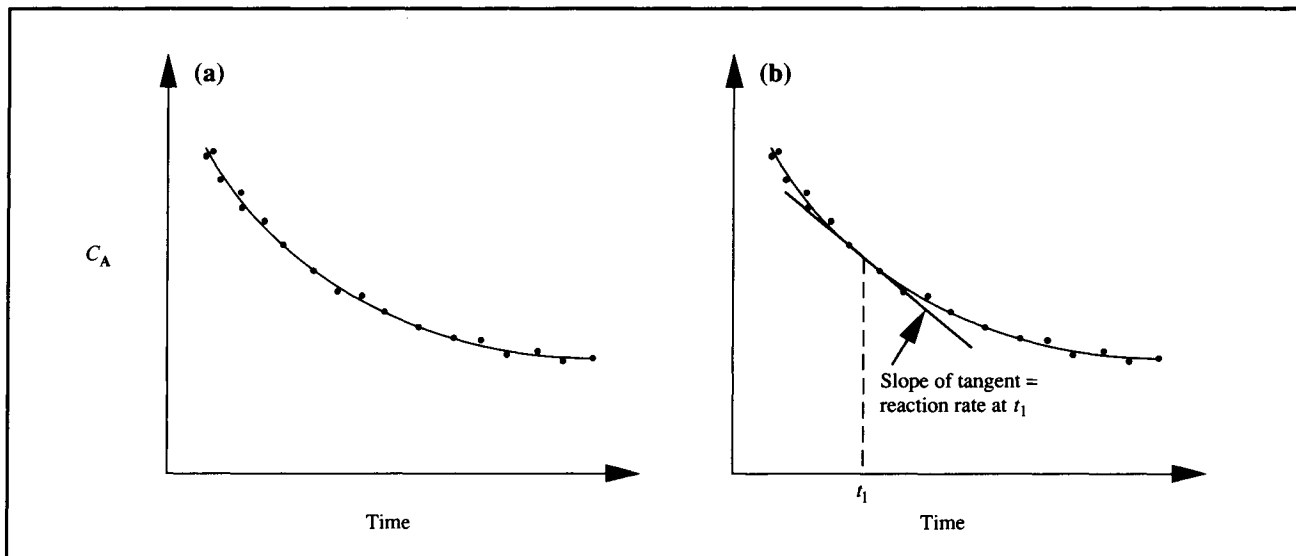
$$\ln k = \ln A - \frac{E}{RT} \quad (11.22)$$

Thus, a plot of $\ln k$ versus $1/T$ gives a straight line with slope $-E/R$. For many reactions the value of E is positive and large, indicating a rapid increase in reaction rate with temperature.

11.2 Calculation of Reaction Rates From Experimental Data

As outlined in Section 11.1.3, the volumetric rate of reaction in a closed system can be found by measuring the rate of change in the mass of reactant present, provided the reactant is involved in only one reaction. Most kinetic studies of biological reactions are carried out in closed systems with a constant volume of reaction mixture; therefore, Eq. (11.17) can be used to evaluate the volumetric reaction rate. The concentration of a particular reactant or product is measured as a function of time. For a reactant such as A in Eq. (11.12), the results will be similar to those shown in Figure 11.1(a); the concentration will decrease with time. The volumetric rate of reaction is equal to dC_A/dt , which can be evaluated as the slope of a

Figure 11.1 (a) Change in reactant concentration with time during reaction. (b) Graphical differentiation of concentration data by drawing a tangent.



smooth curve drawn through the data points. The slope of the curve in Figure 11.1(a) changes with time; the reaction rate is greater at the beginning of the experiment than at the end.

One obvious way to determine reaction rate is to draw tangents to the curve of Figure 11.1(a) at various times and evaluate the slopes of the tangents; this is shown in Figure 11.1(b). If you have ever attempted this you will know that it can be extremely difficult, even though correct in principle. Drawing tangents to curves is a highly subjective procedure prone to great inaccuracy, even with special drawing devices designed for the purpose. The results depend strongly on the way the data are smoothed and the appearance of the curve at the points chosen. More reliable techniques are available for *graphical differentiation* of rate data. Graphical differentiation is valid only if the data can be presumed to differentiate smoothly.

11.2.1 Average Rate–Equal Area Method

This technique for determining rates is based on the *average rate–equal area construction*, and will be illustrated using data for oxygen uptake by immobilised cells. Results from measurement of oxygen concentration in a closed system as a function of time are listed in the first two columns of Table 11.1.

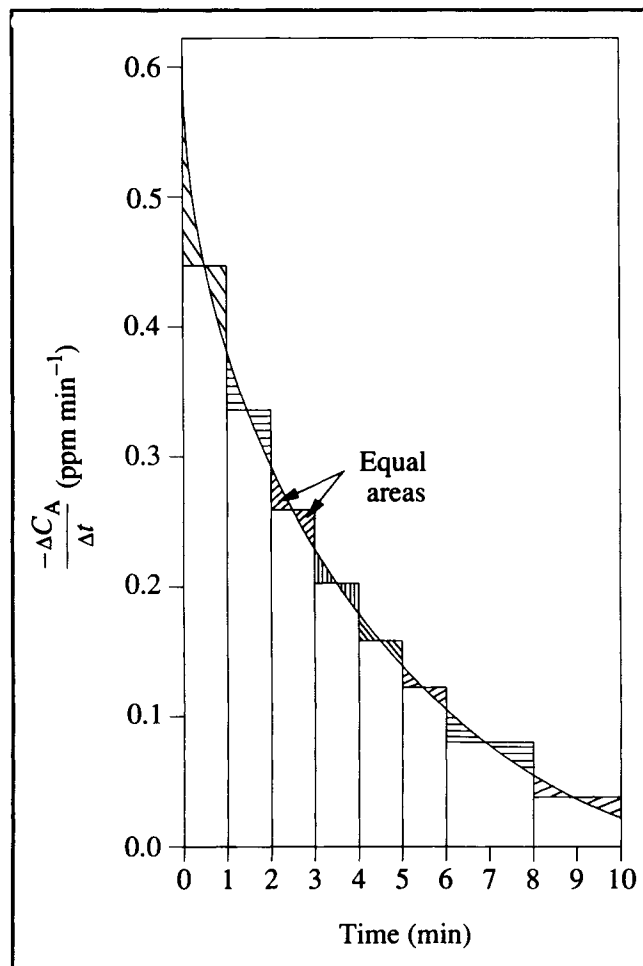
- (i) Tabulate values of ΔC_A and Δt for each time interval as shown in Table 11.1. ΔC_A values are negative because C_A decreases over each interval.

Table 11.1 Graphical differentiation using the average rate–equal area construction

Time (t , min)	Oxygen concentration (C_A , ppm)	ΔC_A	Δt	$\Delta C_A / \Delta t$	dC_A / dt
0.0	8.00	–0.45	1.0	–0.45	–0.59
1.0	7.55	–0.33	1.0	–0.33	–0.38
2.0	7.22	–0.26	1.0	–0.26	–0.29
3.0	6.96	–0.20	1.0	–0.20	–0.23
4.0	6.76	–0.15	1.0	–0.15	–0.18
5.0	6.61	–0.12	1.0	–0.12	–0.14
6.0	6.49	–0.16	2.0	–0.08	–0.11
8.0	6.33	–0.08	2.0	–0.04	–0.06
10.0	6.25				–0.02

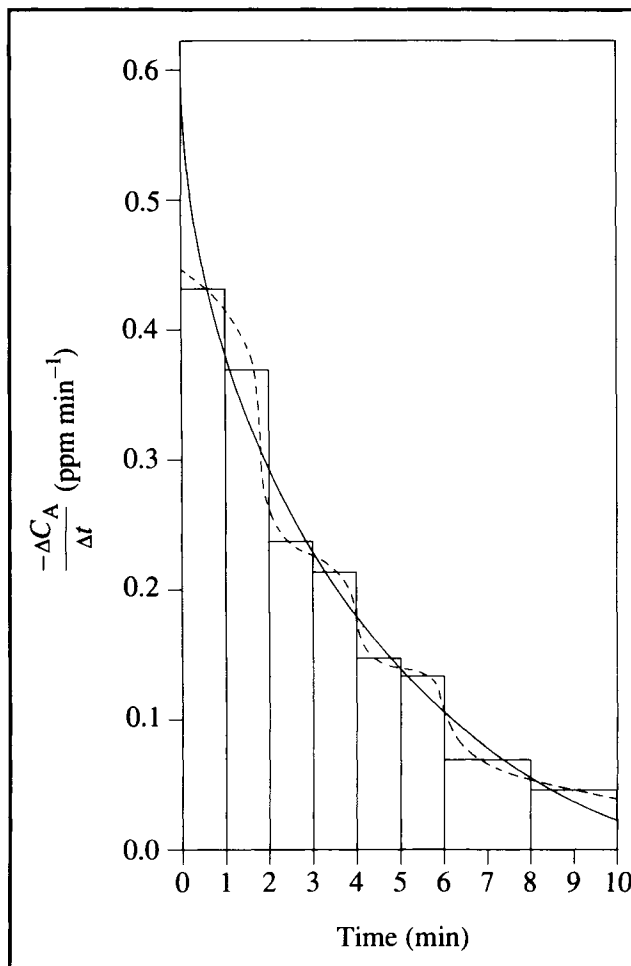
- (ii) Calculate average oxygen uptake rates, $\Delta C_A / \Delta t$, for each time interval.
- (iii) Plot $\Delta C_A / \Delta t$ on linear graph paper. Over each time interval a horizontal line is drawn to represent $\Delta C_A / \Delta t$ for that interval; this is shown in Figure 11.2.
- (iv) Draw a smooth curve to cut the horizontal lines in such a manner that the shaded areas above and below the curve are equal for each time interval. The curve thus developed gives values of dC_A / dt for all points in time. Results for dC_A / dt at the times of sampling can be read from the curve and are tabulated in Table 11.1.

Figure 11.2 Graphical differentiation using the average rate–equal area construction.



A disadvantage of the average rate–equal area method is that it is not easily applied if the data show scatter. If the concentration measurements are not very accurate, the horizontal lines representing $\Delta C_A / \Delta t$ may be located as shown in Figure 11.3. If we were to draw a curve equalising areas at each $\Delta C_A / \Delta t$ line, the rate curve would show complex behaviour oscillating up and down as indicated by the dashed line in Figure 11.3. Experience suggests that this is not a realistic representation of reaction rate. Because of the inaccuracies in measured data, we need several concentration measurements to define a change in rate. The data of Figure 11.3 are better represented using a smooth curve to equalise as far as possible the areas above and below adjacent groups of horizontal lines. For data showing even greater scatter, it may be necessary to average consecutive pairs of $\Delta C_A / \Delta t$ values to simplify graphical analysis.

Figure 11.3 Average rate–equal area method for data with experimental error.



A second graphical differentiation technique for evaluating dC_A/dt is described below.

11.2.2 Mid-Point Slope Method

In this method, the raw data are smoothed and values tabulated at intervals. The mid-point slope method is illustrated using the data of Table 11.1.

- (i) Plot the raw data and smooth by hand. This is shown in Figure 11.4.
- (ii) Mark off the smoothed curve at time intervals of ϵ . ϵ should be chosen so that the number of intervals is less than the number of data points measured; the less accurate the data the fewer should be the intervals. In this example, ϵ is taken as 1.0 min until $t=6$ min; thereafter ϵ

Figure 11.4 Graphical differentiation using the mid-point slope method.

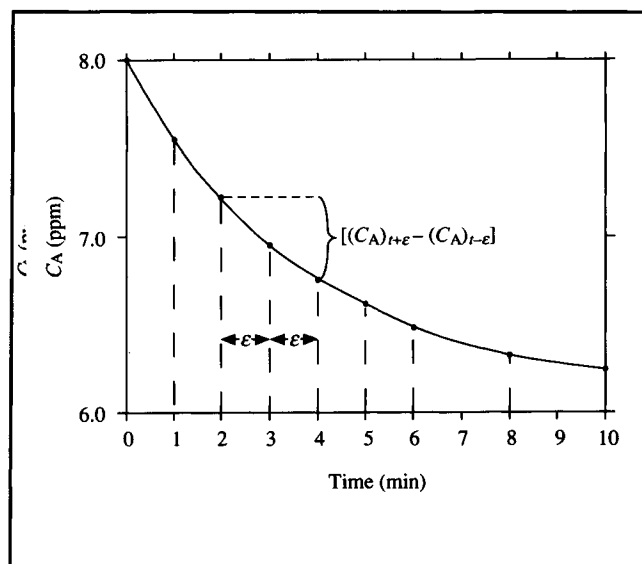


Table 11.2 Graphical differentiation using the mid-point slope method

Time (<i>t</i> , min)	Oxygen concentration (<i>C_A</i> , ppm)	ϵ	$[(C_A)_{t+\epsilon} - (C_A)_{t-\epsilon}]$	dC_A/dt
0.0	8.00	1.0	—	—
1.0	7.55	1.0	−0.78	−0.39
2.0	7.22	1.0	−0.59	−0.30
3.0	6.96	1.0	−0.46	−0.23
4.0	6.76	1.0	−0.35	−0.18
5.0	6.61	1.0	−0.27	−0.14
6.0	6.49	1.0	−0.22	−0.11
8.0	6.33	2.0	−0.24	−0.06
10.0	6.25	2.0	—	—

= 2.0 min. The intervals are marked in Figure 11.4 as dashed lines. Values of ϵ are entered in Table 11.2.

- (iii) In the mid-point slope method, rates are calculated mid-way between two adjacent intervals of size ϵ . Therefore, the first rate determination is made for $t = 1$ min. Calculate the differences $[(C_A)_{t+\epsilon} - (C_A)_{t-\epsilon}]$ from Figure 11.4, where $(C_A)_{t+\epsilon}$ denotes the concentration of A at time $t + \epsilon$, and $(C_A)_{t-\epsilon}$ denotes the concentration at time $t - \epsilon$. A difference calculation is illustrated in Figure 11.4 for $t = 3$ min. Note that the concentrations are not taken from the list of original data but are read from the

smoothed curve. When $t = 6$ min, $\epsilon = 1.0$; concentrations for the difference calculation are read from the curve at $t - \epsilon = 5$ min and $t + \epsilon = 7$ min. For the last rate determination at $t = 8$ min, $\epsilon = 2.0$ and the concentrations are read from the curve at $t - \epsilon = 6$ min and $t + \epsilon = 10$ min.

- (iv) The slope or rate is determined using the central-difference formula:

$$\frac{dC_A}{dt} = \frac{[(C_A)_{t+\epsilon} - (C_A)_{t-\epsilon}]}{2\epsilon} \quad (11.23)$$

These results are listed in Table 11.2.

Values of dC_A/dt calculated using the two differentiation methods compare favourably. Application of both methods allows checking of the results.

11.3 General Reaction Kinetics For Biological Systems

The kinetics of many biological reactions are either zero-order, first-order or a combination of these called Michaelis–Menten kinetics. Kinetic expressions for biological systems are examined in this section.

11.3.1 Zero-Order Kinetics

If a reaction obeys zero-order kinetics, the reaction rate is independent of reactant concentration. The kinetic expression is:

$$r_A = k_0 \quad (11.24)$$

where r_A is the volumetric rate of reaction with respect to A and k_0 is the zero-order rate constant. k_0 as defined in Eq. (11.24) is a volumetric rate constant with units of, e.g. $\text{kgmol m}^{-3} \text{s}^{-1}$. Because the volumetric rate of a catalytic reaction depends on the amount of catalyst present, when Eq. (11.24) is used to represent the rate of a cell or enzyme reaction, the value of k_0 includes the effect of catalyst concentration as well as the specific rate of reaction. We could write:

$$k_0 = k'_0 e \quad \text{or} \quad k_0 = k''_0 x \quad (11.25)$$

where k'_0 is the specific zero-order rate constant for enzyme reaction and e is the concentration of enzyme. Correspondingly, for cell reaction, k''_0 is the specific zero-order rate constant and x is cell concentration.

Let us assume we have collected concentration data for a particular reaction, and wish to determine the appropriate kinetic constant. If the reaction takes place in a closed, constant-volume system, rate of reaction can be evaluated directly as the rate of change in reactant concentration using the methods for graphical differentiation described in Section 11.2. From Eq. (11.24), if the reaction is zero-order the rate will be constant and equal to k_0 at all times during the reaction. Because the kinetic expression for zero-order reactions is relatively simple, rather than differentiate the concentration data it is easier to integrate Eq. (11.24) with $r_A = -dC_A/dt$ to obtain

an equation for C_A as a function of time. The experimental data can then be checked directly against the integrated equation. Integrating Eq. (11.24) with initial condition $C_A = C_{A0}$ at $t = 0$ gives:

$$C_A = \int -r_A dt = C_{A0} - k_0 t. \quad (11.26)$$

Therefore, when the reaction is zero order, a plot of C_A versus time gives a straight line with slope $-k_0$. Application of Eq. (11.26) is illustrated in Example 11.3.

Example 11.3 Kinetics of oxygen uptake

Serratia marcescens is cultured in minimal medium in a small stirred fermenter. Oxygen consumption is measured at a cell concentration of 22.7 g l^{-1} dry weight.

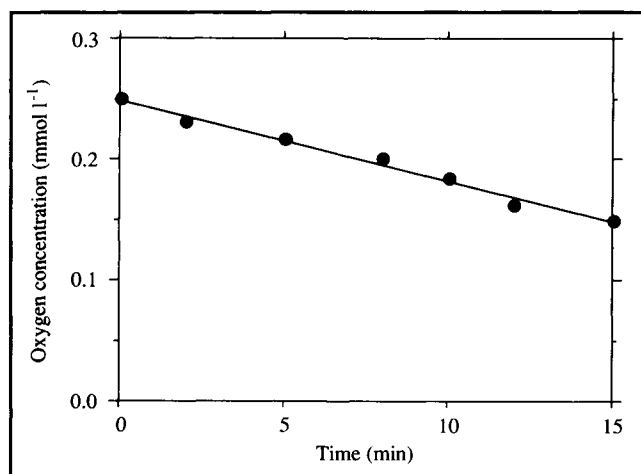
Time (min)	Oxygen concentration (mmol l^{-1})
0	0.25
2	0.23
5	0.21
8	0.20
10	0.18
12	0.16
15	0.15

- Determine the rate constant for oxygen uptake.
- If the cell concentration is reduced to 12 g l^{-1} , what is the value of the rate constant?

Solution:

- As indicated in Section 9.5.1, microbial oxygen consumption is a zero-order reaction over a wide range of oxygen concentrations above C_{crit} . To test if the measured data can be fitted using the zero-order model of Eq. (11.26), plot oxygen concentration as a function of time as shown in Figure 11E3.1.

Figure 11E3.1 Kinetic analysis of oxygen uptake.



The zero-order model fits the data well. The slope is $-6.7 \times 10^{-3} \text{ mmol l}^{-1} \text{ min}^{-1}$; therefore, $k_0 = 6.7 \times 10^{-3} \text{ mmol l}^{-1} \text{ min}^{-1}$.

(b) For cells of the same age cultured under the same conditions, from Eq. (11.25), k_0 can be expected to be directly proportional to the number of cells present. Therefore, at a cell concentration of 12 g l^{-1} :

$$k_0 = \frac{12 \text{ g l}^{-1}}{22.7 \text{ g l}^{-1}} (6.7 \times 10^{-3} \text{ mmol l}^{-1} \text{ min}^{-1}) = 3.5 \times 10^{-3} \text{ mmol l}^{-1} \text{ min}^{-1}.$$

11.3.2 First-Order Kinetics

If a reaction obeys first-order kinetics, the relationship between reaction rate and reactant concentration is as follows:

$$r_A = k_1 C_A \quad (11.27)$$

where r_A is the volumetric rate of reaction and k_1 is the *first-order rate constant* with dimensions T^{-1} . Like the zero-order constant of the previous section, the value of k_1 depends on the catalyst concentration.

Let us assume we follow the progress of a particular reaction in a closed, constant-volume system by measuring the concentration of reactant A as a function of time. Under these conditions, $r_A = -\text{d}C_A/\text{d}t$. To determine whether the reaction

follows first-order kinetics, we first integrate Eq. (11.27) with $r_A = -\text{d}C_A/\text{d}t$, and then check the measured concentration data against the resulting equation. Separating variables and integrating Eq. (11.27) with initial condition $C_A = C_{A0}$ at $t = 0$ gives:

$$C_A = C_{A0} e^{-k_1 t}. \quad (11.28)$$

Taking natural logarithms of both sides:

$$\ln C_A = \ln C_{A0} - k_1 t. \quad (11.29)$$

Therefore, for first-order reaction, a plot of $\ln C_A$ versus time gives a straight line with slope $-k_1$.

Example 11.4 Kinetics of gluconic acid production

Aspergillus niger is used to produce gluconic acid. Product synthesis is monitored in a fermenter; gluconic acid concentration is measured as a function of time for the first 39 h of culture.

Time (h)	Acid concentration (g l ⁻¹)
0	3.6
16	22
24	51
28	66
32	97
39	167

- Determine the rate constant.
- Estimate the product concentration after 20 h.

Solution:

(a) Test whether gluconic acid production can be modelled as a first-order reaction. If product concentration is measured rather than reactant concentration, in a closed reactor:

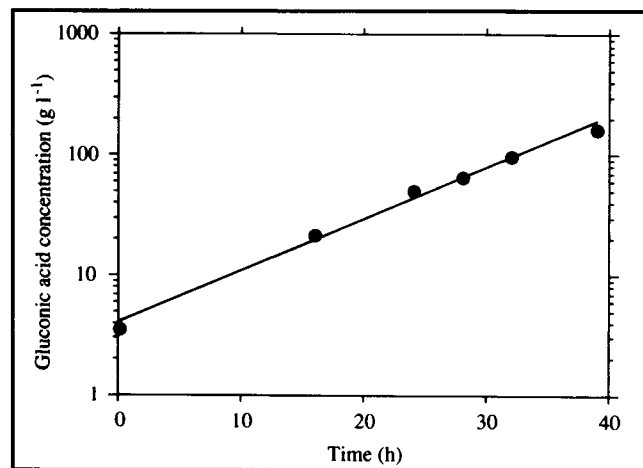
$$r_A = \frac{dC_A}{dt} = k_1 C_A$$

where A denotes gluconic acid. Integrating this equation and taking natural logarithms gives:

$$\ln C_A = \ln C_{A0} + k_1 t.$$

Therefore, a semi-log plot of gluconic acid concentration versus time will give a straight line with slope k_1 . As shown in Figure 11E4.1, the first-order model fits the data well.

Figure 11E4.1 Kinetic analysis of gluconic acid production.



The slope and intercept are evaluated as described in Section 3.4.2; $k_1 = 0.10 \text{ h}^{-1}$, $C_{A0} = 4.1 \text{ g l}^{-1}$.

(b) The kinetic equation is:

$$C_A = 4.1 e^{0.10t}$$

where C_A has units g l^{-1} and t has units h. Therefore, at $t = 20 \text{ h}$, $C_A = 30 \text{ g l}^{-1}$.

11.3.3 Michaelis–Menten Kinetics

The kinetics of most enzyme reactions are reasonably well represented by the *Michaelis–Menten equation*:

$$r_A = \frac{v_{\max} C_A}{K_m + C_A} \quad (11.30)$$

where r_A is the volumetric rate of reaction, C_A is the concentration of reactant A, $v_{\max}$ is the *maximum rate of reaction at infinite reactant concentration*, and K_m is the *Michaelis constant* for reactant A. $v_{\max}$ has the same dimensions as r_A ; K_m has the same dimensions as C_A . Typical units for $v_{\max}$ are $\text{kgmol m}^{-3} \text{ s}^{-1}$; typical units for K_m are kgmol m^{-3} . As defined in Eq. (11.30), $v_{\max}$ is a volumetric rate proportional to the amount of active enzyme present. The Michaelis constant K_m is equal to the reactant concentration at which $r_A = v_{\max}/2$.

K_m values for some enzyme–substrate systems are listed in Table 11.3. K_m and other enzyme properties depend on the source of the enzyme.

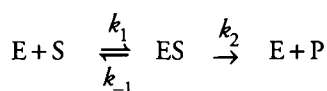
If we adopt conventional symbols for biological reactions and call reactant A the *substrate*, Eq. (11.30) can be rewritten in the familiar form:

$$v = \frac{v_{\max} s}{K_m + s} \quad (11.31)$$

where v is the volumetric rate of reaction and s is the substrate concentration. The biochemical basis of the Michaelis–Menten equation will not be covered here; discussion of enzyme reaction models and assumptions involved in derivation of Eq. (11.31) can be found in biochemistry texts [2, 3]. Suffice it to say here that the simplest reaction sequence which accounts for the kinetic properties of many enzymes is:

Table 11.3 Michaelis constants for some enzyme-substrate systems (From B. Atkinson and F. Mavituna, 1991, Biochemical Engineering and Biotechnology Handbook, 2nd edn, Macmillan, Basingstoke)

Enzyme	Source	Substrate	K_m (mM)
Alcohol dehydrogenase	<i>Saccharomyces cerevisiae</i>	Ethanol	13.0
α -Amylase	<i>Bacillus stearothermophilus</i>	Starch	1.0
	Porcine pancreas	Starch	0.4
β -Amylase	Sweet potato	Amylose	0.07
Aspartase	<i>Bacillus cadaveris</i>	L-Aspartate	30.0
β -Galactosidase	<i>Escherichia coli</i>	Lactose	3.85
Glucose oxidase	<i>Aspergillus niger</i>	D-Glucose	33.0
	<i>Penicillium notatum</i>	D-Glucose	9.6
Histidase	<i>Pseudomonas fluorescens</i>	L-Histidine	8.9
Invertase	<i>Saccharomyces cerevisiae</i>	Sucrose	9.1
	<i>Neurospora crassa</i>	Sucrose	6.1
Lactate dehydrogenase	<i>Bacillus subtilis</i>	Lactate	30.0
Penicillinase	<i>Bacillus licheniformis</i>	Benzylpenicillin	0.049
Urease	Jack bean	Urea	10.5



(11.32)

or

$$v \approx v_{\max}$$

(11.35)

where E is enzyme, S is substrate and P is product. ES is the *enzyme-substrate complex*. Binding of substrate to the enzyme in the first step is considered reversible with forward reaction constant k_1 and reverse reaction constant k_{-1} . Decomposition of the enzyme-substrate complex to give the product is an irreversible reaction with rate constant k_2 ; k_2 is known as the *turnover number*. Analysis of this reaction sequence yields the relationship:

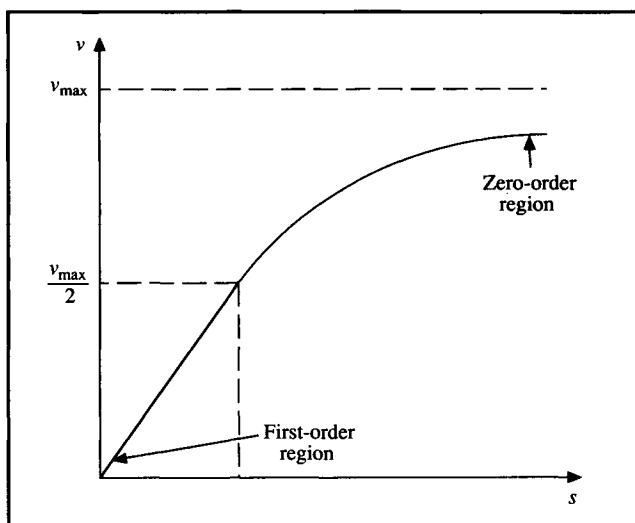
$$v_{\max} = k_2 e_a \quad (11.33)$$

where e_a is the concentration of active enzyme. As expected in catalytic reactions, enzyme E is recovered at the end of the reaction.

An essential feature of Michaelis-Menten kinetics is that the catalyst becomes saturated at high substrate concentrations. Figure 11.5 shows the form of Eq. (11.31); reaction rate v does not increase indefinitely with substrate concentration but approaches a limit, $v_{\max}$. At high substrate concentrations $s \gg K_m$, K_m in the denominator of Eq. (11.31) is negligibly small compared with s so we can write:

$$v \approx \frac{v_{\max} s}{s} \quad (11.34)$$

Therefore, at high substrate concentrations, the reaction rate approaches a constant value independent of substrate concentration; in this concentration range, the reaction is essentially *zero order* with respect to substrate. On the other hand, at low substrate concentrations $s \ll K_m$, the value of s in the denominator of Eq. (11.31) is negligible compared with K_m , and Eq. (11.31) can be simplified to:

Figure 11.5 Michaelis-Menten plot.

$$v \approx \frac{v_{\max}}{K_m} s. \quad (11.36)$$

The ratio of constants $v_{\max}/K_m$ is, in effect, a first-order rate coefficient for the reaction. Therefore, at low substrate concentrations there is an approximate linear dependence of reaction rate on s ; in this concentration range Michaelis–Menten reactions are essentially *first order* with respect to substrate.

The Michaelis–Menten equation is a satisfactory description of the kinetics of many industrial enzymes, although there are exceptions such as glucose isomerase and amyloglucosidase. More complex kinetic expressions must be applied if there are multiple substrates or inhibition effects [2–4]. Procedures for checking whether a particular reaction follows Michaelis–Menten kinetics and for evaluating $v_{\max}$ and K_m from experimental data are described in Section 11.4.

11.3.4 Effect of Conditions on Enzyme Reaction Rate

Rate of enzyme reaction is influenced by other conditions besides substrate concentration, such as temperature and pH. For enzymes with single rate-controlling steps, the effect of temperature is reasonably well described using the Arrhenius expression of Eq. (11.21) with $v_{\max}$ substituted for k . An example showing the relationship between temperature and rate of sucrose inversion by yeast invertase is given in Figure 11.6. Activation energies for enzyme reactions are of the order 40–80 kJ mol⁻¹ [5]; as a rough guide, this means that a 10°C rise in temperature between 20°C and 30°C will increase the rate of reaction by a factor of 2–3.

Although an Arrhenius-type relationship between temperature and rate of reaction is observed for enzymes, the temperature range over which Eq. (11.21) is applicable is quite limited. Many proteins start to denature at 45–50°C; if the temperature is raised higher than this, thermal deactivation occurs and the reaction velocity quickly drops. Figure 11.7 illustrates how the Arrhenius relationship breaks down at high temperatures. In this experiment, the Arrhenius rate-law was obeyed between temperatures of about 0°C ($T = 273.15$ K; $1/T = 3.66 \times 10^{-3}$ K⁻¹) and about 53°C ($T = 326.15$ K; $1/T = 3.07 \times 10^{-3}$ K⁻¹). With further increases in temperature the reaction rate declined rapidly due to thermal deactivation. Enzyme stability and rate of deactivation are important factors affecting overall catalytic performance in reactors. This topic is discussed further in the Section 11.5.

Figure 11.6 Arrhenius plot for inversion of sucrose by yeast invertase. (From I.W. Sizer, 1943, Effects of temperature on enzyme kinetics. *Adv. Enzymol.* 3, 35–62.)

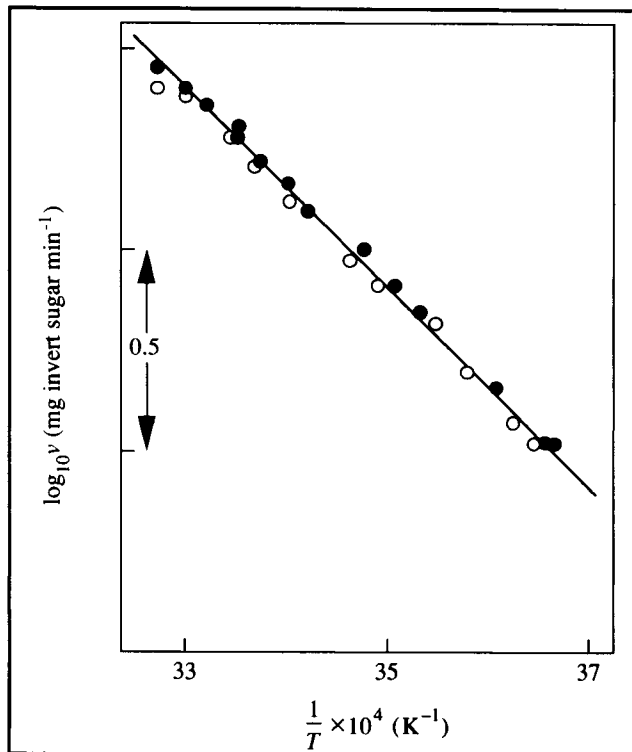


Figure 11.7 Arrhenius plot for catalase. The enzyme breaks down at high temperatures. (From I.W. Sizer, 1944, Temperature activation and inactivation of the crystalline catalase–hydrogen peroxide system. *J. Biol. Chem.* 154, 461–473.)

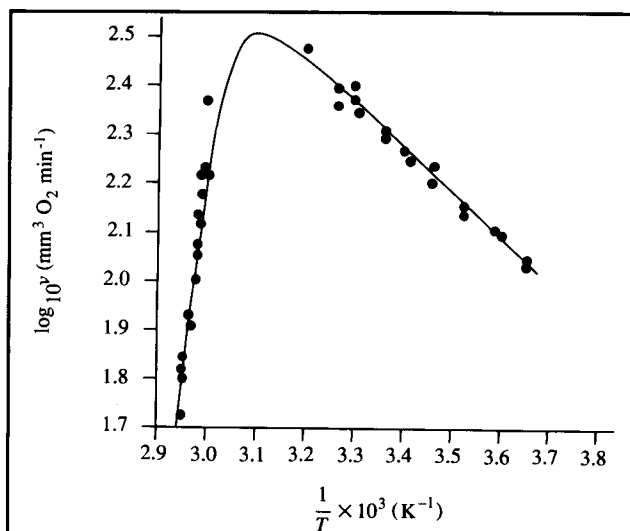
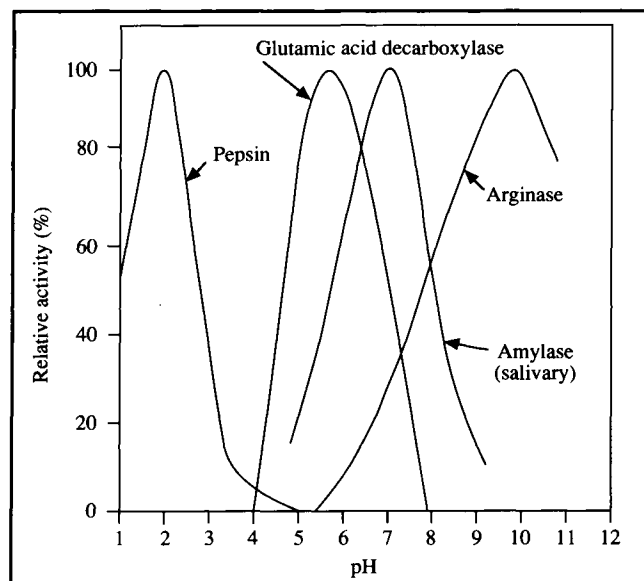


Figure 11.8 Effect of pH on enzyme activity. (From J.S. Fruton and S. Simmonds, 1958, *General Biochemistry*, 2nd edn, John Wiley, New York.)



pH has a pronounced effect on enzyme kinetics, as illustrated in Figure 11.8. The reaction rate is maximum at some optimal pH and declines sharply if the pH is moved either side of the optimum value. Kinetic equations have been developed to describe the effect of pH on enzyme activity; however, the influence of pH is usually determined experimentally. Ionic strength and water activity also have considerable influence on rate of enzyme reaction but few correlations are available for prediction of these effects.

11.4 Determining Enzyme Kinetic Constants From Batch Data

To fully specify the kinetics of Michaelis–Menten reactions, two rate constants, $v_{\max}$ and K_m , must be evaluated. Estimating kinetic parameters for Michaelis–Menten reactions is not as straightforward as for zero- and first-order reactions. Several graphical methods are available; unfortunately some do not give accurate results.

The first step in kinetic analysis of enzyme reactions is to obtain data for rate of reaction v as a function of substrate concentration s . Rates of reaction can be determined from batch concentration data as described in Section 11.2. Typically, only *initial rate data* are used. This means that several batch experiments are carried out with different initial substrate concentrations; from each set of data the reaction rate is evaluated

at time zero. Initial rates and corresponding initial substrate concentrations are used as (v, s) pairs which can then be plotted in various ways for determination of $v_{\max}$ and K_m . Initial rate data are preferred for enzyme reactions because experimental conditions such as enzyme and substrate concentrations are known most accurately at the start of the reaction.

11.4.1 Michaelis–Menten Plot

This simple procedure involves plotting (v, s) values directly as shown in Figure 11.5. $v_{\max}$ and K_m can be estimated roughly from this graph; $v_{\max}$ is the rate as $s \rightarrow \infty$ and K_m is the value of s at $v = v_{\max}/2$. The accuracy of this method is usually poor because of the difficulty of extrapolating to $v_{\max}$.

11.4.2 Lineweaver–Burk Plot

This method uses a linearisation procedure to give a straight-line plot from which $v_{\max}$ and K_m can be determined. Inverting Eq. (11.31) gives:

$$\frac{1}{v} = \frac{K_m}{v_{\max} s} + \frac{1}{v_{\max}} \quad (11.37)$$

so that a plot of $1/v$ versus $1/s$ should give a straight line with slope $K_m/v_{\max}$ and intercept $1/v_{\max}$. This double-reciprocal plot is known as the *Lineweaver–Burk plot*, and is frequently found in the literature on enzyme kinetics. However, the linearisation process used in this method distorts the experimental error in v (see Section 3.3.4) so that these errors are amplified at low substrate concentrations. As a consequence, the Lineweaver–Burk plot often gives inaccurate results and is therefore not recommended [3].

11.4.3 Eadie–Hofstee Plot

If Eq. (11.37) is multiplied by

$$v \left(\frac{v_{\max}}{K_m} \right)$$

and then rearranged, another linearised form of the Michaelis–Menten equation is obtained:

$$\frac{v}{s} = \frac{v_{\max}}{K_m} - \frac{v}{K_m} \quad (11.38)$$

According to Eq. (11.38), a plot of v/s versus v gives a straight line with slope $-1/K_m$ and intercept $v_{\max}/K_m$; this is called the *Eadie–Hofstee plot*. As with the Lineweaver–Burk plot, the Eadie–Hofstee linearisation distorts errors in the data so that the method has reduced accuracy.

11.4.4 Langmuir Plot

Multiplying Eq. (11.37) by s produces the linearised form of the Michaelis–Menten equation according to Langmuir:

$$\frac{s}{v} = \frac{K_m}{v_{\max}} + \frac{s}{v_{\max}}. \quad (11.39)$$

Therefore, a *Langmuir plot* of $1/v$ versus s should give a straight line with slope $1/v_{\max}$ and intercept $K_m/v_{\max}$. Linearisation of data for the Langmuir plot minimises distortions in experimental error. Accordingly, its use for evaluation of $v_{\max}$ and K_m is recommended [6].

11.4.5 Direct Linear Plot

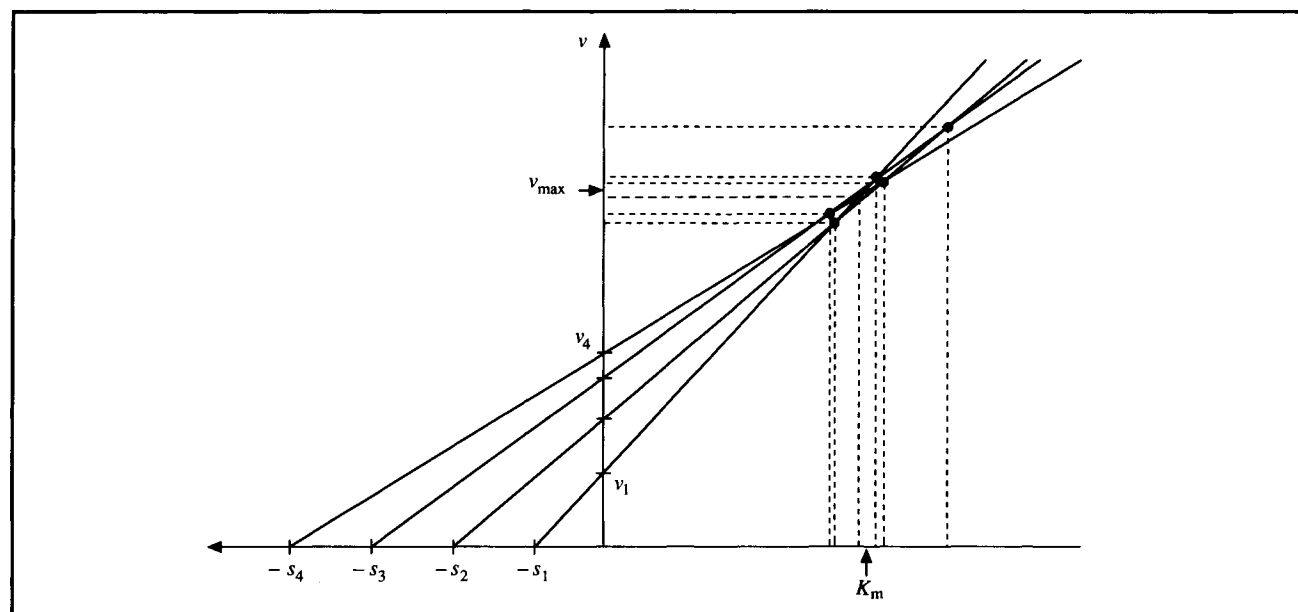
A different method for plotting enzyme kinetic data has been proposed by Eisenthal and Cornish-Bowden [7]. For each

observation, reaction rate v is plotted on the vertical axis against s on the negative horizontal axis. This is shown in Figure 11.9 for four pairs of (v, s) data. A straight line is then drawn to join corresponding $(-s, v)$ points. In the absence of experimental error, lines for each $(-s, v)$ pair intersect at a unique point, $(K_m, v_{\max})$. When real data containing errors are plotted, a family of intersection points is obtained. Each intersection gives one estimate of $v_{\max}$ and K_m ; the median or middle $v_{\max}$ and K_m values are taken as the kinetic parameters for the reaction. This method is relatively insensitive to individual erroneous readings which may be far from the correct values. However a disadvantage of the procedure is that deviations from Michaelis–Menten behaviour are not easily detected. It is recommended therefore for enzymes which are known to obey Michaelis–Menten kinetics.

11.5 Kinetics of Enzyme Deactivation

Enzymes are protein molecules of complex configuration that can be destabilised by relatively weak forces. In the course of enzyme-catalysed reactions, enzyme deactivation occurs at a rate which is dependent on the structure of the enzyme and the reaction conditions. Environmental factors affecting enzyme stability include temperature, pH, ionic strength, mechanical forces and presence of denaturants such as solvents, detergents

Figure 11.9 Direct linear plot for determination of enzyme kinetic parameters. (From R. Eisenthal and A. Cornish-Bowden, 1974, *The direct linear plot: a new graphical procedure for estimating enzyme kinetic parameters. Biochem. J.* 139, 715-720.)



and heavy metals. Because the amount of active enzyme can decline considerably during reaction, in many applications the kinetics of enzyme deactivation are just as important as the kinetics of the reaction itself.

In the simplest model of enzyme deactivation, active enzyme E_a undergoes irreversible transformation to an inactive form E_i :



Rate of deactivation is generally considered to be first order in active enzyme concentration:

$$r_d = k_d e_a \quad (11.41)$$

where r_d is the volumetric rate of deactivation, e_a is the active enzyme concentration and k_d is the *deactivation rate constant*. In a closed system where enzyme deactivation is the only process affecting the concentration of active enzyme:

$$\frac{-de_a}{dt} = r_d = k_d e_a \quad (11.42)$$

Integration of Eq. (11.42) gives an expression for active enzyme concentration as a function of time:

$$e_a = e_{a0} e^{-k_d t} \quad (11.43)$$

where e_{a0} is the concentration of active enzyme at time zero. According to Eq. (11.43), concentration of active enzyme decreases exponentially with time; the greatest rate of enzyme deactivation occurs when e_a is high.

As indicated in Eq. (11.33), the value of $v_{\max}$ for enzyme reaction depends on the amount of active enzyme present. Therefore, as e_a declines due to deactivation, $v_{\max}$ is also diminished. We can estimate the variation of $v_{\max}$ with time

by substituting into Eq. (11.33) the expression for e_a from Eq. (11.43):

$$v_{\max} = k_2 e_{a0} e^{-k_d t} = v_{\max 0} e^{-k_d t} \quad (11.44)$$

where $v_{\max 0}$ is the initial value of $v_{\max}$ before deactivation occurs.

Stability of enzymes is frequently reported in terms of *half-life*. Half-life is the time required for half the enzyme activity to be lost as a result of deactivation; after one half-life, the active enzyme concentration equals $e_{a0}/2$. Substituting $e_a = e_{a0}/2$ into Eq. (11.43), taking logarithms and rearranging yields the following expression:

$$t_h = \frac{\ln 2}{k_d} \quad (11.45)$$

where t_h is the enzyme half-life.

Rate of enzyme deactivation is strongly dependent on temperature. This dependency is generally well described using the Arrhenius equation:

$$k_d = A e^{-E_d/RT} \quad (11.46)$$

where A is the Arrhenius constant or frequency factor, E_d is the activation energy for enzyme deactivation, R is the ideal gas constant, and T is absolute temperature. According to Eq. (11.46), as T increases, rate of enzyme deactivation increases exponentially. Values of E_d are high, of the order 170–400 kJ gmol⁻¹ for many enzymes [5]. Accordingly, a temperature rise of 10°C between 30°C and 40°C will increase the rate of enzyme deactivation by a factor between 10 and 150. The stimulatory effect of increasing temperature on rate of enzyme reaction has already been described in Section 11.3.4. However, as shown here, raising the temperature also reduces the amount of active enzyme present. It is clear that temperature has a critical effect on enzyme kinetics.

Example 11.5 Enzyme half-life

Amyloglucosidase from *Endomycopsis bispora* is immobilised in polyacrylamide gel. Activities of immobilised and soluble enzyme are compared at 80°C. Initial rate data measured at a fixed substrate concentration are listed below.

Time (min)	Enzyme activity ($\mu\text{mol ml}^{-1} \text{min}^{-1}$)	
	Soluble enzyme	Immobilised enzyme
0	0.86	0.45
3	0.79	0.44
6	0.70	0.43
9	0.65	0.43
15	0.58	0.41
20	0.46	0.40
25	0.41	0.39
30	—	0.38
40	—	0.37

What is the half-life for each form of enzyme?

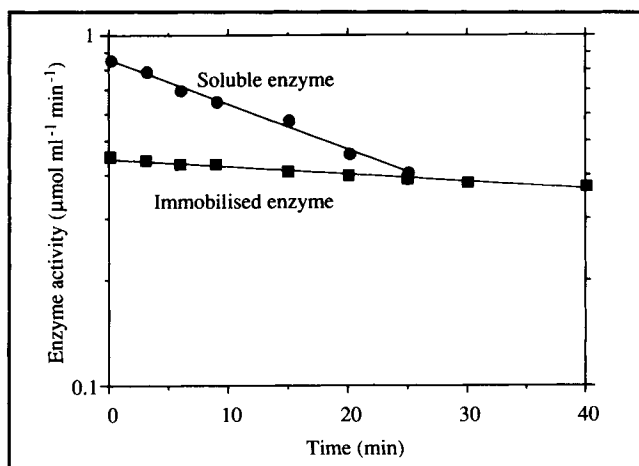
Solution:

From Eq. (11.31), at any fixed substrate concentration, the rate of enzyme reaction v is directly proportional to v_{max} . Therefore, k_d can be determined from Eq. (11.44) using enzyme activity v instead of v_{max} . Taking natural logarithms gives:

$$\ln v = \ln v_0 - k_d t$$

where v_0 is the initial enzyme activity before deactivation. So, if deactivation follows a first-order model, a semi-log plot of reaction rate versus time should give a straight line with slope $-k_d$. The data are plotted in Figure 11E5.1.

Figure 11E5.1 Kinetic analysis of enzyme deactivation.



From the slopes, k_d for soluble enzyme is 0.03 min^{-1} ; k_d for immobilised enzyme is 0.005 min^{-1} . Applying Eq. (11.45) for half-life:

$$t_h (\text{soluble}) = \frac{\ln 2}{0.03 \text{ min}^{-1}} = 23 \text{ min}$$

$$t_h (\text{immobilised}) = \frac{\ln 2}{0.005 \text{ min}^{-1}} = 139 \text{ min.}$$

Immobilisation significantly enhances stability of the enzyme.

11.6 Yields in Cell Culture

The basic concept of reaction yield was introduced in Section 11.1.2 for simple one-step reactions. When we consider processes such as cell growth, we are in effect lumping together many individual enzyme and chemical conversions. Despite this complexity, yield principles can be applied to cell metabolism to relate flow of substrate in metabolic pathways to formation of biomass and other products. Yields which are frequently reported and of particular importance are expressed using *yield coefficients* or *yield factors*. Several yield coefficients, such as yield of biomass from substrate, yield of biomass from oxygen, and yield of product from substrate, are in common use. Yield coefficients allow us to quantify the nutrient requirements and production characteristics of an organism.

Some metabolic yield coefficients: the biomass yield Y_{XS} , the product yield Y_{PS} , and the respiratory quotient RQ , were introduced in Chapter 4. Definition of yield coefficients can be generalised as follows:

$$Y_{FG} = \frac{-\Delta F}{\Delta G} \quad (11.47)$$

where Y_{FG} is the yield factor, F and G are substances involved in metabolism, ΔF is the mass or moles of F produced, and ΔG is the mass or moles of G consumed. The negative sign is required in Eq. (11.47) because ΔG for a consumed substance is negative in value; yield is calculated as a positive quantity. A list of frequently-used yield coefficients is given in Table 11.4. Note that in some cases, such as Y_{PX} , both substances represented by the yield coefficient are products of metabolism. Although the term 'yield' usually refers to the amount of product formed per amount of reactant, yields can also be used to relate other quantities. Some yield coefficients are based on quantities such as ATP formed or heat evolved during metabolism.

Table 11.4 Some metabolic yield coefficients

<i>Symbol</i>	<i>Definition</i>
Y_{XS}	Mass or moles of biomass produced per unit mass or mole of substrate consumed. (Moles of biomass can be calculated from the 'molecular formula' for biomass; see Section 4.6.1)
Y_{PS}	Mass or moles of product formed per unit mass or mole of substrate consumed
Y_{PX}	Mass or moles of product formed per unit mass or mole of biomass formed
Y_{XO}	Mass or moles of biomass formed per unit mass or mole of oxygen consumed
Y_{CS}	Mass or moles of carbon dioxide formed per unit mass or mole of substrate consumed
RQ	Moles of carbon dioxide formed per mole of oxygen consumed. This yield is called the <i>respiratory quotient</i> .
Y_{ATP}	Mass or moles of biomass formed per mole of ATP formed
Y_{kcal}	Mass or moles of biomass formed per kilocalorie of heat evolved during fermentation

11.6.1 Overall and Instantaneous Yields

A problem with application of Eq. (11.47) is that values of ΔF and ΔG depend on the time period over which they are measured. In batch culture, ΔF and ΔG can be calculated as the difference between initial and final values; this gives an *overall yield* representing some sort of average value for the entire culture period. Alternatively, ΔF and ΔG can be determined between two other points in time; this calculation might produce a different value of Y_{FG} . Yields can vary during culture, and it is sometimes necessary to evaluate the *instantaneous yield* at a particular point in time. For a closed, constant-volume reactor in which the reaction between F and G is the only

reaction involving these components, if r_F and r_G are volumetric rates of production and consumption of F and G, respectively, instantaneous yield can be calculated as follows:

$$Y_{FG} = \lim_{\Delta G \rightarrow 0} \frac{-\Delta F}{\Delta G} = \frac{-dF}{dG} = \frac{-dF/dt}{dG/dt} = \frac{r_F}{r_G} \quad (11.48)$$

For example, Y_{XS} at a particular instant in time is defined as:

$$Y_{XS} = \frac{r_X}{r_S} = \frac{\text{growth rate}}{\text{substrate consumption rate}} \quad (11.49)$$

When yields for fermentation are reported, the time or time period to which they refer should be stated.

11.6.2 Theoretical and Observed Yields

As described in Section 11.1.2, it is necessary to distinguish between theoretical and observed yields. This is particularly important for cell metabolism because there are always many reactions occurring at the same time; theoretical and observed yields are therefore very likely to differ. Consider the example of biomass yield from substrate, Y_{XS} . If the total mass of substrate consumed is S_T , some proportion of S_T equal to S_G will be used for growth while the remainder, S_R , is channelled into other products and metabolic activities not related to growth. Therefore, the observed biomass yield based on total substrate consumption is:

$$Y'_{XS} = \frac{-\Delta X}{\Delta S_T} = \frac{-\Delta X}{\Delta S_G + \Delta S_R} \quad (11.50)$$

where ΔX is the amount of biomass produced and Y'_{XS} is the *observed biomass yield from substrate*. Values of observed biomass yields for several organisms and substrates are listed in Table 11.5. In comparison, the *true* or *theoretical biomass yield* from substrate is:

$$Y_{XS} = \frac{-\Delta X}{\Delta S_G} \quad (11.51)$$

as ΔS_G is the mass of substrate actually directed into biomass production. Because of the complexity of metabolism, ΔS_G is usually unknown and the observed yield is the only yield available. Theoretical yields are sometimes referred to as *maximum possible yields* because they represent the yield in the absence of competing reactions.

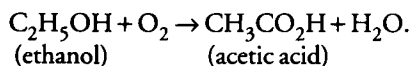
Table 11.5 Observed biomass yields for several microorganisms and substrates

(From S.J. Pirt, 1975, Principles of Microbe and Cell Cultivation, Blackwell Scientific, Oxford)

Microorganism	Substrate	Observed biomass yield Y'_{XS} (g g ⁻¹)
<i>Aerobacter cloacae</i>	Glucose	0.44
<i>Penicillium chrysogenum</i>	Glucose	0.43
<i>Candida utilis</i>	Glucose	0.51
	Acetic acid	0.36
	Ethanol	0.68
<i>Candida intermedia</i>	n-Alkanes (C ₁₆ –C ₂₂)	0.81
<i>Pseudomonas</i> sp.	Methanol	0.41
<i>Methylococcus</i> sp.	Methane	1.01

Example 11.6 Yields in acetic acid production

The equation for aerobic production of acetic acid from ethanol is:



Acetobacter aceti bacteria are added to vigorously-aerated medium containing 10 g l⁻¹ ethanol. After some time, the ethanol concentration is 2 g l⁻¹ and 7.5 g l⁻¹ acetic acid is produced. How does the overall yield of acetic acid from ethanol compare with the theoretical yield?

Solution:

Using a basis of 1 litre, the observed yield over the entire culture period is obtained from application of Eq. (11.10):

$$Y'_{PS} = \frac{7.5 \text{ g}}{(10-2) \text{ g}} = 0.94 \text{ g g}^{-1}.$$

Theoretical yield is based on the mass of ethanol actually used for synthesis of acetic acid. From the stoichiometric equation:

$$Y'_{PS} = \frac{1 \text{ gmol acetic acid}}{1 \text{ gmol ethanol}} = \frac{60 \text{ g}}{46 \text{ g}} = 1.30 \text{ g g}^{-1}.$$

The observed yield is 72% theoretical.

11.7 Cell Growth Kinetics

The kinetics of cell growth are expressed using equations similar to those presented in Section 11.3. From a mathematical point of view there is little difference between the kinetic equations for enzymes and cells; after all, cell metabolism depends on the integrated action of a multitude of enzymes.

11.7.1 Batch Growth

Several phases of cell growth are observed in batch culture; a typical growth curve is shown in Figure 11.10. The different phases of growth are more readily distinguished when the natural logarithm of viable cell concentration is plotted against time; alternatively, a semi-log plot can be used. Rate of growth varies depending on the growth phase. During the lag phase immediately after inoculation, rate of growth is essentially zero. Cells use the lag phase to adapt to their new environment; new enzymes or structural components may be synthesised. Following the lag period, growth starts in the acceleration phase and continues through the growth and decline phases. If growth is exponential, the growth phase appears as a straight line on a semi-log plot. As nutrients in the culture medium become depleted or inhibitory products accumulate, growth slows down and the cells enter the decline phase. After this transition period, the stationary phase is reached during which no further growth occurs. Some cultures exhibit a death phase as the cells lose viability or are destroyed by lysis. Table 11.6 provides a summary of growth and metabolic activity during the phases of batch culture.

During the growth and decline phases, rate of cell growth is described by the equation:

$$r_X = \mu x \tag{11.52}$$

where r_X is the volumetric rate of biomass production with units of, for example, $\text{kg m}^{-3} \text{ s}^{-1}$, x is viable cell concentration

Figure 11.10 Typical batch growth curve.

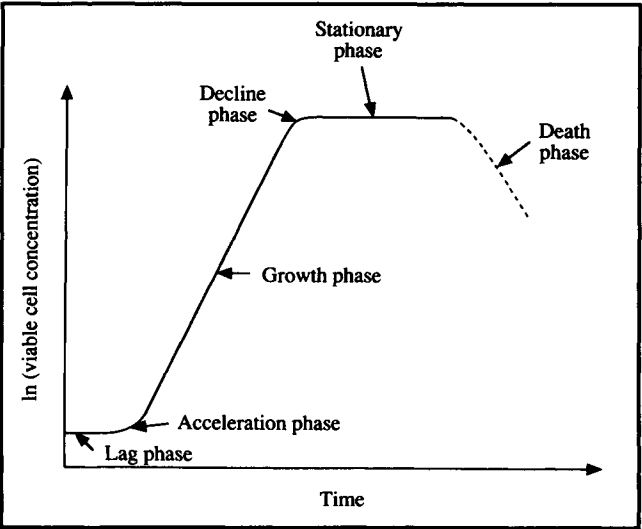


Table 11.6 Summary of batch cell growth

Phase	Description	Specific growth rate
Lag	Cells adapt to the new environment; no or very little growth	$\mu \approx 0$
Acceleration	Growth starts	$\mu < \mu_{\max}$
Growth	Growth achieves its maximum rate	$\mu \approx \mu_{\max}$
Decline	Growth slows due to nutrient exhaustion or build-up of inhibitory products	$\mu < \mu_{\max}$
Stationary	Growth ceases	$\mu = 0$
Death	Cells lose viability and lyse	$\mu < 0$

with units of, for example, kg m^{-3} , and μ is the *specific growth rate*. Specific growth rate has dimensions T^{-1} . Eq. (11.52) has the same form as (11.27); cell growth is therefore considered a *first-order autocatalytic reaction*. In a closed system where growth is the only process affecting cell concentration, $r_X = \frac{dx}{dt}$, and integration of Eq. (11.52) gives an expression for x as a function of time. If μ is constant we can integrate directly with initial condition $x = x_0$ at $t = 0$ to give:

$$x = x_0 e^{\mu t} \quad (11.53)$$

where x_0 is the viable cell concentration at time zero. Eq. (11.53) represents *exponential growth*. Taking natural logarithms:

$$\ln x = \ln x_0 + \mu t. \quad (11.54)$$

According to Eq. (11.54), a plot of $\ln x$ versus time gives a straight line with slope μ . Because the relationship of Eq. (11.54) is strictly valid only if μ is unchanging, a plot of $\ln x$ versus t is often used to assess whether the specific growth rate is constant. As shown in Figure 11.10, μ is usually constant during the growth phase. It is always advisable to prepare a semi-log plot of cell concentration before identifying phases of growth. As shown in Figure 3.6, if cell concentrations are plotted on linear coordinates, growth often appears slow at the beginning of the culture. We might be tempted to conclude there was a lag phase of 1–2 hours for the culture represented in Figure 3.6(a). However, when the same data are plotted using logarithms as shown in Figure 3.6(b), it is clear that the culture did not experience a lag phase. Growth always appears much slower at the beginning of culture because the number of cells present is small.

Cell growth rates are often expressed in terms of the *doubling time* t_d . An expression for doubling time can be derived from Eq. (11.53). Starting with a cell concentration of x_0 , the concentration at $t = t_d$ is $2x_0$. Substituting these values into Eq. (11.53):

$$2x_0 = x_0 e^{\mu t_d} \quad (11.55)$$

and cancelling x_0 gives:

$$2 = e^{\mu t_d}. \quad (11.56)$$

Taking the natural logarithm of both sides:

$$\ln 2 = \mu t_d \quad (11.57)$$

or

$$t_d = \frac{\ln 2}{\mu}. \quad (11.58)$$

11.7.2 Balanced Growth

In an environment favourable for growth, cells regulate their metabolism and adjust the rates of various internal reactions so that a condition of *balanced growth* occurs. During balanced growth, composition of the biomass remains constant. Balanced growth means that the cell is able to modulate the effect of external conditions and keep the cell composition steady despite changes in environmental conditions.

For biomass composition to remain constant during growth, the specific rate of production of each component in the culture must be equal to the cell specific growth-rate μ :

$$r_Z = \mu z \quad (11.59)$$

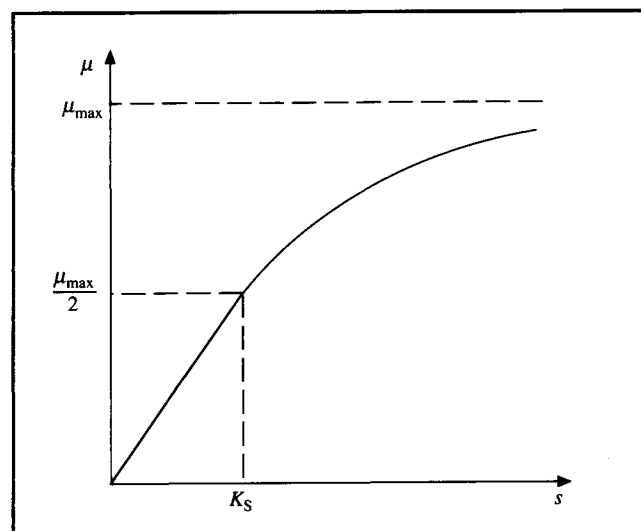
where Z is a cellular constituent such as protein, RNA, polysaccharide, etc., r_Z is the volumetric rate of production of Z , and z is the concentration of Z in the reactor volume. Therefore, during balanced growth the doubling time for each cell component must be equal to t_d for growth. Balanced growth cannot be achieved if environmental changes affect rate of growth. In most cultures, balanced growth occurs at the same time as exponential growth.

11.7.3 Effect of Substrate Concentration

During the growth and decline phases of batch culture, the specific growth rate of cells is dependent on the concentration of nutrients in the medium. Often, a single substrate exerts a dominant influence on rate of growth; this component is known as the *growth-rate-limiting substrate* or, more simply, the *growth-limiting substrate*. The growth-limiting substrate is often the carbon or nitrogen source, although in some cases it is oxygen or another oxidant such as nitrate. During balanced growth, the specific growth rate is related to the concentration of growth-limiting substrate by the *Monod equation*, a homologue of the Michaelis–Menten expression:

$$\mu = \frac{\mu_{\max} s}{K_s + s}. \quad (11.60)$$

Figure 11.11 Relationship between specific growth rate and concentration of growth-limiting substrate in cell culture.



In Eq. (11.60), s is the concentration of growth-limiting substrate, $\mu_{\max}$ is the *maximum specific growth rate*, and K_S is the *substrate constant*. $\mu_{\max}$ has dimensions T^{-1} ; K_S has the same dimensions as substrate concentration. The form of Eq. (11.60) is shown in Figure 11.11. $\mu_{\max}$ and K_S are intrinsic parameters of the cell–substrate system; values of K_S for several organisms are listed in Table 11.7.

Typical values of K_S are very small, of the order of mg per litre for carbohydrate substrates and μg per litre for other compounds such as amino acids. The level of growth-limiting substrate in culture media is normally much greater than K_S . As a result, growth can be approximated using zero-order kinetics with growth rate independent of substrate concentration until s reaches very low values. By analogy with Michaelis–Menten kinetics described in Section 11.3.3, $\mu \approx \mu_{\max}$ provided s is greater than about $10 K_S$. Because K_S is usually very small compared with the starting substrate concentration, s remains $> 10 K_S$ during most of the culture period. This explains why μ remains constant and equal to $\mu_{\max}$ in batch culture until the medium is virtually exhausted of substrate. When s finally falls below $10 K_S$, transition from growth to stationary phase can be very abrupt as the low level of residual substrate is rapidly consumed by the large number of cells present.

The Monod equation is by far the most frequently-used expression relating growth rate to substrate concentration. However, it is valid only for balanced growth and should not be applied when growth conditions are changing rapidly.

Table 11.7 K_S values for several organisms

(From S.J. Pirt, 1975, *Principles of Microbe and Cell Cultivation*, Blackwell Scientific, Oxford; and D.I.C. Wang, C.L. Cooney, A.L. Demain, P. Dunnill, A.E. Humphrey and M.D. Lilly, 1979, *Fermentation and Enzyme Technology*, John Wiley, New York)

Microorganism (genus)	Limiting substrate	K_S (mg l ⁻¹)
<i>Saccharomyces</i>	Glucose	25
<i>Escherichia</i>	Glucose	4.0
	Lactose	20
	Phosphate	1.6
<i>Aspergillus</i>	Glucose	5.0
<i>Candida</i>	Glycerol	4.5
	Oxygen	0.042–0.45
<i>Pseudomonas</i>	Methanol	0.7
	Methane	0.4
<i>Klebsiella</i>	Carbon dioxide	0.4
	Magnesium	0.56
	Potassium	0.39
	Sulphate	2.7
<i>Hansenula</i>	Methanol	120.0
	Ribose	3.0
<i>Cryptococcus</i>	Thiamine	1.4×10^{-7}

There are also other restrictions; for example, the Monod equation has been found to have limited applicability at extremely low substrate levels. When growth is inhibited by high substrate or product concentrations, extra terms can be added to the Monod equation to account for these effects. Several other kinetic expressions have been developed for cell growth; these provide better correlations with experimental data in certain situations [8–11].

11.8 Growth Kinetics With Plasmid Instability

A potential problem in culture of recombinant organisms is plasmid loss or inactivation. Plasmid instability occurs in individual cells which, by reproducing, can generate a large plasmid-free population in the reactor and reduce the overall rate of synthesis of plasmid-encoded products. Plasmid instability occurs as a result of DNA mutation or defective plasmid segregation. For segregational stability, the total number of plasmids present in the culture must double once per generation, and the plasmid copies must be equally distributed between mother and daughter cells.

A simple model has been developed for batch culture to describe changes in the fraction of plasmid-bearing cells as a function of time [12]. The important parameters in this model are the probability of plasmid loss per generation of cells, and the difference in the growth rates of plasmid-bearing and plasmid-free cells. Exponential growth of the host cells is assumed. If x^+ is the concentration of plasmid-carrying cells and x^- is the concentration of plasmid-free cells, the rates at which the two cell populations grow are:

$$r_{X^+} = (1 - p) \mu^+ x^+ \quad (11.61)$$

and

$$r_{X^-} = p \mu^+ x^+ + \mu^- x^- \quad (11.62)$$

where r_{X^+} is the rate of growth of the plasmid-bearing population, r_{X^-} is the rate of growth of the plasmid-free population, p is the probability of plasmid loss per cell division ($p \ll 1$), μ^+ is the specific growth rate of plasmid-carrying cells, and μ^- is the specific growth rate of plasmid-free cells. The model assumes that all plasmid-containing cells are identical in growth rate and probability of plasmid loss; this is the same as assuming that all plasmid-containing cells have the same copy number. By comparing Eq. (11.61) with Eq. (11.52) we can see that the rate of growth of the plasmid-bearing population is reduced by $p \mu^+ x^+$. This is because some of the progeny of plasmid-bearing cells do not contain plasmid and do not join the plasmid-bearing population. On the other hand, growth of the plasmid-free population has two contributions as indicated in Eq. (11.62). Existing plasmid-free cells grow with specific growth rate μ^- as usual; in addition, this population is supplemented by generation of plasmid-free cells due to defective plasmid segregation by plasmid-carrying cells.

At any time, the fraction of cells in the culture with plasmid is:

$$F = \frac{x^+}{x^+ + x^-} \quad (11.63)$$

In batch culture where rates of growth can be determined by monitoring cell concentration, $r_{X^+} = dx^+/dt$ and $r_{X^-} = dx^-/dt$. Therefore, Eqs (11.61) and (11.62) can be integrated simul-

taneously with initial condition $x^+ = x_0^+$ and $x^- = x_0^-$ at $t = 0$. After n generations of plasmid-containing cells:

$$F = \frac{1 - \alpha - p}{1 - \alpha - 2^{n(\alpha+p-1)} p} \quad (11.64)$$

where

$$\alpha = \frac{\mu^-}{\mu^+} \quad (11.65)$$

and

$$n = \frac{\mu^+ t}{\ln 2} \quad (11.66)$$

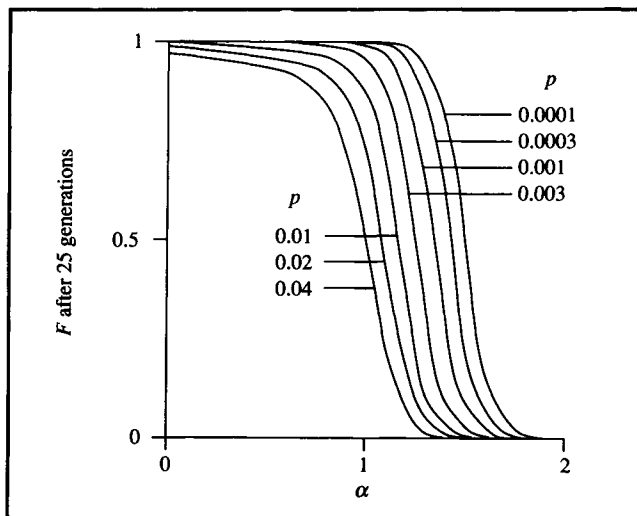
The value of F depends on α , the ratio of the specific growth rates of plasmid-free and plasmid-carrying cells. In the absence of selection pressure, presence of plasmid usually reduces the growth rate of organisms due to the additional metabolic requirements imposed by the plasmid DNA. Therefore α is usually > 1 . In general, the difference between μ^+ and μ^- becomes more pronounced as the size of the

Table 11.8 Relative growth rates of plasmid-free and plasmid-carrying cells

(From T. Imanaka and S. Aiba, 1981, *A perspective on the application of genetic engineering: stability of recombinant plasmid*. Ann. N.Y. Acad. Sci. 369, 1-14)

Organism	Plasmid	$\alpha = \frac{\mu^-}{\mu^+}$
<i>Escherichia coli</i> C600	F' lac	0.99-1.10
<i>E. coli</i> K12 EC1055	R1 drd-19	1.03-1.12
<i>E. coli</i> K12 IR713	TP120 (various)	1.50-2.31
<i>E. coli</i> JC7623	Col E1	1.29
	Col E1 derivative TnA insertion (various)	1.15-1.54
	Col E1 deletion mutant (various)	1.06-1.65
<i>Pseudomonas aeruginosa</i> PA01	TOL	2.00

Figure 11.12 Fraction of plasmid-carrying cells in batch culture after 25 generations. (From T. Imanaka and S. Aiba, 1981, A perspective on the application of genetic engineering: stability of recombinant plasmid. *Ann. N.Y. Acad. Sci.* 369, 1–14.)



plasmid or copy number increases. Some values of α from the literature are listed in Table 11.8; typically $1.0 < \alpha < 2.0$. Under selection pressure α may equal zero; if the plasmid encodes biosynthetic enzymes for essential nutrients, loss of plasmid may result in $\mu^- = 0$. When this is the case, F remains close to 1 as the plasmid-free population cannot reproduce. F also depends on p , the probability of plasmid loss per generation, which can be as high as 0.1 if segregation occurs. When mutation or random insertions or deletions are the only cause of plasmid instability, p is usually much lower at about 10^{-6} . Plasmid fragmentation within a host cell can occur with higher frequency if the cloning vector is inherently unstable.

Batch culture of microorganisms usually requires 25 cell generations or more. Results for F after 25 generations have been calculated from Eq. (11.64) and are shown in Figure 11.12 as function of p and α . F deteriorates substantially as α increases from 1.0 to 2.0. Cultures with $p < 0.01$ and $\alpha < 1$ are relatively stable, with F after 25 generations remaining close to 1. Further application of Eq. (11.64) is illustrated in Example 11.7.

Example 11.7 Plasmid instability in batch culture

A plasmid-containing strain of *E. coli* is used to produce recombinant protein in a 250-litre fermenter. The probability of plasmid loss per generation is 0.005. The specific growth rate of plasmid-free cells is 1.4 h^{-1} ; the specific growth-rate of plasmid-bearing cells is 1.2 h^{-1} . Estimate the fraction of plasmid-bearing cells after 18 h growth if the inoculum contains only cells with plasmid.

Solution:

The number of generations of plasmid-carrying cells in 18 h is calculated from Eq. (11.66):

$$n = \frac{(1.2 \text{ h}^{-1}) 18 \text{ h}}{\ln 2} = 31.$$

Substituting this into Eq. (11.64) with $p = 0.005$ and $\alpha = 1.4 \text{ h}^{-1} / 1.2 \text{ h}^{-1} = 1.17$:

$$F = \frac{1 - 1.17 - 0.005}{1 - 1.17 - 2^{31(1.17 + 0.005 - 1)} 0.005} = 0.45.$$

Therefore, after 18 h only 45% of the cells contain plasmid.

Alternative models for growth with plasmid instability have been developed [13]; some include equations for substrate utilisation and product formation [14]. A weakness in the simple model presented here is the assumption that all plasmid-containing cells are the same. In reality there are dif-

ferences in copy number and therefore specific growth rate between plasmid-carrying cells; probability of plasmid loss also varies from cell to cell. More complex models that recognise the segregated nature of plasmid populations are available [15, 16].

11.9 Production Kinetics in Cell Culture

In this section we consider the kinetics of production of low-molecular-weight compounds, such as ethanol, amino acids, antibiotics and vitamins, which are excreted from cells in culture. As shown in Table 11.9, fermentation products can be classified according to the relationship between product synthesis and energy generation in the cell [10, 17]. Compounds in the first category are formed directly as end- or by-products of energy metabolism; these materials are synthesised in pathways which produce ATP. The second class of product is partly linked to energy generation but requires additional energy for synthesis. Formation of other products such as antibiotics involves reactions far removed from energy metabolism.

Table 11.9 Classification of low-molecular-weight fermentation products

<i>Class of metabolite</i>	<i>Examples</i>
Products directly associated with generation of energy in the cell	Ethanol, acetic acid, gluconic acid, acetone, butanol, lactic acid, other products of anaerobic fermentation
Products indirectly associated with energy generation	Amino acids and their products, citric acid, nucleotides
Products for which there is no clear direct or indirect coupling to energy generation	Penicillin, streptomycin, vitamins

Irrespective of the class of product, rate of product formation in cell culture can be expressed as a function of biomass concentration:

$$r_p = q_p x \quad (11.67)$$

where r_p is the volumetric rate of product formation with units of, for example, $\text{kg m}^{-3} \text{s}^{-1}$, x is biomass concentration, and q_p is the *specific rate of product formation* with dimensions T^{-1} . q_p can be evaluated at any time during fermentation as the ratio of production rate and biomass concentration; q_p is not necessarily constant during batch culture. Depending on whether the product is linked to energy metabolism or not, we can develop equations for q_p as a function of growth rate and other metabolic parameters.

11.9.1 Product Formation Directly Coupled With Energy Metabolism

For products formed in pathways which generate ATP, rate of production is related to cellular energy demand. Growth is usually the major energy-requiring function of cells; therefore, if production is coupled to energy metabolism, product will be formed whenever there is growth. However, ATP is also required for other activities called *maintenance*. Examples of maintenance functions include cell motility, turnover of cellular components and adjustment of membrane potential and internal pH. Maintenance activities are carried out by living cells even in the absence of growth. Products synthesised in energy pathways will be produced whenever maintenance functions are carried out because ATP is required. Kinetic expressions for product formation must account for growth-associated and maintenance-associated production, as in the following equation:

$$r_p = Y_{pX} r_X + m_p x. \quad (11.68)$$

In Eq. (11.68), r_X is the volumetric rate of biomass formation, Y_{pX} is the theoretical or true yield of product from biomass, m_p is the *specific rate of product formation due to maintenance*, and x is biomass concentration. m_p has dimensions T^{-1} and typical units $\text{kg product (kg biomass)}^{-1} \text{s}^{-1}$. Eq. (11.68) states that rate of product formation depends partly on rate of growth but also partly on cell concentration. From Eq. (11.52), r_X is equal to μx ; therefore:

$$r_p = (Y_{pX} \mu + m_p) x. \quad (11.69)$$

Comparison of Eqs (11.67) and (11.69) shows that, for products coupled to energy metabolism, q_p is equal to a combination of growth-associated and non-growth-associated terms:

$$q_p = Y_{pX} \mu + m_p. \quad (11.70)$$

11.9.2 Product Formation Indirectly Coupled With Energy Metabolism

When product is synthesised partly in metabolic pathways used for energy generation and partly in other pathways requiring energy, the relationship between product formation and growth can be complicated. We will not attempt to develop equations for q_p for this type of product. A generalised

treatment of indirectly-coupled product formation is given by Roels and Kossen [10].

11.9.3 Product Formation Not Coupled With Energy Metabolism

Production not involving energy metabolism is difficult to relate to growth because growth and product synthesis are somewhat dissociated. However in some cases, rate of formation of non-growth-associated product is directly proportional to biomass concentration, so that production rate as defined in Eq. (11.67) can be applied with constant q_p . Sometimes q_p is a complex function of growth rate and must be expressed using empirical equations derived from experiment. An example is penicillin synthesis; equations for rate of penicillin production as a function of biomass concentration and specific growth rate have been derived by Heijnen *et al.* [18].

11.10 Kinetics of Substrate Uptake in Cell Culture

Cells consume substrate from the external environment and channel it into different metabolic pathways. Some substrate may be directed into growth and product synthesis; another fraction is used to generate energy for maintenance activities. Substrate requirements for maintenance vary considerably depending on the organism and culture conditions; a complete account of substrate uptake should include a maintenance component. The specific rate of substrate uptake for maintenance activities is known as the *maintenance coefficient*, m_s . The dimensions of m_s are T^{-1} ; typical units are $\text{kg substrate (kg biomass)}^{-1} \text{ s}^{-1}$. Some examples of mainte-

nance coefficients for various microorganisms are listed in Table 11.10. Ionic strength greatly influences the value of m_s ; significant amounts of energy are needed to maintain concentration gradients across cell membranes. The physiological significance of m_s has been the subject of much debate; there are indications that m_s for a particular organism may not be constant at all possible growth rates.

Rate of substrate uptake can be expressed as a function of biomass concentration by an equation analogous to Eq. (11.67):

$$r_s = q_s x \quad (11.71)$$

where r_s is the volumetric rate of substrate consumption with units of, for example, $\text{kg m}^{-3} \text{ s}^{-1}$, q_s is the *specific rate of substrate uptake*, and x is biomass concentration. Like q_p , q_s has dimensions T^{-1} . In this section, we will develop equations for q_s as a function of growth rate and other relevant metabolic parameters.

11.10.1 Substrate Uptake in the Absence of Product Formation

In some cultures there is no extracellular product formation; for example, biomass itself is the product in manufacture of bakers' yeast and single-cell protein. In the absence of product formation, we assume that all substrate entering the cell is used for growth and maintenance functions. Rates of these cell activities are related as follows:

$$r_s = \frac{r_x}{Y_{XS}} + m_s x. \quad (11.72)$$

Table 11.10 Maintenance coefficients for several microorganisms with glucose as energy source

(From S.J. Pirt, 1975, *Principles of Microbe and Cell Cultivation*, Blackwell Scientific, Oxford)

Microorganism	Growth conditions	m_s (kg substrate (kg cells) $^{-1} \text{ h}^{-1}$)
<i>Saccharomyces cerevisiae</i>	Anaerobic	0.036
	Anaerobic, 1.0 M NaCl	0.360
<i>Azotobacter vinelandii</i>	Nitrogen fixing, 0.2 atm dissolved-oxygen tension	1.5
	Nitrogen fixing, 0.02 atm dissolved-oxygen tension	0.15
<i>Klebsiella aerogenes</i>	Anaerobic, tryptophan-limited, 2 g l $^{-1}$ NH $_4$ Cl	2.88
	Anaerobic, tryptophan-limited, 4 g l $^{-1}$ NH $_4$ Cl	3.69
<i>Lactobacillus casei</i>		0.135
<i>Aerobacter cloacae</i>	Aerobic, glucose-limited	0.094
<i>Penicillium chrysogenum</i>	Aerobic	0.022

In Eq. (11.72), r_X is the volumetric rate of biomass production, Y_{XS} is the true yield of biomass from substrate, m_S is the maintenance coefficient, and x is biomass concentration. Eq. (11.72) states that rate of substrate uptake depends partly on the rate of growth but also varies with cell concentration. When r_X is expressed using Eq. (11.52), Eq. (11.72) becomes:

$$r_S = -\left(\frac{\mu}{Y_{XS}} + m_S\right) x. \quad (11.73)$$

If we now express μ as a function of substrate concentration using Eq. (11.60), Eq. (11.73) becomes:

$$r_S = -\left[\frac{\mu_{\max} s}{Y_{XS}(K_S + s)} + m_S\right] x. \quad (11.74)$$

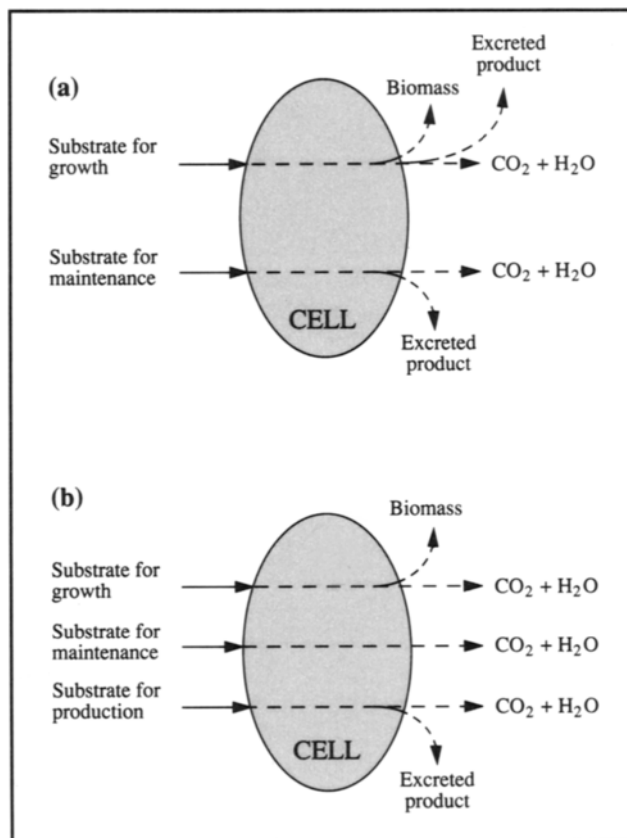
When s is zero, Eq. (11.74) predicts that substrate consumption will proceed at a rate equal to $m_S x$. Substrate uptake in the absence of substrate is impossible; this feature of Eq. (11.74) is therefore unrealistic. The problem arises because of implicit assumptions we have made about the nature of maintenance activities. It can be shown however that Eq. (11.74) is a realistic description of substrate uptake as long as there is external substrate available; when the substrate is exhausted maintenance energy is generally supplied by endogenous metabolism.

11.10.2 Substrate Uptake With Product Formation

Patterns of substrate flow in cells synthesising products depend on whether product formation is coupled to energy metabolism. When products are formed in energy-generating pathways, e.g. in an anaerobic culture, product synthesis is an unavoidable consequence of cell growth and maintenance. Accordingly, as illustrated in Figure 11.13(a), there is no separate flow of substrate into the cell for product synthesis; product is formed from the substrate taken up to support growth and maintenance. Substrate consumed for maintenance does not contribute to growth; it therefore constitutes a separate substrate flow into the cell. In contrast, when production is not linked or only partly linked to energy metabolism, all or some of the substrate required for product synthesis is additional to, and separate from, that needed for growth and maintenance. Flow of substrate in this case is illustrated in Figure 11.13(b).

When products are directly linked to energy generation, equations for rate of substrate consumption do not include a

Figure 11.13 Substrate uptake with product formation: (a) production directly coupled to energy metabolism; (b) production not directly coupled to energy metabolism.



separate term for production; substrate requirements for product formation are already accounted for in terms for growth- and maintenance-associated substrate uptake. Accordingly, equations presented in the previous section for substrate consumption in the absence of product formation apply; rate of substrate uptake is related to growth and maintenance requirements by Eqs (11.73) and (11.74).

In cultures where product synthesis is only indirectly coupled to energy metabolism, rate of substrate consumption is a function of three factors: growth rate, rate of product formation and rate of substrate uptake for maintenance. These different cell functions can be related using yield and maintenance coefficients:

$$r_S = \frac{r_X}{Y_{XS}} + \frac{r_P}{Y_{PS}} + m_S x \quad (11.75)$$

where r_s is the volumetric rate of substrate consumption, r_x is the volumetric rate of biomass production, r_p is the volumetric rate of product formation, Y_{XS} is the true yield of biomass from substrate, Y_{PS} is the true yield of product from substrate, m_s is the maintenance coefficient, and x is biomass concentration. If we express r_x and r_p using Eqs (11.52) and (11.67):

$$r_s = \left(\frac{\mu}{Y_{XS}} + \frac{q_p}{Y_{PS}} + m_s \right) x. \quad (11.76)$$

11.11 Effect of Culture Conditions on Cell Kinetics

Temperature has a marked effect on metabolic rate. Temperature has a direct influence on reaction rate according to the Arrhenius law; it can also change the configuration of cell constituents, especially proteins and membrane components. In general, the effect of temperature on growth is similar to that already described in Section 11.3.4 for enzymes. There is an approximate two fold increase in specific growth rate for every 10°C rise in temperature, until structural breakdown of cell proteins and lipids starts to occur. Like other rate constants, the maintenance coefficient m_s has an Arrhenius-type temperature dependence [19]; this can have a significant kinetic effect on cultures where turnover of macromolecules is an important contribution to maintenance requirements. In contrast, temperature has only a minor effect on the biomass yield coefficient, Y_{XS} [19]. Other cellular responses to temperature are described elsewhere [1, 20, 21].

Growth rate depends on medium pH in much the same way as enzyme activity (Section 11.3.4); maximum growth rate is usually maintained over 1–2 pH units but declines with further variation. pH also affects the profile of product synthesis in anaerobic culture and can change maintenance-energy requirements [1, 20, 21].

11.12 Determining Cell Kinetic Parameters From Batch Data

In order to apply the equations presented in Sections 11.6–11.10 to real fermentations, we must know the kinetic

and yield parameters for the system and have information about rates of growth, substrate uptake and product formation. Batch culture is the most frequently-applied method for investigating kinetic behaviour, but it is not always the best. Methods for determining reaction parameters from batch data are described below.

11.12.1 Rates of Growth, Product Formation and Substrate Uptake

Determining growth rates in cell culture requires measurement of cell concentration. Many different experimental procedures are applied for biomass estimation [20, 22]. Direct measurement can be made of cell number, dry or wet cell mass, packed cell volume or culture turbidity; alternatively, indirect estimates are obtained from measurements of product formation, heat evolution or cell composition. Cell viability is usually evaluated using plating or staining techniques. Each method for biomass estimation will give somewhat different results. For example, rate of growth determined using cell dry weight may differ from that obtained from cell number because dry weight in the culture can increase without a corresponding increase cell number.

Irrespective of how cell concentration is measured, the techniques described in Section 11.2 for graphical differentiation of concentration data are suitable for determining volumetric growth rates in batch culture. The results will depend to some extent on how the data are smoothed. For reasons described in Section 11.7.3, there is usually a relatively abrupt change in growth rate between growth and stationary phases; this feature requires extra care for accurate differentiation of batch growth curves. As discussed in Section 3.3.1, an advantage of hand-smoothing is that it allows us to judge the significance of individual points. Once the volumetric growth rate r_x is known, the specific growth rate μ is obtained by dividing r_x by the cell concentration.

For the growth phase of batch culture, an alternative method can be applied to calculate μ . Assuming growth can be represented by the first-order model of Eq. (11.52), the integrated relationship of Eq. (11.53) or (11.54) allows us to obtain μ directly. During the growth phase when μ is essentially constant, a plot of $\ln x$ versus time gives a straight line with slope μ . This is illustrated in Example 11.8.

Example 11.8 Hybridoma doubling time

A mouse-mouse hybridoma cell line is used to produce monoclonal antibody. Growth in batch culture is monitored with the following results.

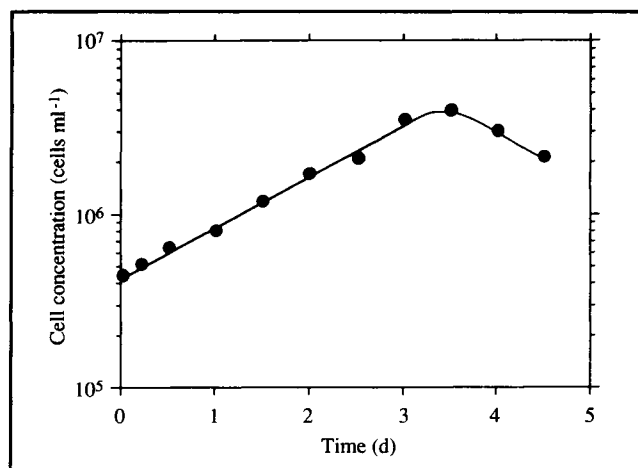
Time (d)	Cell concentration (cells ml ⁻¹ × 10 ⁻⁶)
0.0	0.45
0.2	0.52
0.5	0.65
1.0	0.81
1.5	1.22
2.0	1.77
2.5	2.13
3.0	3.55
3.5	4.02
4.0	3.77
4.5	2.20

- Determine the specific growth rate during the growth phase.
- What is the culture doubling time?

Solution:

- The data are plotted as a semi-log graph in Figure 11E8.1.

Figure 11E8.1 Calculation of specific growth rate for hybridoma cells.



No lag phase is evident. As Eq. (11.54) applies only when μ is constant, i.e. during the exponential growth phase, we must determine which data points belong to the exponential growth phase. In Figure 11E8.1, the final three points appear to belong to the decline and death phases of the culture. Fitting a straight line to the remaining data gives a slope of 0.67. Therefore, $\mu = 0.67 \text{ d}^{-1}$.

- From Eq. (11.58):

$$t_d = \frac{\ln 2}{0.67 \text{ d}^{-1}} = 1.0 \text{ d.}$$

This doubling time applies only during the growth phase of the culture.

Volumetric rates of substrate uptake and product formation, r_S and r_P , can be evaluated by graphical differentiation of substrate- and product-concentration data, respectively. Specific product-formation rate q_P and specific substrate-uptake rate q_S are obtained by dividing the respective volumetric rates by cell concentration.

11.12.2 $\mu_{\max}$ and K_S

The Monod equation for specific growth rate, Eq. (11.60), is analogous mathematically to the Michaelis–Menten expression for enzyme kinetics. In principle therefore, the techniques described in Section 11.4 for determining $v_{\max}$ and K_m for enzyme reaction can be applied for evaluation of $\mu_{\max}$ and K_S . However, because values of K_S in cell culture are usually very low, accurate determination of this parameter from batch data is difficult. Better estimation of K_S can be made using continuous culture of cells as discussed in Chapter 13. On the other hand, measurement of $\mu_{\max}$ from batch data is relatively straightforward. As described in Section 11.7.3, if all nutrients are present in excess, the specific growth rate during exponential growth is equal to the maximum specific growth rate. Therefore, the specific growth rate calculated in Example 11.8 is equal to $\mu_{\max}$.

11.13 Effect of Maintenance on Yields

True yields such as Y_{XS} , Y_{PX} and Y_{PS} are often difficult to evaluate. Although true yields are essentially stoichiometric coefficients, the stoichiometry of biomass production and product formation is only known for relatively simple fermentations.

If the metabolic pathways are complex, stoichiometric calculations become too complicated. However, theoretical yields can be related to observed yields such as Y'_{XS} , Y'_{PX} and Y'_{PS} , which are more easily determined.

11.13.1 Observed Yields

Expressions for observed yield coefficients can be obtained by applying Eq. (11.48):

$$Y'_{XS} = \frac{-dX}{dS} = \frac{r_X}{r_S} \quad (11.77)$$

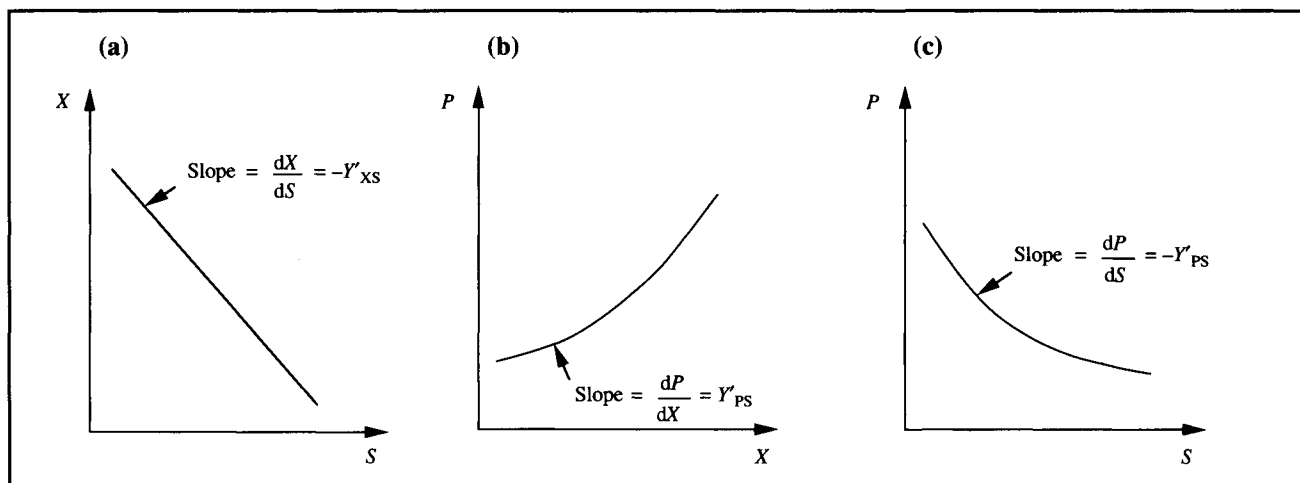
$$Y'_{PX} = \frac{dP}{dX} = \frac{r_P}{r_X} \quad (11.78)$$

and

$$Y'_{PS} = \frac{-dP}{dS} = \frac{r_P}{r_S} \quad (11.79)$$

where X , S and P are masses of cells, substrate and product, respectively, and r_X , r_S and r_P are observed rates evaluated from experimental data. Therefore, yield coefficients can be determined by plotting X , S or P against each other and evaluating the slope as illustrated in Figure 11.14. Alternatively, observed yield coefficients at a particular instant in time can be calculated

Figure 11.14 Evaluation of observed yields in batch culture from cell, substrate and product concentrations.



as the ratio of rates evaluated at that instant. Observed yields are not necessarily constant throughout batch culture; in some cases they exhibit significant dependence on environmental parameters such as substrate concentration and growth rate. Nevertheless, for many cultures, the observed biomass yield Y'_{XS} is approximately constant. Because of the errors in experimental data, considerable uncertainty is usually associated with measured yield coefficients.

11.13.2 Biomass Yield From Substrate

Equations for true biomass yield can be determined for systems without extracellular product formation or when product synthesis is directly coupled to energy metabolism. Substituting expressions for r_X and r_S from Eqs (11.52) and (11.73) into Eq. (11.77) gives:

$$Y'_{XS} = \frac{\mu}{\left(\frac{\mu}{Y_{XS}} + m_S\right)} \quad (11.80)$$

Inverting Eq. (11.80) produces the expression:

$$\frac{1}{Y'_{XS}} = \frac{1}{Y_{XS}} + \frac{m_S}{\mu} \quad (11.81)$$

Therefore, if Y_{XS} and m_S are relatively constant, a plot of $1/Y'_{XS}$ versus $1/\mu$ gives a straight line with slope m_S and intercept $1/Y_{XS}$. Eq. (11.81) is not generally applied to batch growth data; under typical batch conditions, μ does not vary from $\mu_{\max}$ for much of the culture period so it is difficult to plot Y'_{XS} as a function of specific growth rate. We will revisit Eq. (11.81) when we consider continuous cell culture in Chapter 13. As a rule of thumb, true biomass yield from glucose under aerobic conditions is around 0.5 g g^{-1} .

In processes such as production of bakers' yeast and single-cell protein where the required product is biomass, it is desirable to maximise the actual or observed yield of cells from substrate. The true yield Y_{XS} is limited by stoichiometric considerations. However, from Eq. (11.80), Y'_{XS} can be improved by decreasing the maintenance coefficient or increasing the growth rate. m_S may be reduced by lowering the temperature of fermentation, using a medium of lower ionic strength, or by applying a different organism or strain with lower maintenance-energy requirements. Assuming these changes do not reduce the growth rate, they can be employed to improve the biomass yield.

When the culture produces compounds not directly coupled with energy metabolism, Eqs (11.80) and (11.81) do not apply because a different expression for r_S must be used in Eq. (11.77). Determination of true yields and maintenance coefficients is more difficult in this case because of the number of terms involved.

11.13.3 Product Yield From Biomass

Observed yield of product from biomass Y'_{PX} is defined in Eq. (11.78). When product synthesis is directly coupled to energy metabolism, r_P is given by Eq. (11.69). Substituting this and Eq. (11.52) into Eq. (11.78) gives:

$$Y'_{PX} = Y_{PX} + \frac{m_P}{\mu} \quad (11.82)$$

The extent of deviation of Y'_{PX} from Y_{PX} depends on the relative magnitudes of m_P and μ . To increase the observed yield of product for a particular process, m_P should be increased and μ decreased. Eq. (11.82) does not apply to products not directly coupled with energy metabolism; we do not have a general expression for r_P in terms of true yield coefficients for this class of product.

11.13.4 Product Yield From Substrate

Observed product yield from substrate Y'_{PS} is defined in Eq. (11.79). For products coupled to energy generation, expressions for r_P and r_S are available from Eqs (11.69) and (11.73). Therefore:

$$Y'_{PS} = \frac{Y_{PX}\mu + m_P}{\left(\frac{\mu}{Y_{XS}} + m_S\right)} \quad (11.83)$$

In many anaerobic fermentations such as ethanol production, yield of product from substrate is a critical factor affecting process economics. At high Y'_{PS} , more ethanol is produced per mass of carbohydrate consumed so that the cost of production is reduced. Growth rate has a strong effect on Y'_{PS} for ethanol. Because Y'_{PS} is low when $\mu = \mu_{\max}$, it is desirable to reduce the specific growth rate of the cells. Low growth rate can be obtained by depriving the cells of some essential nutrient, e.g. a nitrogen source, or by immobilising the cells to prevent growth. Increasing the rate of maintenance activity relative to growth

will also enhance product yield. This can be done by using a medium of high ionic strength, raising the temperature, or selecting a mutant or different organism with high maintenance requirements. Continuous culture provides more opportunity for manipulating rates of growth than batch culture.

The effect of growth rate and maintenance on Y'_{PS} is difficult to determine for products not directly coupled with energy metabolism unless information is available about the effect of these parameters on q_p .

11.14 Kinetics of Cell Death

The kinetics of cell death is an important consideration in design of sterilisation processes and in analysis of fermentations where substantial viability loss is expected. In a lethal environment, cells in a population do not die all at once; deactivation of the culture occurs over a finite period of time depending on the initial number of viable cells and the severity of the conditions imposed. Loss of cell viability can be described mathematically in much the same way as enzyme deactivation; cell death is assumed to be a first-order process:

$$r_d = k_d N \quad (11.84)$$

where r_d is rate of cell death, N is number of viable cells, and k_d is the *specific death constant*. Alternatively, rate of cell death can be expressed using cell concentration rather than cell number:

$$r_d = k_d x \quad (11.85)$$

where k_d is the specific death constant based on cell concentration and x is the concentration of viable cells.

In a closed system with cell death the only process affecting viable cell concentration, rate of cell death is equal to the rate of decrease in cell number. Therefore, from Eq. (11.84):

$$r_d = \frac{-dN}{dt} = k_d N \quad (11.86)$$

If k_d is constant, we can integrate Eq. (11.86) to derive an expression for N as a function of time:

$$N = N_0 e^{-k_d t} \quad (11.87)$$

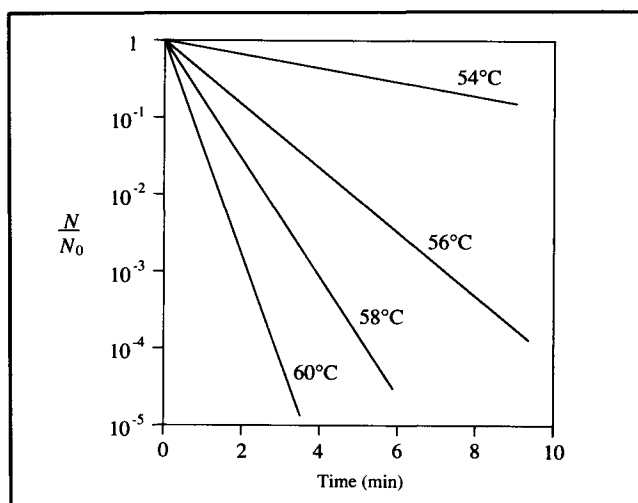
where N_0 is the number of viable cells at time zero. Taking natural logarithms of both sides of Eq. (11.87) gives:

$$\ln N = \ln N_0 - k_d t. \quad (11.88)$$

According to Eq. (11.88), if first-order death kinetics apply, a plot of $\ln N$ versus t gives a straight line with slope $-k_d$. Experimental measurements for many vegetative cells have confirmed the relationship of Eq. (11.88); as an example, data for thermal death of *Escherichia coli* at various temperatures are shown in Figure 11.15. However, first-order death kinetics do not always hold, particularly for bacterial spores immediately after exposure to heat.

Like other kinetic constants, the value of the specific death constant k_d depends on temperature. This effect can be described using the Arrhenius relationship of Eq. (11.46). Typical E_d values for thermal destruction of microorganisms are high, of the order 250–290 kJ gmol⁻¹ [23]. Therefore, small increases in temperature have a significant effect on k_d and rate of death.

Figure 11.15 Relationship between temperature and rate of thermal death for vegetative *Escherichia coli* cells. (From S. Aiba, A.E. Humphrey and N.F. Millis, 1965, *Biochemical Engineering*, Academic Press, New York.)



Example 11.9 Thermal death kinetics

The number of viable spores of a new strain of *Bacillus subtilis* is measured as a function of time at various temperatures.

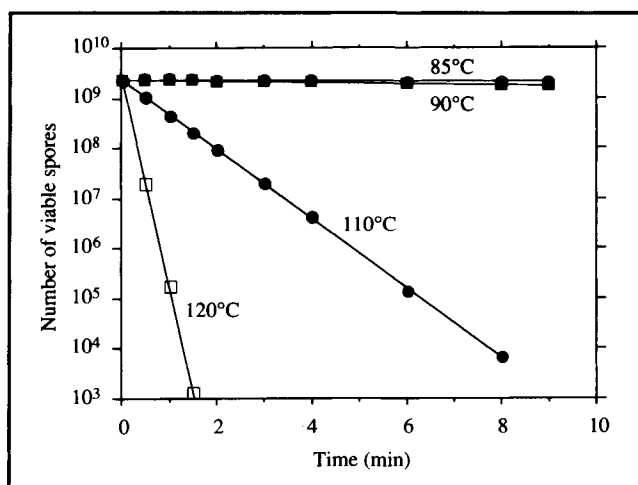
Time (min)	Number of spores at:			
	$T = 85^\circ\text{C}$	$T = 90^\circ\text{C}$	$T = 110^\circ\text{C}$	$T = 120^\circ\text{C}$
0.0	2.40×10^9	2.40×10^9	2.40×10^9	2.40×10^9
0.5	2.39×10^9	2.38×10^9	1.08×10^9	2.05×10^7
1.0	2.37×10^9	2.30×10^9	4.80×10^8	1.75×10^5
1.5	—	2.29×10^9	2.20×10^8	1.30×10^3
2.0	2.33×10^9	2.21×10^9	9.85×10^7	—
3.0	2.32×10^9	2.17×10^9	2.01×10^7	—
4.0	2.28×10^9	2.12×10^9	4.41×10^6	—
6.0	2.20×10^9	1.95×10^9	1.62×10^5	—
8.0	2.19×10^9	1.87×10^9	6.88×10^3	—
9.0	2.16×10^9	1.79×10^9	—	—

- Determine the activation energy for thermal death of *B. subtilis* spores.
- What is the specific death constant at 100°C ?
- Estimate the time required to kill 99% of spores in a sample at 100°C ?

Solution:

- A semi-log plot of number of viable spores versus time is shown in Figure 11E9.1.

Figure 11E9.1 Thermal death of *Bacillus subtilis* spores.



From Eq. (11.88), the slopes of the lines in Figure 11E9.1 are equal to $-k_d$ at the various temperatures. Fitting straight lines to the data gives the following results:

$$k_d(85^\circ\text{C}) = 0.012 \text{ min}^{-1}$$

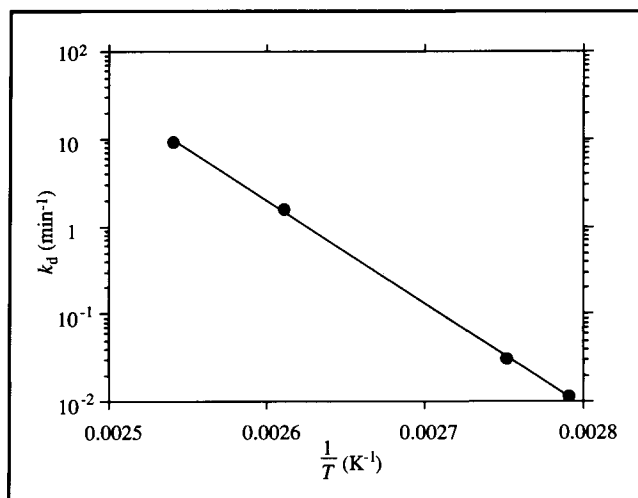
$$k_d(90^\circ\text{C}) = 0.032 \text{ min}^{-1}$$

$$k_d(110^\circ\text{C}) = 1.60 \text{ min}^{-1}$$

$$k_d(120^\circ\text{C}) = 9.61 \text{ min}^{-1}$$

The relationship between k_d and absolute temperature is given by Eq. (11.46). Therefore, a semi-log plot of k_d versus $1/T$ should yield a straight line with slope $= -E_d/R$ where T is absolute temperature. T is converted to degrees Kelvin using the formula of Eq. (2.24); $1/T$ -values in units of K^{-1} are plotted in Figure 11E9.2.

Figure 11E9.2 Calculation of kinetic parameters for thermal death of spores.



The slope is $-27\,030\text{ K}$. From Table 2.5, $R = 8.3144\text{ J K}^{-1}\text{ gmol}^{-1}$. Therefore:

$$\begin{aligned} E_d &= 27\,030\text{ K} (8.3144\text{ J K}^{-1}\text{ gmol}^{-1}) = 2.25 \times 10^5\text{ J gmol}^{-1} \\ &= 225\text{ kJ gmol}^{-1}. \end{aligned}$$

(b) The equation to the line in Figure 11E9.2 is:

$$k_d = 6.52 \times 10^{30} e^{-27\,030/T}.$$

Therefore, at $T = 100^\circ\text{C} = 373.15\text{ K}$, $k_d = 0.23\text{ min}^{-1}$.

(c) From Eq. (11.88):

$$t = \frac{-(\ln N - \ln N_0)}{k_d}$$

or

$$t = \frac{-\ln\left(\frac{N}{N_0}\right)}{k_d}.$$

For N equal to 1% of N_0 , $N/N_0 = 0.01$. At 100°C , $k_d = 0.23\text{ min}^{-1}$ and the time required is:

$$t = \frac{-\ln(0.01)}{0.23\text{ min}^{-1}} = 20\text{ min}.$$

As contaminating organisms are being killed by heat sterilisation, nutrients in the medium may also be destroyed. The sensitivity of nutrient molecules to temperature is described by the Arrhenius equation of Eq. (11.46). Values of the activation energy E_d for thermal destruction of vitamins and amino acids are 84–92 kJ gmol⁻¹; for proteins E_d is about 165 kJ gmol⁻¹ [23]. Because these values are somewhat lower than typical E_d values for microorganisms, raising the temperature has a greater effect on cell death than nutrient destruction. This means that sterilisation at higher temperatures for shorter periods of time has the advantage of killing cells with limited destruction of medium components.

11.15 Summary of Chapter 11

At the end of Chapter 11 you should:

- (i) understand the difference between *reversible* and *irreversible reactions*, and the limitations of equilibrium thermodynamics in representing cell and enzyme reactions;
- (ii) be able to calculate reaction rates from batch concentration data using *graphical differentiation*;
- (iii) be familiar with kinetic relationships for *zero-order*, *first-order* and *Michaelis–Menten reactions*;
- (iv) be able to determine enzyme kinetic parameters $v_{\max}$ and K_m from batch concentration data;
- (v) be able to quantify the effect of temperature on rates of enzyme reaction and deactivation;
- (vi) be able to calculate *yield coefficients* for cell culture;
- (vii) know the basic relationships for cell growth kinetics and be able to evaluate growth, substrate uptake and production rates in batch culture;
- (viii) be able to analyse growth in cultures with plasmid instability;
- (ix) know how *maintenance activities* affect substrate utilisation in cells; and
- (x) be able to describe the kinetics of *cell death*.

Problems

11.1 Reaction equilibrium

Calculate equilibrium constants for the following reactions under standard conditions:

- (a) $\text{glutamine} + \text{H}_2\text{O} \rightarrow \text{glutamate} + \text{NH}_4^+$
 $\Delta G_{\text{rxn}}^\circ = -14.1 \text{ kJ mol}^{-1}$

- (b) $\text{malate} \rightarrow \text{fumarate} + \text{H}_2\text{O}$
 $\Delta G_{\text{rxn}}^\circ = 3.2 \text{ kJ mol}^{-1}$.

Could either of these reactions be considered irreversible?

11.2 Equilibrium yield

The following reaction catalysed by phosphoglucumutase occurs during breakdown of glycogen:



A reaction is started by adding phosphoglucumutase to 0.04 gmol glucose 1-phosphate in 1 litre solution at 25°C. The reaction proceeds to equilibrium at which the concentration of glucose 1-phosphate is 0.002 M and the concentration of glucose 6-phosphate is 0.038 M.

- (a) Calculate the equilibrium constant.
- (b) What is the theoretical yield?
- (c) What is the yield based on amount of reactant supplied?

11.3 Reaction rate

- (a) The volume of a fermenter is doubled while keeping the cell concentration and other fermentation conditions the same.
 - (i) How is the volumetric productivity affected?
 - (ii) How is the specific productivity affected?
 - (iii) How is the total productivity affected?
- (b) If instead of (a) the cell concentration were doubled, what affect would this have on volumetric, specific and total productivities?
- (c) A fermenter produces 100 kg lysine per day.
 - (i) If the volumetric productivity is 0.8 g l⁻¹ h⁻¹, what is the volume of the fermenter?
 - (ii) The cell concentration is 20 g l⁻¹ dry weight. Calculate the specific productivity.

11.4 Enzyme kinetics

Lactase, also known as β -galactosidase, catalyses the hydrolysis of lactose to produce glucose and galactose from milk and whey. Experiments are carried out to determine the kinetic parameters for the enzyme. Initial rate data are listed below.

Lactose concentration (mol l ⁻¹ × 10 ²)	Initial reaction velocity (mol l ⁻¹ min ⁻¹ × 10 ³)
2.50	1.94
2.27	1.91
1.84	1.85
1.35	1.80
1.25	1.78
0.730	1.46
0.460	1.17
0.204	0.779

Evaluate $v_{\max}$ and K_m .

11.5 Effect of temperature on hydrolysis of starch

α -Amylase from malt is used to hydrolyse starch. The dependence of initial reaction rate on temperature is determined experimentally. Results measured at fixed starch and enzyme concentrations are listed below.

Temperature (°C)	Rate of glucose production (mmol m ⁻³ s ⁻¹)
20	0.31
30	0.66
40	1.20
60	6.33

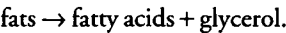
- Determine the activation energy for this reaction.
- α -Amylase is used to break down starch in baby food. It is proposed to carry out the reaction at a relatively high temperature so that the viscosity is reduced. What is the reaction rate at 55°C compared with 25°C?
- Thermal deactivation of this enzyme is described by the equation:

$$k_d = 2.25 \times 10^{27} e^{-41630/RT}$$

where k_d is the deactivation rate constant in h⁻¹, R is the ideal gas constant in cal gmol⁻¹ K⁻¹, and T is temperature in K. What is the half-life of the enzyme at 55°C compared with 25°C? Which of these two operating temperatures is more practical for processing baby food?

11.6 Enzyme reaction and deactivation

Lipase is being investigated as an additive to laundry detergent for removal of stains from fabric. The general reaction is:



The Michaelis constant for pancreatic lipase is 5 mM. At 60°C, lipase is subject to deactivation with a half-life of 8 min. Fat hydrolysis is carried out in a well-mixed batch reactor which simulates a top-loading washing machine. The initial fat concentration is 45 gmol m⁻³. At the beginning of the reaction the rate of hydrolysis is 0.07 mmol l⁻¹ s⁻¹. How long does it take for the enzyme to hydrolyse 80% of the fat present?

11.7 Growth parameters for recombinant *E. coli*

Escherichia coli is being used for production of recombinant porcine growth hormone. The bacteria are grown aerobically in batch culture with glucose as growth-limiting substrate. Cell and substrate concentrations are measured as a function of culture time; the results are listed below.

Time (h)	Cell concentration, x (kg m ⁻³)	Substrate concentration, s (kg m ⁻³)
0.0	0.20	25.0
0.33	0.21	24.8
0.5	0.22	24.8
0.75	0.32	24.6
1.0	0.47	24.3
1.5	1.00	23.3
2.0	2.10	20.7
2.5	4.42	15.7
2.8	6.9	10.2
3.0	9.4	5.2
3.1	10.9	1.65
3.2	11.6	0.2
3.5	11.7	0.0
3.7	11.6	0.0

- Plot μ as a function of time.
- What is the value of $\mu_{\max}$?
- What is the observed biomass yield from substrate? Is Y'_{XS} constant?

11.8 Growth parameters for hairy roots

Hairy roots are produced by genetic transformation of plants using *Agrobacterium rhizogenes*. The following biomass and sugar concentrations were obtained during batch culture of *Atropa belladonna* hairy roots in a bubble-column fermenter.

Time (d)	Biomass concentration (g l ⁻¹ dry weight)	Sugar concentration (g l ⁻¹)
0	0.64	30.0
5	1.95	27.4
10	4.21	23.6
15	5.54	21.0
20	6.98	18.4
25	9.50	14.8
30	10.3	13.3
35	12.0	9.7
40	12.7	8.0
45	13.1	6.8
50	13.5	5.7
55	13.7	5.1

- Plot μ as a function of culture time. When is the growth rate maximum?
- Plot the specific rate of sugar uptake as a function of time.
- What is the observed biomass yield from substrate? Is Y'_{XS} constant?

11.9 Ethanol fermentation by yeast and bacteria

Ethanol is produced by anaerobic fermentation of glucose by *Saccharomyces cerevisiae*. For the particular strain of *S. cerevisiae* employed, the maintenance coefficient is $0.18 \text{ kg kg}^{-1} \text{ h}^{-1}$, Y_{XS} is 0.11 kg kg^{-1} , Y_{PX} is 3.9 kg kg^{-1} and $\mu_{\max}$ is 0.4 h^{-1} . It is decided to investigate the possibility of using *Zymomonas mobilis* bacteria instead of yeast for making ethanol. *Z. mobilis* is known to produce ethanol under anaerobic conditions using a different metabolic pathway to that employed by yeast. Typical values of Y_{XS} are lower than for yeast at about 0.06 kg kg^{-1} ; on the other hand, the maintenance coefficient is higher at $2.2 \text{ kg kg}^{-1} \text{ h}^{-1}$. Y_{PX} for *Z. mobilis* is 7.7 kg kg^{-1} ; $\mu_{\max}$ is 0.3 h^{-1} .

- From stoichiometry, what is the maximum theoretical yield of ethanol from glucose?
- Y'_{PS} is maximum and equal to the theoretical yield when there is zero growth and all substrate entering the cell is

used for maintenance activities. If ethanol is the sole extra-cellular product of energy-yielding metabolism, calculate m_p for each organism.

- S. cerevisiae* and *Z. mobilis* are cultured in batch fermenters. Predict the observed product yield from substrate for the two cultures.
- What is the efficiency of ethanol production by the two organisms? Efficiency is defined as the observed product yield from substrate divided by the maximum or theoretical product yield.
- How does the specific rate of ethanol production by *Z. mobilis* compare with that by *S. cerevisiae*?
- Using Eq. (11.70), compare the proportions of growth-associated and non-growth-associated ethanol production by *Z. mobilis* and *S. cerevisiae*. For which organism is non-growth-associated production more substantial?
- In order to achieve the same volumetric ethanol productivity from the two cultures, what yeast concentration is required compared with the concentration of bacteria?
- At zero growth, the efficiency of ethanol production is the same in both cultures. Under these conditions, if the same concentration of yeast and bacteria are employed, what size fermenter is required for the yeast compared with the bacteria in order to achieve the same total productivity?
- Predict the observed biomass yield from substrate for the two organisms. For which organism is biomass disposal less of a problem?
- Make a recommendation about which organism is better suited for industrial ethanol production, and give your reasons.

11.10 Plasmid loss during culture maintenance

A stock culture of plasmid-containing *Streptococcus cremoris* cells is maintained with regular sub-culturing for a period of 28 d. After this time, the fraction of plasmid-carrying cells is measured and found to be 0.66. The specific growth rate of plasmid-free cells at the storage temperature is 0.033 h^{-1} ; the specific growth rate of plasmid-containing cells is 0.025 h^{-1} . If all the cells initially contained plasmid, estimate the probability per generation of plasmid loss.

11.11 Medium sterilisation

A steam steriliser is used to sterilise liquid medium for fermentation. The initial concentration of contaminating organisms is 10^8 per litre. For design purposes, the final acceptable level of contamination is usually taken to be 10^{-3}

cells; this corresponds to a risk that one batch in a thousand will remain contaminated even after the sterilisation process is complete. For how long should 1 m³ medium be treated if the temperature is:

- (a) 80°C?
- (b) 121°C?
- (c) 140°C?

To be safe, assume that the contaminants present are spores of *Bacillus stearothermophilus*, one of the most heat-resistant microorganisms known. For these spores the activation energy for thermal death is 283 kJ gmol⁻¹ and the Arrhenius constant is 10^{36.2} s⁻¹ [24].

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Suggestions for Further Reading

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Heterogeneous Reactions

In the previous chapter, reaction rate was considered as a function of substrate concentration and temperature. Reaction systems were assumed to be homogeneous; local variations in concentration and rate of conversion were not examined. Yet, in many bioprocesses, concentrations of substrates and products differ from point to point in the reaction mixture. Concentration gradients arise in single-phase systems when mixing is poor; if different phases are present, local variations in composition can be considerable. As described in Chapter 9, concentration gradients occur within phase boundary layers around gas bubbles and solids. More severe gradients are found inside solid biocatalysts such as cell flocs, pellets, biofilms, and immobilised-cell and -enzyme beads.

Reactions occurring in the presence of significant concentration or temperature gradients are called *heterogeneous reactions*. Because biological reactions are not generally associated with large temperature gradients, we confine our attention in this chapter to concentration effects. When heterogeneous reactions occur in solid catalysts, not all reactive molecules are available for immediate conversion. Reaction takes place only after reactants are transported to the site of reaction. Thus, mass-transfer processes can have a considerable influence on the overall conversion rate.

Because rate of reaction is generally dependent on substrate concentration, when concentrations in the system vary, kinetic analysis becomes more complex. The principles of homogeneous reaction and the equations outlined in Chapter 11 remain valid for heterogeneous systems; however, the concentrations used in these equations must be those actually prevailing at the site of reaction. For solid biocatalysts, we must know the concentration of substrate at each point inside the solid in order to determine the local rate of conversion. In most cases these concentrations cannot be measured; fortunately, they can be estimated using diffusion-reaction theory.

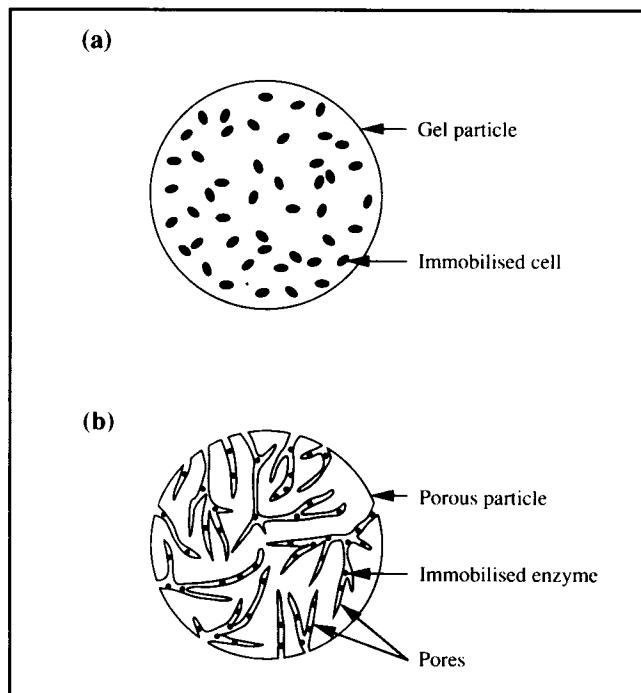
In this chapter, methods are presented for analysing reactions affected by mass transfer. The mathematics required is more sophisticated than is applied elsewhere in this book; however, attention can be directed to the results of the analysis rather than to the mathematical derivations. The practical outcome of this chapter is simple criteria for assessing mass-transfer limitations which can be used directly in experimental design.

12.1 Heterogeneous Reactions in Bioprocessing

Reactions involving solid-phase catalysts are important in bioprocessing. Macroscopic flocs, clumps and pellets are produced naturally by certain bacteria and fungi; mycelial pellets are common in antibiotic fermentations. Some cells grow as biofilms on reactor walls; others form slimes such as in waste treatment processes. Plant cell suspensions invariably contain aggregates; microorganisms in soil crumbs play a crucial role in environmental bioremediation of land. Animal tissues are now being cultured on three-dimensional scaffolds for surgical transplantation and organ repair. More traditionally, many food fermentations involve microorganisms attached to solid particles. In all of these systems, rate of reaction depends on the rate of mass transfer outside or within the solid catalyst.

If cells or enzymes do not spontaneously form clumps or attach to solid surfaces, they can be induced to do so using *immobilisation* techniques. Many procedures are available for artificial immobilisation of cells and enzymes; the results of two commonly-used methods are illustrated in Figure 12.1. As shown in Figure 12.1(a), cells and enzymes can be immobilised by entrapment within gels such as alginate, agarose and carrageenan. Cells or enzymes are mixed with liquified gel before it is hardened or cross-linked and broken into small particles. The gel polymer must be porous and relatively soft to allow diffusion of reactants and products to and from the interior of the particle. As shown in Figure 12.1(b), an alternative to gel immobilisation is entrapment within porous solids

Figure 12.1 Immobilised biocatalysts: (a) cells entrapped in soft gel; (b) enzymes attached to the internal surfaces of a porous solid.

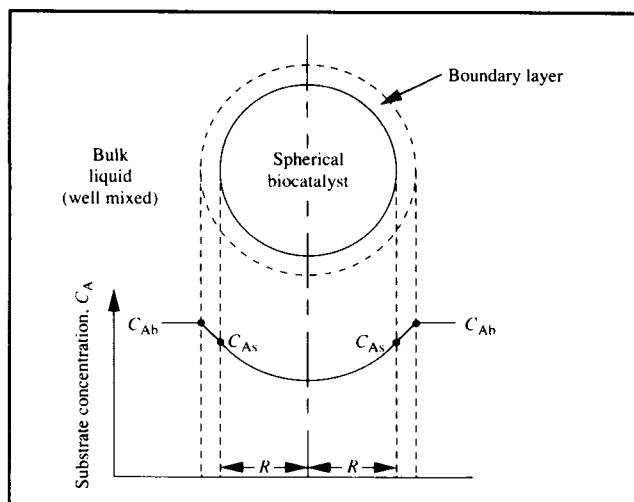


such as ceramics, porous glass and resin beads. Enzymes or cells migrate into the pores of these particles and attach to the internal surfaces; substrate must diffuse through the pores for reaction to occur. In both immobilisation methods, sites of reaction are distributed throughout the particle. Thus, a catalyst particle of higher activity can be formed by increasing the loading of cells or enzyme per volume of matrix.

Immobilised biocatalysts have many advantages in large-scale processing. One of the most important is continuous operation using the same catalytic material. For enzymes, an additional advantage is that immobilisation often enhances stability and increases the enzyme half-life. Further discussion of immobilisation methods and the rationale behind cell and enzyme immobilisation can be found in many articles and books; a selection of references is given at the end of this chapter.

In Chapter 11, enzymes and cells were considered as biological catalysts. In heterogeneous reactions involving a solid phase, the term 'catalyst' is also used to refer to the entire catalytically-active body, such as a particle or biofilm. Engineering analysis of heterogeneous reactions applies equally well to naturally occurring solid catalysts and artificially immobilised cells and enzymes.

Figure 12.2 Typical substrate concentration profile for a spherical biocatalyst.



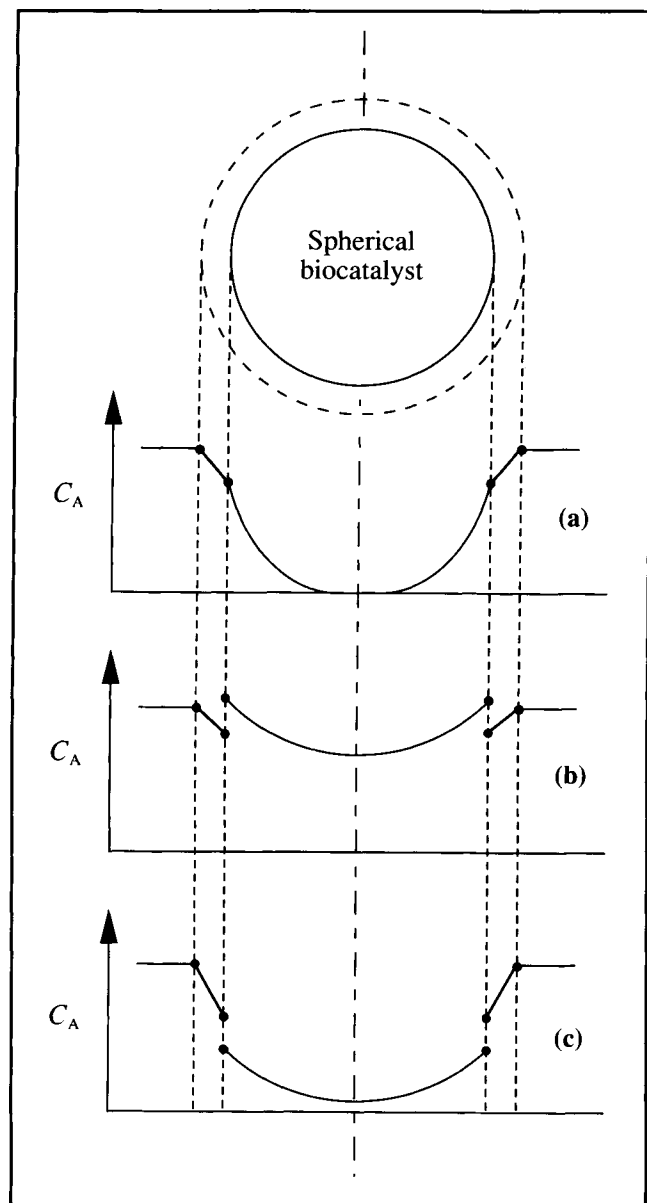
12.2 Concentration Gradients and Reaction Rates in Solid Catalysts

Consider a spherical catalyst of radius R immersed in well-mixed liquid containing substrate A. In the bulk liquid away from the particle the substrate concentration is uniform and equal to C_{Ab} . If the particle were inactive, after some time the concentration of substrate inside the solid would reach a constant value in equilibrium with C_{Ab} . However, when substrate is consumed by reaction, its concentration C_A decreases within the particle as shown in Figure 12.2. If immobilised cells or enzymes are distributed uniformly within the catalyst, the concentration profile is symmetrical with a minimum at the centre. Mass transfer of substrate to reaction sites in the particle is driven by the concentration difference between the bulk solution and particle interior.

In the bulk liquid, substrate is carried rapidly by convective currents. However, as substrate molecules approach the solid they must be transported from the bulk liquid across the relatively stagnant boundary layer to the solid surface; this process is called *external mass transfer*. A concentration gradient develops across the boundary layer from C_{Ab} in the bulk liquid to C_{As} at the solid-liquid interface. If the particle were not porous and all enzyme or cells confined to its outer surface, external mass transfer would be the only transport process required. More often, reaction takes place inside the particle so that *internal mass transfer* through the solid is also required.

Although the form of the concentration gradient shown in Figure 12.2 is typical, other variations are possible. If mass transfer is much slower than reaction, it is possible that all substrate entering the particle will be consumed before reaching the

Figure 12.3 Variations in substrate concentration profile in spherical biocatalysts.



centre. In this case, the concentration falls to zero within the solid as illustrated in Figure 12.3(a). Cells or enzyme near the centre are starved of substrate and the core of the particle becomes inactive. In the examples of Figures 12.3(b) and 12.3(c), the *partition coefficient* for the substrate is not equal to unity. This means that, at equilibrium and in the absence of reaction, the concentration of substrate in the solid is naturally higher or lower than in the liquid. In Figure 12.3(b), the dis-

continuity of concentration at the solid–liquid interface shows that substrate distributes preferentially to the solid phase. Conversely, Figure 12.3(c) shows the concentrations when substrate is attracted more to the liquid than to the solid. The effect of mass transfer on intraparticle concentration can be magnified or diminished by substrate partitioning. Partitioning is important when the substrate and solid are charged or if strong hydrophobic interactions cause repulsion or attraction. Because most materials used for cell and enzyme immobilisation are very porous and contain a high percentage of water, partition effects can often be neglected. In our treatment of heterogeneous reaction, we will assume that partitioning is not significant.

12.2.1 True and Observed Reaction Rates

Because concentrations vary in solid catalysts, local rates of reaction also vary depending on position within the particle. Even for zero-order reactions, reaction rate changes with position if substrate is exhausted. Each cell or enzyme molecule responds to the substrate concentration at its location with a rate of reaction determined by the kinetic parameters of the catalyst. This local rate of reaction is known as the *true rate* or *intrinsic rate*. Like any reaction rate, intrinsic rates can be expressed using total, volumetric or specific bases as described in Section 11.1.3. The relationship between true reaction rate and local substrate concentration follows the principles outlined in Chapter 11 for homogeneous reactions.

True local reaction rates are difficult to measure in solid catalysts without altering the reaction conditions. It is possible, however, to measure the overall reaction rate for the entire catalyst. In a closed system, the rate of disappearance of substrate from the bulk liquid must equal the overall rate of conversion by reaction; in heterogeneous systems this is also called the *observed rate*. It is important to remember that the observed rate is not usually equal to the true activity of any cell or enzyme in the particle. Because intraparticle substrate levels are reduced inside solid catalysts, we expect the observed rate to be less than if the entire particle were exposed to the bulk liquid. The relationship between observed rate and bulk substrate concentration is not as simple as in homogeneous reactions. Kinetic equations for heterogeneous reactions also involve mass-transfer parameters.

True reaction rates depend on the kinetic parameters of the cells or enzyme. For example, rate of reaction by an immobilised enzyme obeying Michaelis–Menten kinetics depends on the values of $v_{\max}$ and K_m for the enzyme in its immobilised state. These parameters are sometimes called *true kinetic parameters* or *intrinsic kinetic parameters*. Because kinetic parameters

can be altered during immobilisation as a result of cell or enzyme damage, configurational change and steric hindrance, values measured before immobilisation may not apply. Unfortunately, true kinetic parameters for immobilised biocatalysts can be difficult to determine because measured reaction rates incorporate mass transfer effects. The problem of evaluating true kinetic parameters is discussed further in Section 12.9.

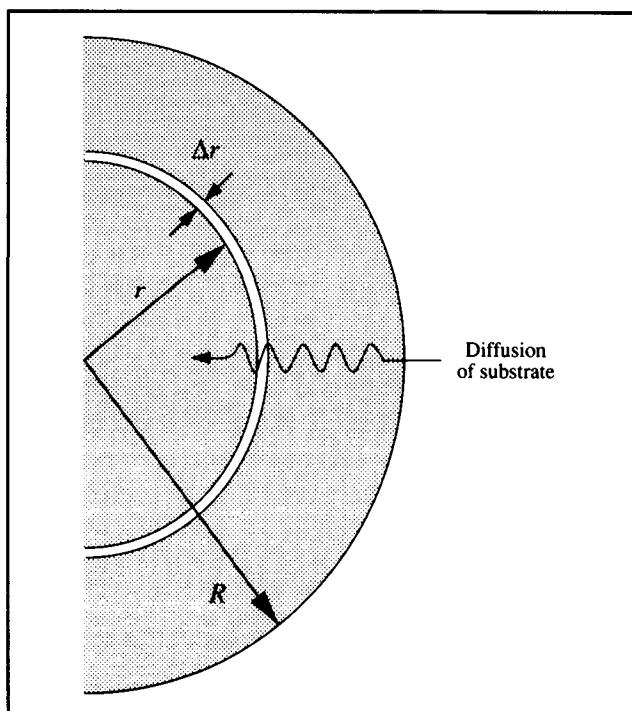
12.2.2 Interaction Between Mass Transfer and Reaction

Rates of reaction and substrate mass transfer are not independent in heterogeneous systems. Rate of mass transfer depends on the concentration gradient established in the system; this in turn depends on the rate of substrate depletion by reaction. On the other hand, rate of reaction depends on the availability of substrate; this of course depends on the rate of mass transfer. One of the objectives in analysing heterogeneous reactions is to determine the relative influences of mass transfer and reaction on observed reaction rates. One can conceive, for example, that if a reaction proceeds slowly even in the presence of adequate substrate, it is likely that mass transfer will be rapid enough to meet the reaction demand. In this case, the observed rate would be determined more directly by the reaction process than mass transfer. Conversely, if the reaction tends to be very rapid, it is likely that mass transfer will be too slow to supply substrate at the rate required. The observed rate would then reflect strongly the rate of mass transfer. As will be shown in the remainder of this chapter, there are mathematical criteria for assessing the extent to which mass transfer influences the observed reaction rate. Reactions which are significantly affected are called *mass-transfer limited* or *diffusion-limited* reactions. It is also possible to distinguish the relative influence of internal and external mass transfer. Improvement of mass transfer and the elimination of mass-transfer limitations are desired objectives in heterogeneous catalysis. Once the effect and location of major mass-transfer resistances are identified, it is then possible to devise strategies for their elimination.

12.3 Internal Mass Transfer and Reaction

Let us now concentrate on the processes occurring within a solid biocatalyst; external mass transfer will be examined later in the chapter. The exact equations and procedures used in this analysis depend on the geometry of the system and the reaction kinetics. First, let us consider the case of cells or enzymes immobilised in a spherical particle.

Figure 12.4 Shell mass balance on a spherical particle.



12.3.1 Steady-State Shell Mass Balance

Mathematical analysis of heterogeneous reactions involves a technique called the *shell mass balance*. In this section, we will perform a shell mass balance on a spherical catalyst particle of radius R . Imagine a thin spherical shell of thickness Δr located at radius r from the centre, as shown in Figure 12.4. It may be helpful to think of this shell as the thin wall of a ping-pong ball encased inside and concentric with a larger cricket ball of radius R . Substrate diffusing into the sphere must cross the shell to reach the centre.

A mass balance of substrate is performed around the shell by considering the processes of mass transfer and reaction occurring at radius r . The system considered for the mass balance is the shell only; the remainder of the sphere is ignored for the moment. Substrate diffuses into the shell at radius $(r + \Delta r)$ and leaves at radius r ; within the shell, immobilised cells or enzyme consume substrate by reaction. Flow of mass through the shell can be analysed using the general mass-balance equation derived in Chapter 4:

$$\left\{ \begin{array}{c} \text{mass in} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} - \left\{ \begin{array}{c} \text{mass out} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} + \left\{ \begin{array}{c} \text{mass} \\ \text{generated} \\ \text{within} \\ \text{system} \end{array} \right\} - \left\{ \begin{array}{c} \text{mass} \\ \text{consumed} \\ \text{within} \\ \text{system} \end{array} \right\} = \left\{ \begin{array}{c} \text{mass} \\ \text{accumulated} \\ \text{within} \\ \text{system} \end{array} \right\} \quad (4.1)$$

Before application of Eq. (4.1), certain assumptions must be made so that each term in the equation can be expressed mathematically [1].

- (i) *The particle is isothermal.* Kinetic parameters for enzyme and cell reactions are strong functions of temperature. If temperature in the particle varies, different values of the kinetic parameters must be applied. However, as temperature gradients generated by immobilised cells and enzymes are generally negligible, assuming constant temperature throughout the particle is reasonable and greatly simplifies the mathematical analysis.
- (ii) *Mass transfer occurs by diffusion only.* We will assume that the particle is impermeable to flow, so that convection within the pores is negligible. This assumption is valid for many solid-phase biocatalysts. However, some anomalies have been reported [2, 3]; depending on pore size, pressure gradients can induce convection of liquid through the particle and significantly enhance nutrient supply. When convective transport occurs, the analysis of mass transfer and reaction presented in this chapter must be modified [4–6].
- (iii) *Diffusion can be described using Fick's law with constant effective diffusivity.* We will assume that diffusive transport through the particle is governed by Fick's law (Section 9.1.1). Interaction of substrate with other concentration gradients and phenomena affecting transport of charged species are ignored. Fick's law will be applied using the *effective diffusivity* of substrate in the solid, $\mathcal{D}_{Ac}$. The value of $\mathcal{D}_{Ac}$ is a complex function of the molecular-diffusion characteristics of the substrate, the tortuosity of the diffusion path within the solid, and the fraction of the particle volume available for diffusion. We will assume that $\mathcal{D}_{Ac}$ is constant and independent of substrate concentration in the particle; this means that $\mathcal{D}_{Ac}$ does not change with position.
- (iv) *The particle is homogeneous.* Immobilised enzymes or cells are assumed to be distributed uniformly within the particle. Properties of the immobilisation matrix should also be uniform.
- (v) *The substrate partition coefficient is unity.* This assumption is valid for most substrates and particles, and ensures there is no discontinuity of concentration at the solid–liquid interface.
- (vi) *The particle is at steady state.* This assumption is usually valid if there is no change in activity of the catalyst, for example, due to enzyme deactivation, cell growth or differentiation. It is not valid when the system exhibits rapid transients such as when cells quickly consume and store substrates for subsequent metabolism.

- (vii) *Substrate concentration varies with a single spatial variable.*

For the sphere of Figure 12.4, we will assume that concentration varies only in the radial direction, and that substrate diffuses radially through the particle from the external surface towards the centre.

Eq. (4.1) is applied according to these assumptions. Substrate is transported into and out of the shell by diffusion; therefore, the first and second terms are expressed using Fick's law with constant effective diffusivity. The third term is zero as no substrate is generated. Substrate is consumed by reaction inside the shell at a rate equal to the volumetric rate of reaction r_A multiplied by the volume of the shell. According to assumption (vi) listed above, the system is at steady state. Thus, its composition and mass must be unchanging, substrate cannot accumulate in the shell, and the right-hand side of Eq. (4.1) is zero. After substituting the appropriate expressions and applying calculus to reduce the dimensions of the shell to an infinitesimal thickness, the result of the shell mass balance is a second-order differential equation for substrate concentration as a function of radius in the particle.

For a shell mass-balance on substrate A, the terms of Eq. (4.1) are expressed as follows:

$$\text{Rate of input by diffusion:} \quad \left(\mathcal{D}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_{r+\Delta r}$$

$$\text{Rate of output by diffusion:} \quad \left(\mathcal{D}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_r$$

$$\text{Rate of generation:} \quad 0$$

$$\text{Rate of consumption by reaction:} \quad r_A 4\pi r^2 \Delta r$$

$$\text{Rate of accumulation at steady state:} \quad 0.$$

$\mathcal{D}_{Ac}$ is the effective diffusivity of substrate A, C_A is the concentration of A in the particle, r is distance measured radially from the centre, Δr is thickness of the shell, and r_A is the rate of reaction per unit volume particle. Each of the above terms has dimensions MT^{-1} or NT^{-1} with units of, for example, $kg\ h^{-1}$ or $gmol\ s^{-1}$. The first two terms are derived from Fick's law of Eq. (9.1); the area of the spherical shell available for diffusion is $4\pi r^2$. The term

$$\left(\mathcal{D}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_{r+\Delta r}$$

means $\left(\mathcal{G}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right)$ evaluated at radius $(r + \Delta r)$;

$$\left(\mathcal{G}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_r$$

means $\left(\mathcal{G}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right)$ evaluated at r .

The shell volume is $4\pi r^2 \Delta r$.

From Eq. (4.1) we obtain the following steady-state mass-balance equation:

$$\left(\mathcal{G}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_{r+\Delta r} - \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} 4\pi r^2 \right) \Big|_r - r_A 4\pi r^2 \Delta r = 0. \quad (12.1)$$

Dividing each term by $4\pi \Delta r$ gives:

$$\frac{\left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right) \Big|_{r+\Delta r} - \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right) \Big|_r}{\Delta r} - r_A r^2 = 0. \quad (12.2)$$

Eq. (12.2) can be written in the form:

$$\frac{\Delta \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right)}{\Delta r} - r_A r^2 = 0 \quad (12.3)$$

where

$$\Delta \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right) \text{ means the change in } \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right)$$

across Δr .

Eq. (12.3) is valid for a spherical shell of thickness Δr . To develop an equation which applies to any *point* in the sphere, we must shrink Δr to zero. As Δr appears only in the first term of Eq. (12.3), taking the limit of Eq. (12.3) as $\Delta r \rightarrow 0$ gives:

$$\lim_{\Delta r \rightarrow 0} \frac{\Delta \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right)}{\Delta r} - r_A r^2 = 0. \quad (12.4)$$

Invoking the definition of the derivative from Section D.2 of the Appendix, Eq. (12.4) is identical to the second-order differential equation:

$$\frac{d}{dr} \left(\mathcal{G}_{Ac} \frac{dC_A}{dr} r^2 \right) - r_A r^2 = 0. \quad (12.5)$$

According to assumption (iii), $\mathcal{G}_{Ac}$ is independent of r and can be moved outside the differential:

$$\mathcal{G}_{Ac} \frac{d}{dr} \left(\frac{dC_A}{dr} r^2 \right) - r_A r^2 = 0. \quad (12.6)$$

In Eq. (12.6) we have a differential equation representing diffusion and reaction in a spherical biocatalyst. That Eq. (12.6) is a second-order differential equation becomes clear if the first term is written in its expanded form:

$$\mathcal{G}_{Ac} \left(\frac{d^2 C_A}{dr^2} r^2 + 2r \frac{dC_A}{dr} \right) - r_A r^2 = 0. \quad (12.7)$$

Eq. (12.7) can be solved by integration to yield an expression for the concentration profile in the particle: C_A as a function of r . However, we cannot integrate Eq. (12.7) as it stands because the reaction rate r_A is in most cases a function of C_A . Let us consider solutions of Eq. (12.7) with r_A representing first-order, zero-order and Michaelis–Menten kinetics.

12.3.2 Concentration Profile: First-Order Kinetics and Spherical Geometry

For first-order kinetics, Eq. (12.7) becomes:

$$\mathcal{G}_{Ac} \left(\frac{d^2 C_A}{dr^2} r^2 + 2r \frac{dC_A}{dr} \right) - k_1 C_A r^2 = 0 \quad (12.8)$$

where k_1 is the intrinsic first-order rate constant with dimensions T^{-1} . For biocatalytic reactions, k_1 depends on the

density of cells or enzyme in the particle. According to assumptions (i), (iii) and (iv) in Section 12.3.1, k_1 and $\mathcal{D}_{Ac}$ for a given particle can be considered constant. Accordingly, as the only variables in Eq. (12.8) are C_A and r , the equation is ready for integration. Because Eq. (12.8) is a second-order differential equation we need two boundary conditions. These are:

$$C_A = C_{As} \quad \text{at } r = R \quad (12.9)$$

$$\frac{dC_A}{dr} = 0 \quad \text{at } r = 0 \quad (12.10)$$

where C_{As} is the concentration of substrate at the outer surface of the particle. For the present we will assume C_{As} is known or can be measured. Eq. (12.10) is called the *symmetry condition*. As indicated in Figures 12.2 and 12.3, the substrate concentration profile is symmetrical about the centre of the sphere.

Therefore, the substrate concentration is minimum with slope $dC_A/dr = 0$ at $r = 0$. Integration of Eq. (12.8) with boundary conditions Eqs (12.9) and (12.10) gives the following expression for substrate concentration as a function of radius [7]:

$$C_A = C_{As} \frac{R}{r} \frac{\sinh(r\sqrt{k_1/\mathcal{D}_{Ac}})}{\sinh(R\sqrt{k_1/\mathcal{D}_{Ac}})} \quad (12.11)$$

In Eq. (12.11), $\sinh$ is the abbreviation for *hyperbolic sine*; $\sinh x$ is defined as:

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (12.12)$$

Eq. (12.11) may appear complex, but contains simple exponential terms relating C_A and r , $\mathcal{D}_{Ac}$ representing rate of mass transfer, and k_1 representing rate of reaction.

Example 12.1 Concentration profile for immobilised enzyme

Enzyme is immobilised in 8 mm diameter agarose beads at a concentration of $0.018 \text{ kg protein m}^{-3}$ gel. Ten beads are immersed in a well-mixed solution containing $3.2 \times 10^{-3} \text{ kg m}^{-3}$ substrate. The effective diffusivity of substrate in agarose gel is $2.1 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. Kinetics of the enzyme can be approximated as first order with specific rate constant $3.11 \times 10^5 \text{ s}^{-1}$ per kg protein. Mass transfer effects outside the particles are negligible. Plot the steady-state substrate concentration profile as a function of particle radius.

Solution:

$R = 4 \times 10^{-3} \text{ m}$; $\mathcal{D}_{Ac} = 2.1 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. In the absence of external mass-transfer effects, $C_{As} = 3.2 \times 10^{-3} \text{ kg m}^{-3}$.

$$\text{Volume per bead} = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi (4 \times 10^{-3} \text{ m})^3 = 2.68 \times 10^{-7} \text{ m}^3.$$

Therefore, 10 beads have volume $2.68 \times 10^{-6} \text{ m}^3$. The amount of enzyme present is:

$$2.68 \times 10^{-6} \text{ m}^3 (0.018 \text{ kg m}^{-3}) = 4.83 \times 10^{-8} \text{ kg}.$$

Therefore:

$$k_1 = 3.11 \times 10^5 \text{ s}^{-1} \text{ kg}^{-1} (4.83 \times 10^{-8} \text{ kg}) = 0.015 \text{ s}^{-1}$$

and:

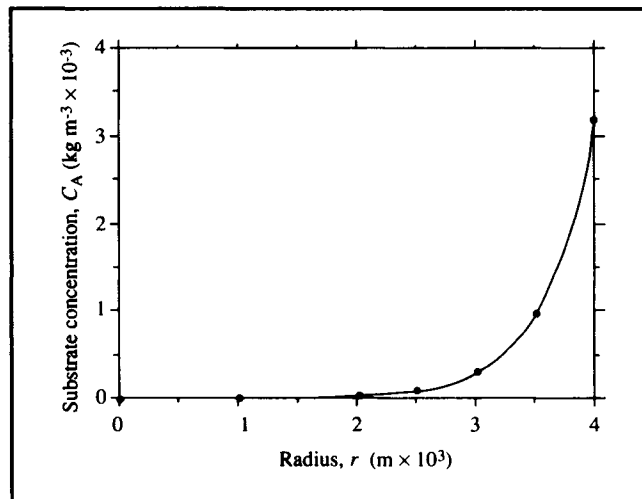
$$R \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}} = 10.693.$$

The denominator of Eq. (12.11) is:

$$\sinh \left(R \sqrt{k_1 / \mathcal{D}_{Ac}} \right) = \frac{e^{10.693} - e^{-10.693}}{2} = 2.202 \times 10^4.$$

C_A is calculated as a function of r from Eq. (12.11) and plotted in Figure 12E1.1. Substrate concentration drops rapidly inside the particle to reach virtually zero 2 mm from the centre.

Figure 12E1.1 Substrate concentration profile in an immobilised-enzyme bead.



12.3.3 Concentration Profile: Zero-Order Kinetics and Spherical Geometry

From Eq. (12.7), the differential equation for zero-order kinetics is:

$$\mathcal{D}_{Ac} \left(\frac{d^2 C_A}{dr^2} r^2 + 2r \frac{dC_A}{dr} \right) - k_0 r^2 = 0 \quad (12.13)$$

where k_0 is the intrinsic zero-order rate constant with units of, for example, $gmol\ s^{-1}\ m^{-3}$ particle. Like k_1 for first-order reactions, k_0 varies with cell or enzyme density in the catalyst.

Zero-order reactions are unique in that, provided substrate is present, reaction rate is independent of substrate concentration. In solving Eq. (12.13) we must account for the possibility that substrate becomes depleted within the particle. As illustrated in Figure 12.5, if we assume this occurs at some radius R_0 , the rate of reaction for $0 < r \leq R_0$ is zero. Everywhere else inside the particle, i.e., $r > R_0$, the volumetric reaction rate is constant and equal to k_0 irrespective of substrate concentration. For this situation the boundary conditions are:

$$C_A = C_{As} \quad \text{at } r = R \quad (12.9)$$

$$\frac{dC_A}{dr} = 0 \quad \text{at } r = R_0. \quad (12.14)$$

Solution of Eq. (12.13) with these boundary conditions gives the following expression for C_A as a function of r [7]:

$$C_A = C_{As} + \frac{k_0 R^2}{6\mathcal{D}_{Ac}} \left(\frac{r^2}{R^2} - 1 + \frac{2R_0^3}{rR^2} - \frac{2R_0^3}{R^3} \right). \quad (12.15)$$

Eq. (12.15) is difficult to apply in practice because R_0 is generally not known. However, the equation can be simplified if C_A remains > 0 everywhere so that R_0 no longer exists. Substituting $R_0 = 0$ into Eq. (12.15) gives:

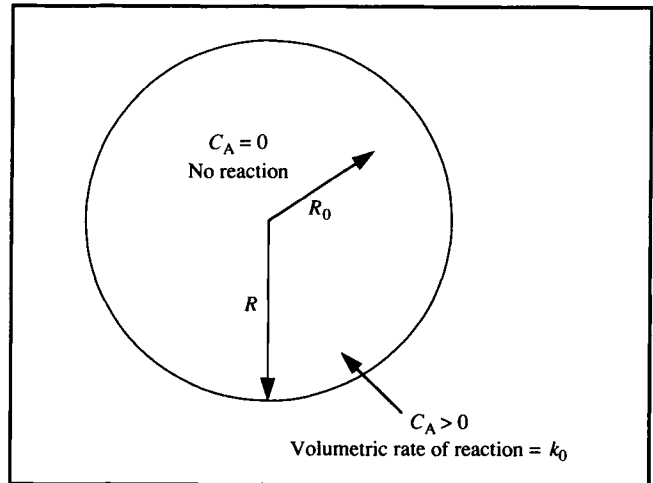
$$C_A = C_{As} + \frac{k_0}{6\mathcal{D}_{Ac}} (r^2 - R^2). \quad (12.16)$$

In bioprocess applications, it is important that the core of catalyst particles does not become starved of substrate. The likelihood of this happening increases with size of the particle. For zero-order reactions we can calculate the maximum particle radius for which C_A remains > 0 . In such a particle, substrate is depleted just at the centre point. Therefore, calculating R from Eq. (12.16) with $C_A = r = 0$:

$$R_{\max} = \sqrt{\frac{6\mathcal{D}_{Ac}C_{As}}{k_0}} \quad (12.17)$$

where $R_{\max}$ is the maximum particle radius for $C_A > 0$.

Figure 12.5 Concentration and reaction zones in a spherical particle with zero-order reaction. Substrate is depleted at radius R_0 .



Example 12.2 Maximum particle size for zero-order reaction

Non-viable yeast cells are immobilised in alginate beads. The beads are stirred in glucose medium under anaerobic conditions. The effective diffusivity of glucose in the beads depends on cell density according to the relationship:

$$\mathcal{D}_{Ac} = 6.33 - 7.17y_C$$

where $\mathcal{D}_{Ac}$ is effective diffusivity $\times 10^{10} \text{ m}^2 \text{ s}^{-1}$ and y_C is the weight fraction of yeast in the gel. Rate of glucose uptake can be assumed to be zero order; the rate constant at a yeast density in alginate of 15 wt% is $0.5 \text{ g l}^{-1} \text{ min}^{-1}$. For maximum reaction rate, the concentration of glucose inside the particles should remain above zero.

- Plot the maximum allowable particle size as a function of bulk glucose concentration between 5 g l^{-1} and 60 g l^{-1} .
- For 30 g l^{-1} glucose, plot $R_{\max}$ as a function of cell loading between 10 and 45 wt%.

Solution:

- At $y_C = 0.15$, $\mathcal{D}_{Ac} = 5.25 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$. Converting k_0 to units of $\text{kg m}^{-3} \text{ s}^{-1}$:

$$\begin{aligned} k_0 &= 0.5 \text{ g l}^{-1} \text{ min}^{-1} \cdot \left| \frac{1 \text{ kg}}{1000 \text{ g}} \right| \cdot \left| \frac{1000 \text{ l}}{1 \text{ m}^3} \right| \cdot \left| \frac{1 \text{ min}}{60 \text{ s}} \right| \\ &= 8.33 \times 10^{-3} \text{ kg m}^{-3} \text{ s}^{-1}. \end{aligned}$$

Assume C_{As} is equal to the bulk glucose concentration; C_{As} in g l^{-1} is the same as kg m^{-3} . $R_{\max}$ is calculated from Eq. (12.17).

$C_{As} (\text{kg m}^{-3})$	$R_{\max} (\text{m})$
5	1.38×10^{-3}
15	2.38×10^{-3}
25	3.07×10^{-3}
45	4.13×10^{-3}
60	4.76×10^{-3}

These results are plotted in Figure 12E2.1. At low external glucose concentrations, particles are restricted to small radii. The driving force for diffusion increases with C_{As} so that larger particles may be used.

Figure 12E2.1 Maximum particle radius as a function of external substrate concentration.

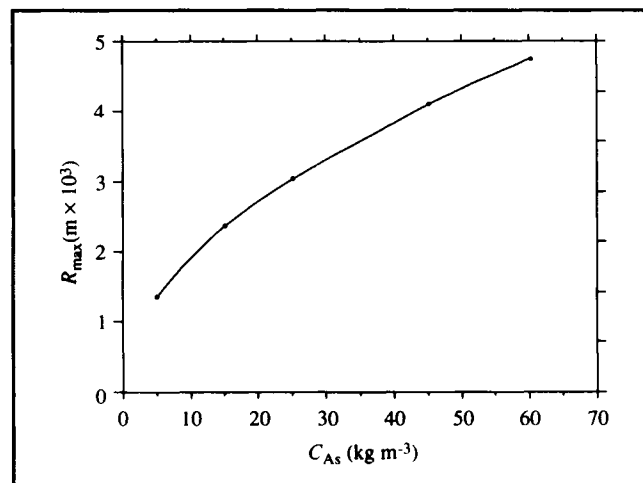
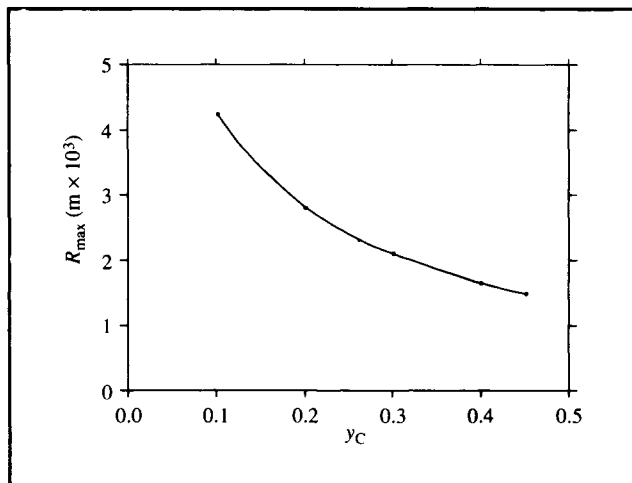


Figure 12E2.2 Maximum particle radius as a function of cell density.



(b) $C_{As} = 30 \text{ kg m}^{-3}$. As y_C varies, values of $\mathcal{D}_{Ac}$ and k_0 are affected. Changes in $\mathcal{D}_{Ac}$ can be calculated from the equation provided. We assume k_0 is directly proportional to cell density as described in Eq. (11.25), i.e. there is no steric hindrance or interaction between cells as y_C increases. Results as a function of y_C are listed below.

y_C	$\mathcal{D}_{Ac}$ ($\text{m}^2 \text{s}^{-1}$)	k_0 ($\text{kg m}^{-3} \text{s}^{-1}$)	$R_{\max}$ (m)
0.1	5.61×10^{-10}	5.55×10^{-3}	4.27×10^{-3}
0.2	4.90×10^{-10}	1.11×10^{-2}	2.82×10^{-3}
0.3	4.18×10^{-10}	1.67×10^{-2}	2.12×10^{-3}
0.4	3.46×10^{-10}	2.22×10^{-2}	1.67×10^{-3}
0.45	3.10×10^{-10}	2.50×10^{-2}	1.50×10^{-3}

The results are plotted in Figure 12E2.2. As y_C increases, $\mathcal{D}_{Ac}$ declines and k_0 increases. Lower $\mathcal{D}_{Ac}$ reduces the rate of diffusion into the particles; higher k_0 increases the demand for substrate. Therefore, increasing the cell density exacerbates mass-transfer restrictions. To ensure adequate supply of substrate under these conditions, the particle size must be reduced.

12.3.4 Concentration Profile: Michaelis–Menten Kinetics and Spherical Geometry

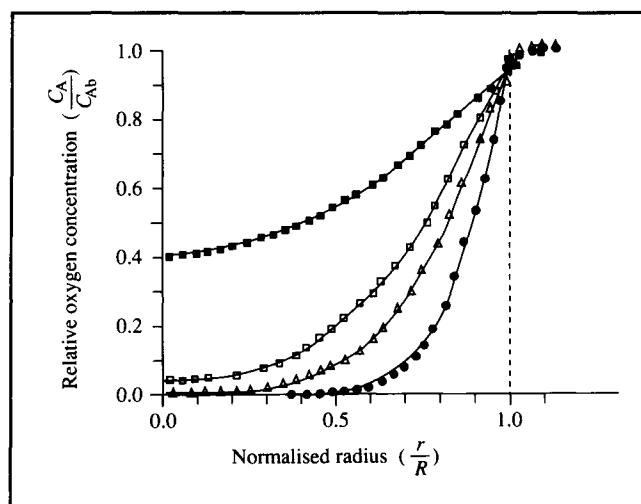
If reaction in the particle follows Michaelis–Menten kinetics, r_A takes the form of Eq. (11.30). Eq. (12.7) becomes:

$$\mathcal{D}_{Ac} \left(\frac{d^2 C_A}{dr^2} + 2r \frac{dC_A}{dr} \right) - \frac{v_{\max} C_A}{K_m + C_A} r^2 = 0 \quad (12.18)$$

where $v_{\max}$ and K_m are intrinsic kinetic parameters for the reaction. $v_{\max}$ has units of, for example, $\text{kg s}^{-1} \text{m}^{-3}$ particle; its value depends on the concentration of cells or enzyme in the particle.

Owing to the non-linearity of the Michaelis–Menten expression, simple analytical integration of Eq. (12.18) is not possible. However, results for C_A as a function of r can be obtained using numerical methods, usually by computer. Because Michaelis–Menten kinetics lie somewhere between zero- and first-order kinetics (see Section 11.3.3), explicit

Figure 12.6 Measured and calculated oxygen concentrations in a spherical agarose bead containing immobilised enzyme. Particle diameter = 4 mm; $C_{Ab} = 0.2 \text{ mol m}^{-3}$. Enzyme loadings are: $0.0025 \text{ kg m}^{-3} \text{ gel}$ (■); $0.005 \text{ kg m}^{-3} \text{ gel}$ (□); $0.0125 \text{ kg m}^{-3} \text{ gel}$ (△); and $0.025 \text{ kg m}^{-3} \text{ gel}$ (●). Measured concentrations are shown using symbols; calculated profiles are shown as lines. (From C.M. Hooijmans, S.G.M. Geraats and K.Ch.A.M. Luyben, 1990, Use of an oxygen microsensor for the determination of intrinsic kinetic parameters of an immobilised oxygen reducing enzyme. *Biotechnol. Bioeng.* 35, 1078–1087.)



solutions found in Sections 12.3.2 and 12.3.3 can be used to estimate the extreme limits for Michaelis–Menten reactions.

Concentration profiles calculated from the equations presented in this section have been verified experimentally in several studies. Using special microelectrodes with tip diameters of the order $1 \mu\text{m}$, it is possible to measure concentrations of oxygen and ions inside soft solids and cell slimes. As an example, oxygen concentrations measured in immobilised-enzyme beads are shown in Figure 12.6. The experimental data are very close to the calculated concentration profiles. Similar results have been found in other systems [8–10].

12.3.5 Concentration Profiles in Other Geometries

Our attention so far has been focussed on spherical catalysts. However, equations similar to Eq. (12.7) can be obtained from shell mass balances on other geometries. Of all other shapes, the one of most interest in bioprocessing is the flat plate. A typical substrate concentration profile for this geometry without external boundary-layer effects is illustrated in

Figure 12.7 Substrate concentration profile in an infinite flat plate without boundary-layer effects.

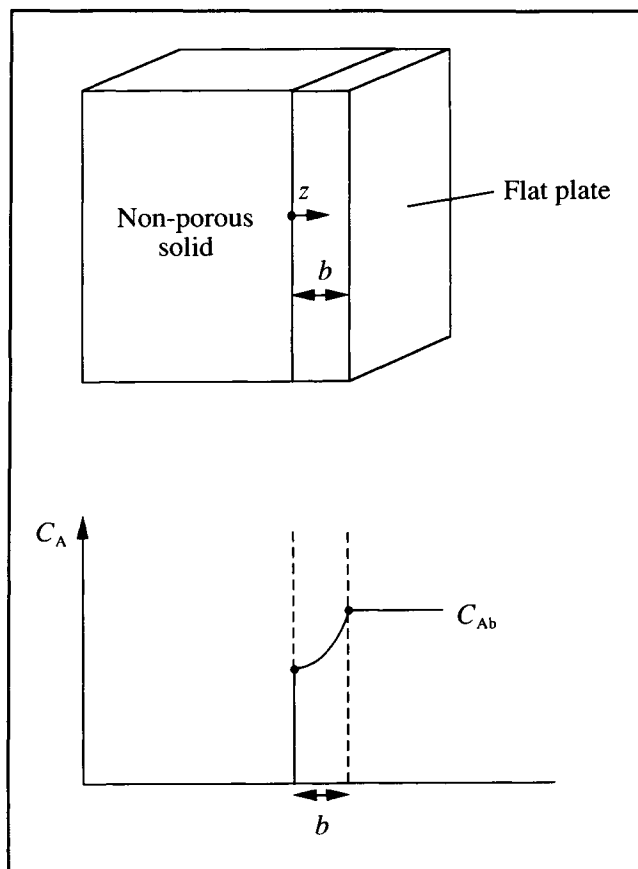


Figure 12.7. Equations for flat-plate geometry are used to analyse reactions in cell films attached to inert solids; the biofilm constitutes the flat plate. Even if the surface supporting the biofilm is curved rather than flat, if the film thickness b is very small compared with the radius of curvature, equations for flat-plate geometry are applicable. To simplify mathematical treatment and keep the problem one-dimensional (as required by assumption (vii) of Section 12.3.1), the flat plate is assumed to have infinite length. In practice, this assumption is reasonable if its length is much greater than its thickness. If not, it must be assumed that the ends of the plate are sealed to eliminate axial concentration gradients.

Another catalyst shape of some relevance to bioprocessing is the hollow cylinder; this is useful in analysis of hollow-fibre membrane reactors. However, because of its relatively limited application, we will not consider this geometry further.

Concentration profiles for spherical and flat-plate geometries and first- and zero-order kinetics are summarised in Table

12.1. Boundary conditions for the flat plate similar to Eqs (12.9) and (12.10) are as follows:

$$C_A = C_{As} \quad \text{at } z = b \quad (12.19)$$

$$\frac{dC_A}{dz} = 0 \quad \text{at } z = 0 \quad (12.20)$$

where C_{As} is the concentration of A at the solid–liquid interface, z is distance measured from the inner surface of the plate and b is the plate thickness.

12.3.6 Prediction of Observed Reaction Rate

Equations for intracatalyst substrate concentration such as those in Table 12.1 allow us to predict overall rates of reaction. Let us consider the situation for spherical particles and first-order, zero-order and Michaelis–Menten kinetics. Analogous equations can be derived for other geometries.

- (i) *First-order kinetics.* Rate of reaction at any point in the sphere depends on the first-order kinetic constant k_1 and the concentration of substrate at that point. The overall rate for the entire particle is equal to the sum of all such rates at every location in the solid. This sum is mathematically equivalent to integrating the expression $k_1 C_A$ over the entire particle volume, taking into account the variation of C_A with radius expressed in Eq. (12.11). The result is an equation for the observed reaction rate $r_{A,obs}$ in a single particle:

$$r_{A,obs} = 4\pi R \mathcal{D}_{Ac} C_{As} [R\sqrt{k_1/\mathcal{D}_{Ac}} \coth(R\sqrt{k_1/\mathcal{D}_{Ac}}) - 1] \quad (12.21)$$

where $\coth$ is the abbreviation for *hyperbolic cotangent* defined by:

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}. \quad (12.22)$$

- (ii) *Zero-order kinetics.* As long as substrate is present, zero-order reactions occur at a fixed rate independent of substrate concentration. Therefore, if $C_A > 0$ everywhere in the particle, the overall rate of reaction is equal to the zero-order rate constant k_0 multiplied by the particle volume:

Table 12.1 Steady-state concentration profiles

First-order reaction: $r_A = k_1 C_A$

$$\text{Sphere}^a \quad C_A = C_{As} \frac{R}{r} \frac{\sinh(r\sqrt{k_1/\mathcal{D}_{Ac}})}{\sinh(R\sqrt{k_1/\mathcal{D}_{Ac}})}$$

$$\text{Flat plate}^b \quad C_A = C_{As} \frac{\cosh(z\sqrt{k_1/\mathcal{D}_{Ac}})}{\cosh(b\sqrt{k_1/\mathcal{D}_{Ac}})}$$

Zero-order reaction: $r_A = k_0$

$$\text{Sphere}^c \quad C_A = C_{As} + \frac{k_0}{6\mathcal{D}_{Ac}} (r^2 - R^2)$$

$$\text{Flat plate}^c \quad C_A = C_{As} + \frac{k_0}{2\mathcal{D}_{Ac}} (z^2 - b^2)$$

^a Sinh is the abbreviation of hyperbolic sine. Sinh x is defined as:

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

^b Cosh is the abbreviation of hyperbolic cosine. Cosh x is defined as:

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

^c For $C_A > 0$ everywhere within the catalyst.

$$r_{A,obs} = \frac{4}{3} \pi R^3 k_0. \quad (12.23)$$

However, if C_A falls to zero at some radius R_0 , the inner volume $\frac{4}{3} \pi R_0^3$ is inactive. In this case, the rate of reaction per particle is equal to k_0 multiplied by the active particle volume:

$$r_{A,obs} = \left(\frac{4}{3} \pi R^3 - \frac{4}{3} \pi R_0^3 \right) k_0 = \frac{4}{3} \pi (R^3 - R_0^3) k_0. \quad (12.24)$$

- (iii) *Michaelis–Menten kinetics.* Observed rates for Michaelis–Menten reactions cannot be expressed

explicitly because we do not have an equation for C_A as a function of radius. $r_{A,obs}$ can be evaluated, however, using numerical methods.

12.4 The Thiele Modulus and Effectiveness Factor

Charts based on the equations of the previous section allow us to determine $r_{A,obs}$ relative to r_{As}^* , the reaction rate that would occur if all cells or enzyme were exposed to the external substrate concentration. Differences between $r_{A,obs}$ and r_{As}^* show immediately the extent to which reaction is affected by internal mass transfer. Comparison of these rates requires application of theory as described in the following sections.

12.4.1 First-Order Kinetics

If a catalyst particle is unaffected by mass transfer, the concentration of substrate inside the particle is constant and equal to the surface concentration, C_{As} . Thus, the rate of first-order reaction without internal mass-transfer effects is equal to $k_1 C_{As}$ multiplied by the particle volume:

$$r_{As}^* = \frac{4}{3} \pi R^3 k_1 C_{As}. \quad (12.25)$$

The extent to which $r_{A,obs}$ is different from r_{As}^* is expressed by means of the *internal effectiveness factor* η_i :

$$\eta_i = \frac{r_{A,obs}}{r_{As}^*} = \frac{(\text{observed rate})}{\left(\text{rate that would occur if } C_A = C_{As} \text{ everywhere in the particle} \right)}. \quad (12.26)$$

In the absence of mass-transfer limitations, $r_{A,obs} = r_{As}^*$ and $\eta_i = 1$; when mass-transfer effects reduce $r_{A,obs}$, $\eta_i < 1$. For calculation of η_i , $r_{A,obs}$ and r_{As}^* should have the same units, for example, $\text{kg s}^{-1} \text{m}^{-3}$, gmol s^{-1} per particle, etc. We can substitute expressions for $r_{A,obs}$ and r_{As}^* from Eqs (12.21) and (12.25) into Eq. (12.26) to derive an expression for η_{i1} , the internal effectiveness factor for first-order reaction:

$$\eta_{i1} = \frac{3\mathcal{D}_{Ac}}{R^2 k_1} \left[R \sqrt{k_1 / \mathcal{D}_{Ac}} \coth(R \sqrt{k_1 / \mathcal{D}_{Ac}}) - 1 \right]. \quad (12.27)$$

Thus, the internal effectiveness factor for first-order reaction depends on only three parameters: R , k_1 and $\mathcal{D}_{Ac}$. These parameters are usually grouped together to form a dimensionless variable called the *Thiele modulus*. There are several definitions of the Thiele modulus in the literature; as it was formulated originally [11], application of the modulus was cumbersome because a separate definition was required for different reaction kinetics and catalyst geometries. Generalised moduli which apply to any catalyst shape and reaction kinetics have since been proposed [12–14]. The *generalised Thiele modulus* ϕ is defined as:

$$\phi = \frac{V_p}{S_x} \frac{r_A|_{C_{As}}}{\sqrt{2}} \left(\int_{C_{A,eq}}^{C_{As}} \mathcal{D}_{Ac} r_A dC_A \right)^{-1/2} \quad (12.28)$$

where V_p is catalyst volume, S_x is external surface area, C_{As} is substrate concentration at the surface of the catalyst, r_A is reaction rate, $r_A|_{C_{As}}$ is the reaction rate when $C_A = C_{As}$, $\mathcal{D}_{Ac}$ is effective diffusivity of substrate and $C_{A,eq}$ is the equilibrium substrate concentration. As explained in Section 11.1.1, fermentations and many enzyme reactions are irreversible so that $C_{A,eq}$ is zero for most biological applications. From geometry, $V_p/S_x = R/3$ for spheres and b for flat plates. Expressions determined from Eq. (12.28) for first-order, zero-order and Michaelis–Menten kinetics are listed in Table 12.2 as ϕ_1 , ϕ_0 and ϕ_m , respectively. ϕ represents a dimensionless combination of the important parameters affecting mass transfer and reaction in heterogeneous systems: catalyst size (R or b), effective diffusivity ($\mathcal{D}_{Ac}$), surface concentration (C_{As}), and intrinsic rate parameters (k_0 , k_1 or $v_{\max}$ and K_m). Only the Thiele modulus for first-order reactions does not depend on substrate concentration.

When parameters R , k_1 and $\mathcal{D}_{Ac}$ in Eq. (12.27) are grouped together as ϕ_1 , the result is:

$$\eta_{i1} = \frac{1}{3\phi_1^2} (3\phi_1 \coth 3\phi_1 - 1) \quad (12.29)$$

where $\coth$ is defined by Eq. (12.22). Eq. (12.29) applies to spherical geometry and first-order reaction; an analogous equation for flat plates is listed in Table 12.3. Plots of η_{i1} versus ϕ_1 for sphere, cylinder and flat-plate catalysts are shown in Figure 12.8. The curves coincide exactly for $\phi_1 \rightarrow 0$ and $\phi_1 \rightarrow \infty$, and fall within 10–15% for the remainder of the

Table 12.2 Generalised Thiele moduliFirst-order reaction: $r_A = k_1 C_A$

$$\phi_1 = \frac{V_p}{S_x} \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}}$$

$$\text{Sphere} \quad \phi_1 = \frac{R}{3} \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}}$$

$$\text{Flat plate} \quad \phi_1 = b \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}}$$

Zero-order reaction: $r_A = k_0$

$$\phi_0 = \frac{1}{\sqrt{2}} \frac{V_p}{S_x} \sqrt{\frac{k_0}{\mathcal{D}_{Ac} C_{As}}}$$

$$\text{Sphere} \quad \phi_0 = \frac{R}{3\sqrt{2}} \sqrt{\frac{k_0}{\mathcal{D}_{Ac} C_{As}}}$$

$$\text{Flat plate} \quad \phi_0 = \frac{b}{\sqrt{2}} \sqrt{\frac{k_0}{\mathcal{D}_{Ac} C_{As}}}$$

Michaelis–Menten reaction: $r_A = \frac{v_{\max} C_A}{K_m + C_A}$

$$\phi_m = \frac{1}{\sqrt{2}} \frac{V_p}{S_x} \sqrt{\frac{v_{\max}}{\mathcal{D}_{Ac} C_{As}}} \left(\frac{1}{1+\beta} \right) \left[1 + \beta \ln \left(\frac{\beta}{1+\beta} \right) \right]^{-1/2}$$

$$\beta = K_m / C_{As}$$

$$\text{Sphere} \quad \phi_m = \frac{R}{3\sqrt{2}} \sqrt{\frac{v_{\max}}{\mathcal{D}_{Ac} C_{As}}} \left(\frac{1}{1+\beta} \right) \left[1 + \beta \ln \left(\frac{\beta}{1+\beta} \right) \right]^{-1/2}$$

$$\text{Flat plate} \quad \phi_m = \frac{b}{\sqrt{2}} \sqrt{\frac{v_{\max}}{\mathcal{D}_{Ac} C_{As}}} \left(\frac{1}{1+\beta} \right) \left[1 + \beta \ln \left(\frac{\beta}{1+\beta} \right) \right]^{-1/2}$$

range. Figure 12.8 can be used to evaluate η_{i1} for any catalyst shape provided ϕ_1 is calculated using Eq. (12.28). Because of the errors involved in estimating the parameters defining ϕ_1 , it has been suggested that effectiveness factor curves be viewed as diffuse bands rather than precise functions [15].

Thus, if the first-order Thiele modulus is known, we can

use Figure 12.8 to find the internal effectiveness factor and Eqs (12.25) and (12.26) to predict the overall reaction rate for the catalyst. At low values of $\phi_1 < 0.3$, $\eta_{i1} \approx 1$ and the rate of reaction is not adversely affected by internal mass transfer. However as ϕ_1 increases above 0.3, η_{i1} falls as mass-transfer limitations come into play. Therefore, the value of the Thiele

Table 12.3 Effectiveness factors (ϕ for each geometry and kinetic order is defined in Table 12.2)First-order reaction: $r_A = k_1 C_A$

$$\text{Sphere}^a \quad \eta_{i1} = \frac{1}{3\phi_1^2} (3\phi_1 \coth 3\phi_1 - 1)$$

$$\text{Flat plate}^b \quad \eta_{i1} = \frac{\tanh \phi_1}{\phi_1}$$

Zero-order reaction: $r_A = k_0$

$$\text{Sphere}^c \quad \eta_{i0} = 1 \quad \text{for } 0 < \phi_0 \leq 0.577$$

$$\left. \begin{aligned} \text{Sphere}^c \quad \eta_{i0} &= 1 - \left[\frac{1}{2} + \cos \left(\frac{\Psi + 4\pi}{3} \right) \right]^3 \\ \text{where } \Psi &= \cos^{-1} \left(\frac{2}{3\phi_0^2} - 1 \right) \end{aligned} \right\} \quad \text{for } \phi_0 > 0.577$$

$$\text{Flat plate} \quad \eta_{i0} = 1 \quad \text{for } 0 < \phi_0 \leq 1$$

$$\eta_{i0} = \frac{1}{\phi_0} \quad \text{for } \phi_0 > 1$$

^a Coth is the abbreviation of hyperbolic cotangent. Coth x is defined as:

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}.$$

^b Tanh is the abbreviation of hyperbolic tangent. Tanh x is defined as:

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

^c Cos is the abbreviation of cosine. The notation $\cos^{-1} x$ (or $\arccos x$) denotes any angle whose cosine is x . Angles used to determine $\cos$ and $\cos^{-1}$ are in radians.

modulus indicates immediately whether the rate of reaction is diminished due to diffusional effects, or whether the catalyst is performing at its maximum rate at the prevailing surface concentration. For strong diffusion limitations at $\phi_1 > 10$, η_{i1} for all geometries can be estimated as:

$$\eta_{i1} \approx \frac{1}{\phi_1}.$$

(12.30)

12.4.2 Zero-Order Kinetics

When substrate is present throughout the catalyst, evaluation of the zero-order internal effectiveness factor η_{i0} is straightforward. Under these conditions, the reaction proceeds at the same rate that would occur if $C_A = C_{As}$ throughout in the particle. Therefore, from Eq. (12.26), $\eta_{i0} = 1$ and:

$$r_{A,\text{obs}} = r_{As}^* = \frac{4}{3} \pi R^3 k_0.$$

(12.31)

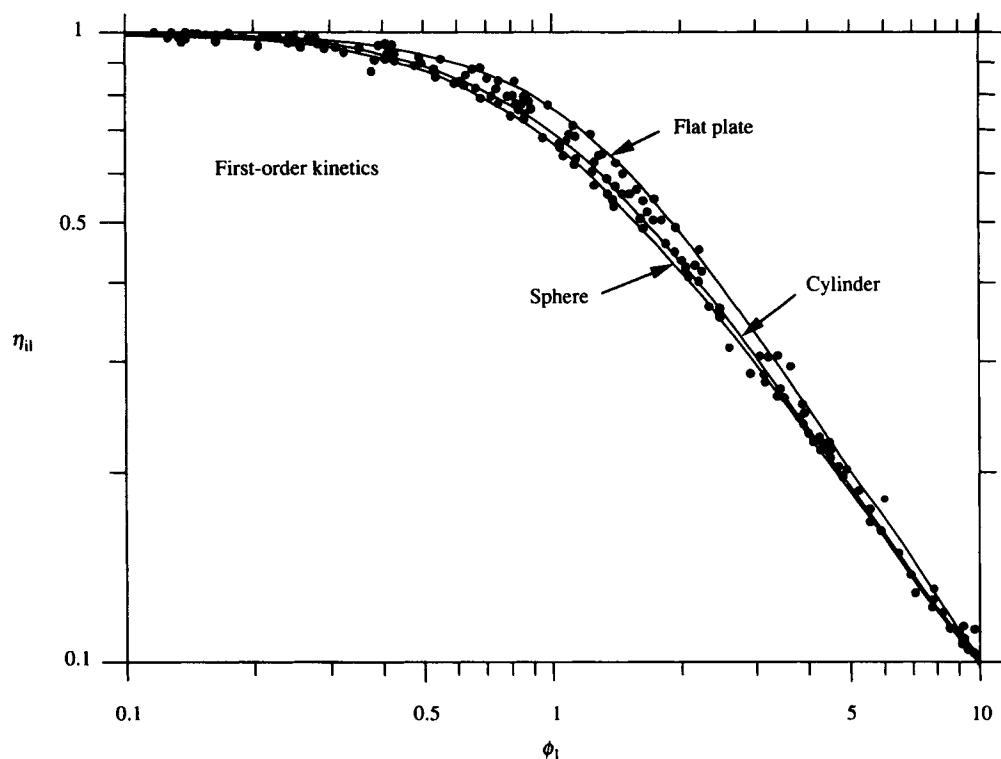


Figure 12.8 Internal effectiveness factor η_{i1} as a function of the generalised Thiele modulus ϕ_1 for first-order kinetics and spherical, cylindrical and flat-plate geometries. The dots represent calculations on finite or hollow cylinders and parallelepipeds. (From R. Aris, 1975, *The Mathematical Theory of Diffusion and Reaction in Permeable Catalysts*, vol. 1, Oxford University Press, London.)

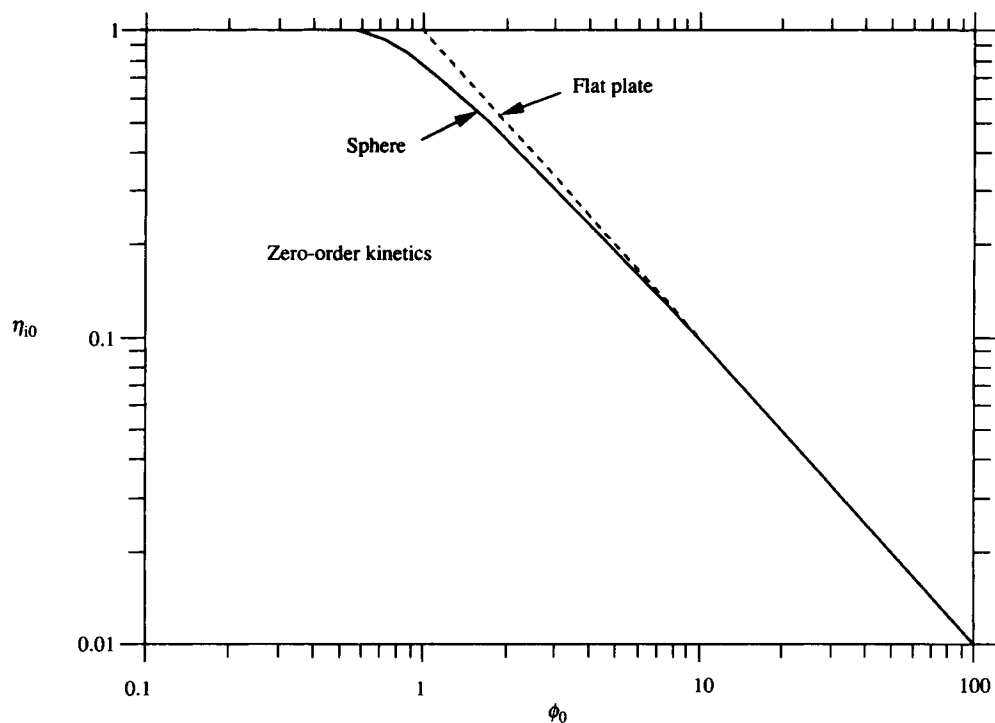


Figure 12.9 Internal effectiveness factor η_{i0} as a function of the generalised Thiele modulus ϕ_0 for zero-order kinetics and spherical and flat-plate geometries.

If C_A falls to zero within the pellet, the effectiveness factor must be evaluated differently. In this case, $r_{A,obs}$ is given by Eq. (12.24) and the internal effectiveness factor is:

$$\eta_{i0} = \frac{\frac{4}{3}\pi(R^3 - R_0^3)k_0}{\frac{4}{3}\pi R^3 k_0} = 1 - \left(\frac{R_0}{R}\right)^3. \quad (12.32)$$

According to the above analysis, to evaluate η_{i0} for zero-order kinetics, first we must know whether or not substrate is depleted in the catalyst, then, if it is, the value of R_0 . Usually this information is unavailable because we cannot easily measure intraparticle concentrations. Fortunately, further mathematical analysis [7] overcomes this problem by representing the system in terms of measurable properties such as R , $\mathcal{D}_{Ac}$, C_{As} and k_0 rather than R_0 . These parameters define the Thiele modulus for zero-order reaction, ϕ_0 . The results are summarised in Table 12.3 and Figure 12.9. C_A remains > 0 and $\eta_{i0} = 1$ for $0 < \phi_0 \leq 0.577$; for $\phi_0 > 0.577$, η_{i0} declines as more

and more of the particle becomes inactive. Effectiveness factors for flat-plate systems are also shown in Table 12.3 and Figure 12.9; in flat films, $\phi_0 = 1$ represents the threshold condition for substrate depletion. η_{i0} curves for spherical and flat-plate geometries coincide exactly at small and large values of ϕ_0 .

12.4.3 Michaelis–Menten Kinetics

For a spherical catalyst, the rate of Michaelis–Menten reaction in the absence of internal mass-transfer effects is:

$$r_{As}^* = \frac{4}{3}\pi R^3 \left(\frac{v_{\max} C_{As}}{K_m + C_{As}} \right). \quad (12.33)$$

Our analysis cannot proceed further, however, because we do not have an equation for $r_{A,obs}$. Accordingly, we cannot develop an analytical expression for η_{im} as a function of ϕ_m . Diffusion-reaction equations for Michaelis–Menten kinetics are generally solved by numerical computation. As an example, the results for flat-plate geometry are shown in Figure 12.10 as a function of β , which is equal to K_m/C_{As} .

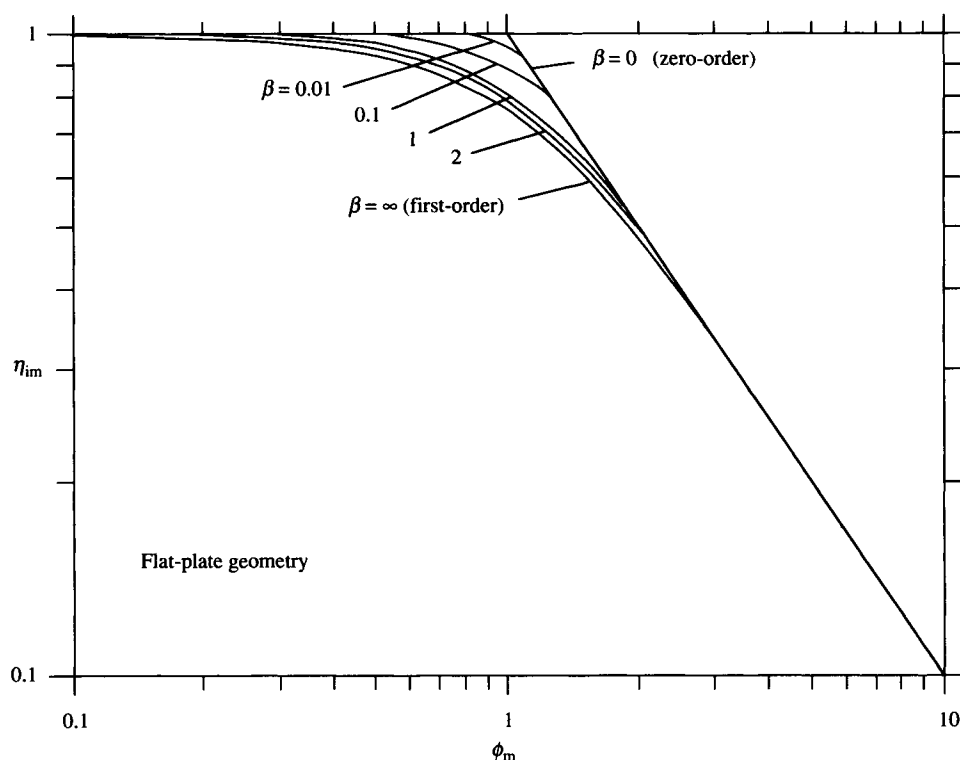


Figure 12.10 Internal effectiveness factor η_{im} as a function of the generalised Thiele modulus ϕ_m and parameter β for Michaelis–Menten kinetics and flat-plate geometry. $\beta = K_m/C_{As}$. (From R. Aris, 1975, *The Mathematical Theory of Diffusion and Reaction in Permeable Catalysts*, vol. 1, Oxford University Press, London.)

We can obtain approximate values for η_{im} by considering the zero- and first-order asymptotes of the Michaelis–Menten equation. As indicated in Figure 12.10, curves for η_{im} fall between the lines for zero- and first-order reactions. The exact position depends on the value of β . As $\beta \rightarrow \infty$, Michaelis–Menten kinetics can be approximated as first order and the internal effectiveness factor evaluated from Figure 12.8 with $k_1 = v_{max}/K_m$. When $\beta \rightarrow 0$, zero-order reaction and Figure 12.9 apply with $k_0 = v_{max}$. Effectiveness factors for β between zero and infinity must be evaluated using numerical methods. Use of the generalised Thiele modulus as defined in Table 12.2 eliminates almost all variation in the internal effectiveness factor with changing β , except in the vicinity of $\phi_m = 1$. As in Figures 12.8 and 12.9, the generalised modulus also brings together effectiveness-factor curves for all shapes of catalyst at the two asymptotes $\phi_m \rightarrow 0$ and $\phi_m \rightarrow \infty$. Therefore, Figure 12.10 is valid for spherical catalysts if ϕ_m is much less or much greater than 1. It should be noted however that variation between geometries in the intermediate region

around $\phi_m = 1$ can be significant [16].

If values of ϕ_m and β are such that Michaelis–Menten kinetics cannot be approximated by either zero- or first-order equations, η_{im} can be estimated using an equation proposed by Moo-Young and Kobayashi [17]:

$$\eta_{im} = \frac{\eta_{i0} + \beta \eta_{i1}}{1 + \beta} \quad (12.34)$$

where $\beta = K_m/C_{As}$, η_{i0} is the zero-order internal effectiveness factor obtained using values of ϕ_0 evaluated with $k_0 = v_{max}$; η_{i1} is the first-order effectiveness factor obtained using ϕ_1 calculated with $k_1 = v_{max}/K_m$. For flat-plate geometry, the largest deviation of Eq. (12.34) from exact values of η_{im} is 0.089; this occurs at $\phi_m = 1$ and $\beta = 0.2$. For spherical geometry, the greatest deviations occur around $\phi_m = 1.7$ and $\beta = 0.3$; the maximum error in this region is 0.09. Further details can be found in the original paper [17].

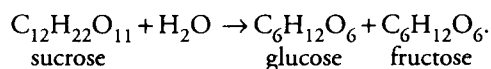
Example 12.3 Reaction rates for free and immobilised enzyme

Invertase is immobilised in ion-exchange resin of average diameter 1 mm. The amount of enzyme in the beads is measured by protein assay as 0.05 kg m^{-3} . 20 cm^3 beads are packed into a small column reactor; 75 ml sucrose solution at a concentration of 16 mM is pumped rapidly through the bed. In another reactor an identical quantity of free enzyme is mixed into the same volume of sucrose solution. Assume the kinetic parameters for free and immobilised enzyme are equal: K_m is 8.8 mM and the turnover number is $2.4 \times 10^{-3} \text{ gmol glucose (g enzyme)}^{-1} \text{ s}^{-1}$. The effective diffusivity of sucrose in the ion-exchange resin is $2 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$.

- What is the rate of reaction by free enzyme?
- What is the rate of reaction by immobilised enzyme?

Solution:

The invertase reaction is:



Convert the data provided to units of gmol, m and s.

$$K_m = \frac{8.8 \times 10^{-3} \text{ gmol}}{\text{litre}} \cdot \left| \frac{1000 \text{ litres}}{1 \text{ m}^3} \right| = 8.8 \text{ gmol m}^{-3}$$

$$\mathcal{D}_{Ac} = 2 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1} \cdot \left| \frac{1 \text{ m}}{100 \text{ cm}} \right|^2 = 2 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$$

$$R = \frac{1 \text{ mm}}{2} \cdot \left| \frac{1 \text{ m}}{10^3 \text{ mm}} \right| = 5 \times 10^{-4} \text{ m}.$$

If flow through the reactor is rapid, we can assume C_{As} is equal to the bulk sucrose concentration C_{Ab} :

$$C_{As} = C_{Ab} = 16 \text{ mM} = \frac{16 \times 10^{-3} \text{ gmol}}{\text{litre}} \cdot \left| \frac{1000 \text{ litres}}{1 \text{ m}^3} \right| = 16 \text{ gmol m}^{-3}.$$

Also:

$$\text{Mass of enzyme} = 20 \text{ cm}^3 \cdot \frac{0.05 \text{ kg}}{\text{m}^3} \cdot \left| \frac{1 \text{ m}}{100 \text{ cm}} \right|^3 = 10^{-6} \text{ kg}.$$

(a) In the free-enzyme reactor:

$$\text{Enzyme concentration} = \frac{10^{-6} \text{ kg}}{75 \text{ cm}^3} \cdot \left| \frac{100 \text{ cm}}{1 \text{ m}} \right|^3 = 1.33 \times 10^{-2} \text{ kg m}^{-3}.$$

Production of 1 gmol glucose requires consumption of 1 gmol sucrose; therefore $k_2 = 2.4 \times 10^{-3} \text{ gmol sucrose (g enzyme)}^{-1} \text{ s}^{-1}$. From Eq. (11.33), $v_{\max}$ is obtained by multiplying the turnover number by the concentration of active enzyme. Assuming all enzyme present is active:

$$\begin{aligned} v_{\max} &= \left(\frac{2.4 \times 10^{-3} \text{ gmol}}{\text{g s}} \right) (1.33 \times 10^{-2} \text{ kg m}^{-3}) \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| \\ &= 3.19 \times 10^{-2} \text{ gmol m}^{-3} \text{ s}^{-1}. \end{aligned}$$

Free-enzyme reaction takes place at uniform sucrose concentration, C_{Ab} . The volumetric rate of reaction is given by the Michaelis–Menten equation:

$$\begin{aligned} v &= \frac{v_{\max} C_{Ab}}{K_m + C_{Ab}} = \frac{(3.19 \times 10^{-2} \text{ gmol m}^{-3} \text{ s}^{-1}) (16 \text{ gmol m}^{-3})}{8.8 \text{ gmol m}^{-3} + 16 \text{ gmol m}^{-3}} \\ &= 2.06 \times 10^{-2} \text{ gmol m}^{-3} \text{ s}^{-1}. \end{aligned}$$

The total rate of reaction is v multiplied by the liquid volume:

$$\begin{aligned} \text{Rate of reaction} &= (2.06 \times 10^{-2} \text{ gmol m}^{-3} \text{ s}^{-1}) (75 \text{ cm}^3) \cdot \left| \frac{1 \text{ m}}{100 \text{ cm}} \right|^3 \\ &= 1.55 \times 10^{-6} \text{ gmol s}^{-1}. \end{aligned}$$

(b) For heterogeneous reactions, $v_{\max}$ is expressed on a catalyst-volume basis. Therefore:

$$\begin{aligned} v_{\max} &= \left(\frac{2.4 \times 10^{-3} \text{ gmol}}{\text{g s}} \right) (0.05 \text{ kg m}^{-3}) \cdot \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| \\ &= 0.12 \text{ gmol s}^{-1} \text{ m}^{-3} \text{ particle}. \end{aligned}$$

To determine the effect of mass transfer we must calculate η_{im} . The method used depends on the values of β and ϕ_m :

$$\beta = \frac{K_m}{C_{As}} = \frac{8.8 \text{ gmol m}^{-3}}{16 \text{ gmol m}^{-3}} = 0.55.$$

From Table 12.2:

$$\phi_m = \frac{R}{3\sqrt{2}} \sqrt{\frac{v_{\max}}{\mathcal{D}_{Ac} C_{As}}} \left(\frac{1}{1+\beta} \right) \left[1 + \beta \ln \left(\frac{\beta}{1+\beta} \right) \right]^{-1/2}$$

$$\phi_m = \frac{5 \times 10^{-4} \text{ m}}{3\sqrt{2}} \sqrt{\frac{0.12 \text{ gmol m}^{-3} \text{ s}^{-1}}{(2 \times 10^{-10} \text{ m}^2 \text{ s}^{-1})(16 \text{ gmol m}^{-3})}} \left(\frac{1}{1+0.55} \right) \left[1 + 0.55 \ln \left(\frac{0.55}{1+0.55} \right) \right]^{-1/2}$$

$$= 0.71.$$

Because both β and ϕ_m have intermediate values, Figure 12.10 cannot be applied for spherical geometry. Instead, we must use Eq. (12.34). From Table 12.2:

$$\phi_0 = \frac{R}{3\sqrt{2}} \sqrt{\frac{k_0}{\mathcal{D}_{Ac} C_{As}}}$$

$$= \frac{R}{3\sqrt{2}} \sqrt{\frac{v_{\max}}{\mathcal{D}_{Ac} C_{As}}}$$

$$= \frac{5 \times 10^{-4} \text{ m}}{3\sqrt{2}} \sqrt{\frac{0.12 \text{ gmol m}^{-3} \text{ s}^{-1}}{(2 \times 10^{-10} \text{ m}^2 \text{ s}^{-1})(16 \text{ gmol m}^{-3})}}$$

$$= 0.72.$$

From Figure 12.9 or Table 12.3, $\eta_{i0} = 0.93$. Similarly:

$$\phi_1 = \frac{R}{3} \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}}$$

$$= \frac{R}{3} \sqrt{\frac{v_{\max}}{K_m \mathcal{D}_{Ac}}}$$

$$= \frac{5 \times 10^{-4} \text{ m}}{3} \sqrt{\frac{0.12 \text{ gmol m}^{-3} \text{ s}^{-1}}{(8.8 \text{ gmol m}^{-3})(2 \times 10^{-10} \text{ m}^2 \text{ s}^{-1})}}$$

$$= 1.4.$$

From Figure 12.8 or Table 12.3, $\eta_{i1} = 0.54$. Substituting these results into Eq. (12.34):

$$\eta_{\text{im}} = \frac{0.93 + 0.55(0.54)}{1 + 0.55} = 0.79.$$

The rate of immobilised-enzyme reaction without diffusional limitations is the same as that for free enzyme: $1.55 \times 10^{-6} \text{ gmol s}^{-1}$. Rate of reaction for the immobilised enzyme is 79% that of free enzyme even though the amount of enzyme present and external substrate concentration are the same:

$$\text{Observed rate} = 0.79 (1.55 \times 10^{-6} \text{ gmol s}^{-1}) = 1.22 \times 10^{-6} \text{ gmol s}^{-1}.$$

12.4.4 The Observable Thiele Modulus

Diffusion-reaction theory as presented in the previous sections allows us to quantify the effect of mass transfer on rate of reaction. However, a drawback to the methods outlined so far is that they are useful only if we know the true kinetic parameters for the reaction: k_0 , k_1 or $v_{\max}$ and K_m . In many cases these values are not known and, as discussed in Section 12.9, can be

difficult to evaluate for biological systems. A way to circumvent this problem is to apply the *observable Thiele modulus* Φ , sometimes called *Weisz's modulus* [18], which is defined as:

$$\Phi = \left(\frac{V_p}{S_x} \right)^2 \frac{r_{A,\text{obs}}}{\mathcal{D}_{Ac} C_{As}} \quad (12.35)$$

where V_p is catalyst volume, S_x is external surface area, $r_{A,obs}$ is the observed reaction rate *per unit volume of catalyst*, $\mathcal{D}_{Ac}$ is effective diffusivity of substrate, and C_{As} is the substrate concentration at the external surface. Expressions for Φ for spheres and flat plates are listed in Table 12.4. Evaluation of the observable Thiele modulus does not rely on prior knowledge of kinetic parameters; Φ is defined in terms of the measured reaction rate, $r_{A,obs}$.

For the observable Thiele modulus to be useful, we need to relate Φ to the internal effectiveness factor η_i . Some mathematical consideration of the equations already presented in this chapter yields the following relationships for first-order, zero-order and Michaelis–Menten kinetics:

$$\text{First order} \quad \Phi = \phi_1^2 \eta_{i1} \quad (12.36)$$

$$\text{Zero order} \quad \Phi = 2\phi_0^2 \eta_{i0} \quad (12.37)$$

$$\text{Michaelis–Menten} \quad \Phi = 2\phi_m^2 \eta_{im} (1 + \beta) \left[1 + \beta \ln \left(\frac{\beta}{1 + \beta} \right) \right] \quad (12.38)$$

Table 12.4 Observable Thiele moduli

Sphere	$\Phi = \left(\frac{R}{3} \right)^2 \frac{r_{A,obs}}{\mathcal{D}_{Ac} C_{As}}$
Flat plate	$\Phi = b^2 \frac{r_{A,obs}}{\mathcal{D}_{Ac} C_{As}}$

Eqs (12.36)–(12.38) apply to all catalyst geometries and allow us to develop plots of Φ versus η_i from relationships between ϕ and η_i developed in the previous sections and represented in Figures 12.8–12.10. Curves for spherical catalysts and first-order, zero-order and Michaelis–Menten kinetics are given in Figure 12.11; results for flat-plate geometry are shown in Figure 12.12. All curves for β between zero and infinity are bracketed by the first- and zero-order lines. At each value of β , curves for all geometries coincide in the asymptotic regions $\Phi \rightarrow 0$ and $\Phi \rightarrow \infty$; at intermediate values of Φ the variation between effectiveness factors for different geometries can be significant. For $\Phi > 10$:

$$\text{First-order kinetics} \quad \eta_{i1} \approx \frac{1}{\Phi} \quad (12.39)$$

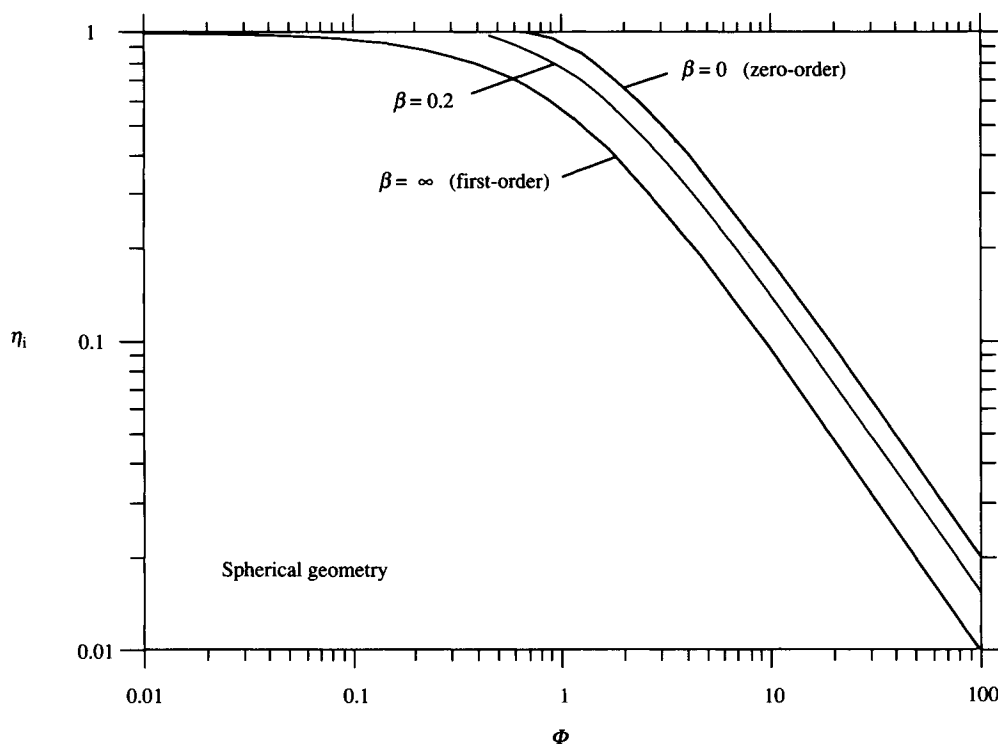


Figure 12.11 Internal effectiveness factor η_i as a function of the observable Thiele modulus Φ for spherical geometry and first-order, zero-order and Michaelis–Menten kinetics. $\beta = K_m / C_{As}$. (From W.H. Pitcher, 1975, Design and operation of immobilized enzyme reactors. In: R.A. Messing, Ed, *Immobilized Enzymes For Industrial Reactors*, pp. 151–199, Academic Press, New York.)

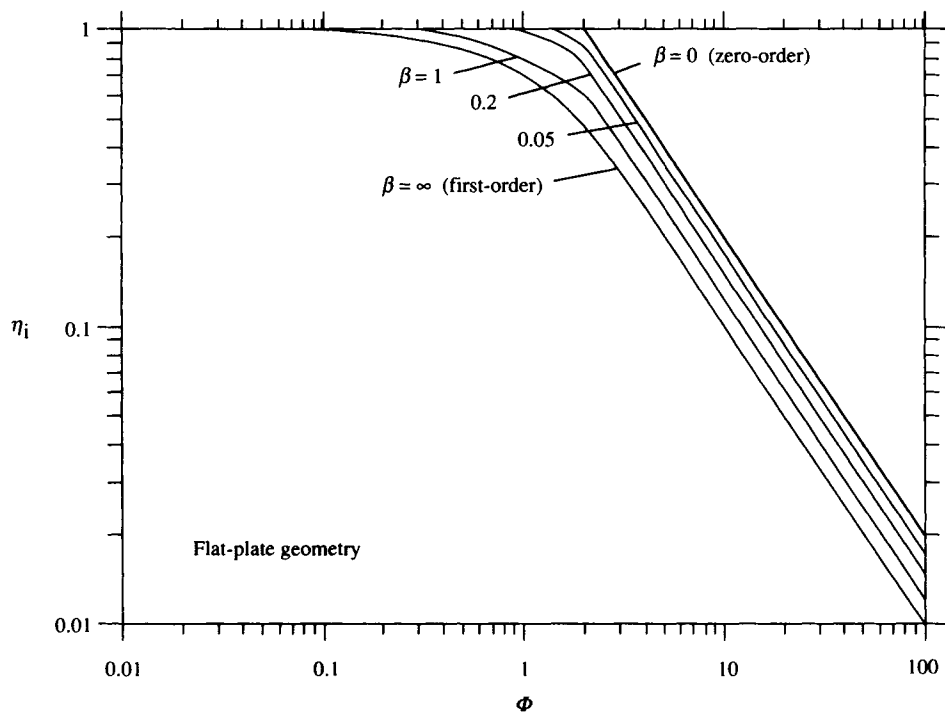


Figure 12.12
Internal effectiveness factor η_i as a function of the observable Thiele modulus Φ for flat-plate geometry and first-order, zero-order and Michaelis–Menten kinetics.
 $\beta = K_m / C_{As}$. (From W.H. Pitcher, 1975, Design and operation of immobilized enzyme reactors. In: R.A. Messing, Ed, *Immobilized Enzymes For Industrial Reactors*, pp. 151–199, Academic Press, New York.)

Zero-order kinetics $\eta_{i0} \approx \frac{2}{\Phi}$.
(12.40)

Although Φ is an observable modulus independent of kinetic parameters such as $v_{\max}$ and K_m , use of Figures 12.11 and 12.12 for Michaelis–Menten reactions requires knowledge of K_m for evaluation of β . This makes application of the observable Thiele modulus difficult for Michaelis–Menten kinetics. However, we know that effectiveness factors for Michaelis–Menten reactions lie between the first- and zero-order curves of Figures 12.11 and 12.12; therefore, we can always estimate the upper and lower bounds of η_{im} .

12.4.5 Weisz's Criteria

The following general observations can be made from Figures 12.11 and 12.12.

If $\Phi < 0.3$, $\eta_i \approx 1$ and internal mass-transfer limitations are insignificant.

If $\Phi > 3$, η_i is substantially < 1 and internal mass-transfer limitations are significant.

The above statements are known as *Weisz's criteria*, and are valid for all geometries and reaction kinetics. For Φ in the intermediate range $0.3 < \Phi < 3$, closer analysis is required to determine the influence of mass transfer on reaction rate.

Example 12.4 Internal oxygen transfer to immobilised cells

Baby hamster kidney cells are immobilised in alginate beads. The average particle diameter is 5 mm. Rate of oxygen consumption at a bulk concentration of $8 \times 10^{-3} \text{ kg O}_2 \text{ m}^{-3}$ is $8.4 \times 10^{-5} \text{ kg s}^{-1} \text{ m}^{-3}$ catalyst. The effective diffusivity of oxygen in the beads is $1.88 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. Assume that the oxygen concentration at the surface of the catalyst is equal to the bulk concentration, and that oxygen uptake follows zero-order kinetics.

- Are internal mass-transfer effects significant?
- What reaction rate would be observed if diffusional resistance were eliminated?

Solution:

- To assess internal mass transfer, calculate the observable Thiele modulus. From Table 12.4 for spherical geometry:

$$\Phi = \left(\frac{R}{3} \right)^2 \frac{r_{A,obs}}{\mathcal{D}_{Ac} C_{As}}$$

With

$$R = \frac{5 \times 10^{-3} \text{ m}}{2} = 2.5 \times 10^{-3} \text{ m}$$

then

$$\Phi = \left(\frac{2.5 \times 10^{-3} \text{ m}}{3} \right)^2 \frac{8.4 \times 10^{-5} \text{ kg s}^{-1} \text{ m}^{-3}}{(1.88 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}) (8 \times 10^{-3} \text{ kg m}^{-3})} = 3.88.$$

From Weisz's criteria, internal mass-transfer effects are significant.

(b) For spherical catalysts and zero-order reaction, from Figure 12.11, at $\Phi = 3.9$, $\eta_{i0} = 0.4$. From Eq. (12.26), without diffusional restrictions the reaction rate would be:

$$r_{As}^* = \frac{r_{A,obs}}{\eta_i} = \frac{8.4 \times 10^{-5} \text{ kg m}^{-3} \text{ s}^{-1}}{0.4} = 2.1 \times 10^{-4} \text{ kg s}^{-1} \text{ m}^{-3} \text{ catalyst.}$$

12.4.6 Minimum Intracatalyst Substrate Concentration

It is sometimes of interest to know the minimum concentration $C_{A,min}$ inside solid catalysts. We can use this information to check, for example, that the concentration does not fall below some critical value for cell metabolism. $C_{A,min}$ is easily estimated for zero-order reactions. If Φ is such that $\eta_i < 1$, $C_{A,min}$ is zero because $\eta_i = 1$ if $C_A > 0$ throughout the particle. For $\eta_i = 1$, simple manipulation of the equations already presented in this chapter allow us to estimate $C_{A,min}$. The results are summarised in Table 12.5.

12.5 External Mass Transfer

Many equations in Sections 12.3 and 12.4 contain the term C_{As} , the concentration of substrate A at the external surface of the catalyst. This term made its way into the analysis in the boundary conditions used for solution of the shell mass balance. It was assumed that C_{As} is a known quantity. However, because surface concentrations are very difficult to measure accurately, we must find ways to estimate C_{As} using theoretical principles.

Reduction in substrate concentration from C_{Ab} in the bulk liquid to C_{As} at the catalyst surface occurs across the boundary layer surrounding the solid. In the absence of the boundary

Table 12.5 Minimum intracatalyst substrate concentration with zero-order kinetics (Φ for each geometry is defined in Table 12.4)

Sphere	$C_{A,min} = 0$	for $\Phi \geq 0.667$
	$C_{A,min} = C_{As} \left(1 - \frac{3}{2} \Phi \right)$	for $\Phi < 0.667$
Flat plate	$C_{A,min} = 0$	for $\Phi \geq 2$
	$C_{A,min} = C_{As} \left(1 - \frac{1}{2} \Phi \right)$	for $\Phi < 2$

layer, $C_{As} = C_{Ab}$, which is easily measured. When the boundary layer is present, C_{As} takes some value less than C_{Ab} . Rate of mass transfer across a liquid boundary layer is represented by the following equation (Section 9.4):

$$N_A = k_S a (C_{Ab} - C_{As}) \quad (12.41)$$

where N_A is the rate of mass transfer, k_S is the *liquid-phase mass-transfer coefficient* with dimensions LT^{-1} , and a is the external surface area of the catalyst. If N_A is expressed per volume of catalyst with units of, for example, $\text{kgmol s}^{-1} \text{m}^{-3}$, to be consistent, a must also be expressed on a catalyst-volume basis with units of, for example, $\text{m}^2 \text{m}^{-3}$ or m^{-1} . Using the previous notation of this chapter, a in Eq. (12.41) is equal to S_x/V_p for the catalyst. At steady state, the rate of substrate transfer across the boundary layer must be equal to the rate of consumption by the catalyst, $r_{A,obs}$. Therefore:

$$r_{A,obs} = k_S \frac{S_x}{V_p} (C_{Ab} - C_{As}) \quad (12.42)$$

where $r_{A,obs}$ is the rate per volume of catalyst. Rearranging gives:

$$\frac{C_{As}}{C_{Ab}} = 1 - \frac{V_p}{S_x} \frac{r_{A,obs}}{k_S C_{Ab}} \quad (12.43)$$

Eq. (12.43) can be used to evaluate C_{As} before applying equations in the previous sections to calculate internal substrate concentrations and effectiveness factors. The magnitude of external mass-transfer effects can be gauged from Eq. (12.43). $C_{As}/C_{Ab} \approx 1$ indicates no external mass-transfer limitations; the substrate concentration at the surface is approximately equal to that in the bulk. On the other hand, $C_{As}/C_{Ab} \ll 1$ indicates a very steep concentration gradient in the boundary layer and severe external mass-transfer effects. We can define from Eq. (12.43) an *observable modulus for external mass transfer*, Ω :

$$\Omega = \frac{V_p}{S_x} \frac{r_{A,obs}}{k_S C_{Ab}} \quad (12.44)$$

Expressions for Ω for different catalyst geometries are listed in Table 12.6. Criteria for assessing external mass transfer are as follows:

If $\Omega \ll 1$, $C_{As} \approx C_{Ab}$ and external mass-transfer effects are insignificant.

Table 12.6 Observable moduli for external mass transfer

Sphere	$\Omega = \frac{R}{3} \frac{r_{A,obs}}{k_S C_{Ab}}$
Flat plat	$\Omega = b \frac{r_{A,obs}}{k_S C_{Ab}}$

Table 12.7 External effectiveness factors

First-order reaction: $r_A = k_1 C_A$

$$\eta_{e1} = \frac{C_{As}}{C_{Ab}}$$

Zero-order reaction: $r_A = k_0$

$$\eta_{e0} = 1$$

Michaelis–Menten reaction: $r_A = \frac{v_{\max} C_A}{K_m + C_A}$

$$\eta_{em} = \frac{C_{As} (K_m + C_{Ab})}{C_{Ab} (K_m + C_{As})}$$

Otherwise, $C_{As} < C_{Ab}$ and external mass-transfer effects are significant.

For reaction affected by both internal and external mass-transfer restrictions, we can define a *total effectiveness factor* η_T :

$$\eta_T = \frac{r_{A,obs}}{r_{Ab}^*} = \frac{(\text{observed rate})}{\left(\frac{\text{rate that would occur if } C_A = C_{Ab}}{\text{everywhere in the particle}} \right)} \quad (12.45)$$

η_T can be related to the internal effectiveness factor η_i . Eq. (12.45) may be written as:

$$\eta_T = \left(\frac{r_{A,obs}}{r_{As}^*} \right) \left(\frac{r_{As}^*}{r_{Ab}^*} \right) = \eta_i \eta_e \quad (12.46)$$

where η_e is the *external effectiveness factor* and η_i is defined in Eq. (12.26). Therefore, η_e has the following meaning:

$$\eta_e = \frac{r_{As}^*}{r_{Ab}^*} = \frac{\left(\frac{\text{rate that would occur if } C_A = C_{As}}{\text{everywhere in the particle}} \right)}{\left(\frac{\text{rate that would occur if } C_A = C_{Ab}}{\text{everywhere in the particle}} \right)} \quad (12.47)$$

Expressions for η_e for first-order, zero-order and Michaelis–Menten kinetics are listed in Table 12.7. For zero-order reactions, as long as C_{As} and $C_{Ab} > 0$, $\eta_{e0} = 1$. However, $\eta_{e0} = 1$ does not imply that an external boundary layer does not exist; even if there is a reduction in concentration across the boundary layer, because r_{As}^* and r_{Ab}^* are independent of

C_A , $\eta_{e0} = 1$. Furthermore, $\eta_{e0} = 1$ does not imply that eliminating the external boundary layer could not improve the reaction rate. Removing the boundary layer would increase the value of C_{As} , thus establishing a greater driving force for internal mass-transfer and reducing the likelihood of C_A falling to zero inside the particle.

Example 12.5 Effect of mass transfer on bacterial denitrification

Denitrifying bacteria are immobilised in gel beads and used in a stirred reactor for removal of nitrate from groundwater. At a nitrate concentration of 3 g m^{-3} , the conversion rate is $0.011 \text{ g s}^{-1} \text{ m}^{-3}$ catalyst. The effective diffusivity of nitrate in the gel is $1.5 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, the beads are 6 mm in diameter, and the liquid–solid mass-transfer coefficient is 10^{-5} m s^{-1} . K_m for the immobilised bacteria is approximately 25 g m^{-3} .

- Does external mass transfer influence the reaction rate?
- Are internal mass-transfer effects significant?
- By how much would the reaction rate be improved if both internal and external mass-transfer resistances were eliminated?

Solution:

(a)

$$R = \frac{6 \times 10^{-3} \text{ m}}{2} = 3 \times 10^{-3} \text{ m}.$$

The effect of external mass transfer is found by calculating Ω . From Table 12.6:

$$\Omega = \frac{R}{3} \frac{r_{A, \text{obs}}}{k_S C_{Ab}} = \frac{3 \times 10^{-3} \text{ m}}{3} \frac{0.011 \text{ g s}^{-1} \text{ m}^{-3}}{(10^{-5} \text{ m s}^{-1}) (3 \text{ g m}^{-3})} = 0.37.$$

As this value of Ω is relatively large, external mass transfer influences the reaction rate. From Eq. (12.43):

$$\frac{C_{As}}{C_{Ab}} = 1 - \Omega = 0.63$$

$$C_{As} = 0.63 C_{Ab} = 0.63 (3 \text{ g m}^{-3}) = 1.9 \text{ g m}^{-3}.$$

- (b) Calculate the observable Thiele modulus with $C_{As} = 1.9 \text{ g m}^{-3}$ using the equation for spheres from Table 12.4:

$$\Phi = \left(\frac{R}{3} \right)^2 \frac{r_{A, \text{obs}}}{\mathcal{D}_{Ac} C_{As}}$$

$$\Phi = \left(\frac{3 \times 10^{-3} \text{ m}}{3} \right)^2 \frac{0.011 \text{ g s}^{-1} \text{ m}^{-3}}{(1.5 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}) (1.9 \text{ g m}^{-3})} = 3.9.$$

From Weisz's criteria, internal mass-transfer effects are significant.

- (c) Because the nitrate concentration is much smaller than K_m , we can assume the reaction is first order. From Table 12.7:

$$\eta_{e1} = \frac{C_{As}}{C_{Ab}} = 0.63.$$

Reaction rate in the absence of mass-transfer effects is r_{Ab}^* , which is related to $r_{A, \text{obs}}$ by Eq. (12.45). Therefore, we can calculate r_{Ab}^* if we know η_{T1} . From Figure 12.11, at $\Phi = 3.9$ $\eta_{i1} = 0.21$. As $\eta_{e1} = 0.63$, from Eq. (12.46):

$$\eta_{T1} = \eta_{i1} \eta_{e1} = 0.21 (0.63) = 0.13.$$

Therefore, from Eq. (12.45):

$$r_{Ab}^* = \frac{r_{A,obs}}{\eta_{T1}} = \frac{0.011 \text{ g s}^{-1} \text{ m}^{-3}}{0.13} = 0.085 \text{ g s}^{-1} \text{ m}^{-3}.$$

The reaction rate would be increased by a factor of about 7.7.

12.6 Liquid–Solid Mass-Transfer Correlations

The mass-transfer coefficient k_S must be known before we can account for external mass-transfer effects. k_S depends on reactor hydrodynamics and liquid properties such as viscosity, density and diffusivity. It is difficult to determine k_S accurately, especially for particles which are neutrally buoyant. However, values can be estimated using correlations from the literature; these are usually accurate under the conditions specified to within 10–20%. Selected correlations are presented below. Further details can be found in chemical engineering texts [19, 20].

Correlations for k_S are expressed using the following dimensionless groups:

$$Re_p = (\text{particle}) \text{ Reynolds number} = \frac{D_p u_{pL} \rho_L}{\mu_L} \quad (12.48)$$

$$Sc = \text{Schmidt number} = \frac{\mu_L}{\rho_L \mathcal{D}_{AL}} \quad (12.49)$$

$$Sh = \text{Sherwood number} = \frac{k_S D_p}{\mathcal{D}_{AL}} \quad (12.50)$$

and

$$Gr = \text{Grashof number} = \frac{g D_p^3 \rho_L (\rho_p - \rho_L)}{\mu_L^2} \quad (12.51)$$

where D_p is particle diameter, u_{pL} is linear velocity of the particle relative to the bulk liquid, ρ_L is liquid density, μ_L is liquid viscosity, $\mathcal{D}_{AL}$ is the molecular diffusivity of component A in the liquid, k_S is the liquid–solid mass-transfer coefficient, g is gravitational acceleration, and ρ_p is particle density. The Sherwood number contains the mass-transfer coefficient and represents the ratio of overall and diffusive mass-transfer rates through the boundary layer. The Schmidt number represents the ratio of momentum diffusivity and mass diffusivity, and is made up of

physical properties of the system. At constant temperature, pressure and composition, Sc is constant for Newtonian fluids. The Grashof number represents the ratio of gravitational forces to viscous forces, and is important when the particles are neutrally buoyant. The form of the correlation used to evaluate Sh and therefore k_S depends on the configuration of the mass-transfer system, the flow conditions and other factors.

12.6.1 Free-Moving Spherical Particles

The equations presented here apply to solid particles suspended in stirred vessels. Rate of mass transfer depends on the velocity of the solid relative to the liquid; this is known as the *slip velocity*. Slip velocity in suspensions is difficult to measure and must therefore be estimated before calculating k_S . The following equations allow evaluation of the particle Reynolds number which incorporates the slip velocity, u_{pL} [7]:

$$\text{For } Gr < 36 \quad Re_p = Gr^{1/18} \quad (12.52)$$

$$\text{For } 36 < Gr < 8 \times 10^4 \quad Re_p = 0.153 Gr^{0.71} \quad (12.53)$$

$$\text{For } 8 \times 10^4 < Gr < 3 \times 10^9 \quad Re_p = 1.74 Gr^{0.5}. \quad (12.54)$$

Once Re_p is known, Sh can be determined using equations such as [7, 19–22]:

$$\text{For } Re_p Sc < 10^4 \quad Sh = \sqrt{4 + 1.21 (Re_p Sc)^{0.67}} \quad (12.55)$$

$$\text{For } Re_p < 10^3 \quad Sh = 2 + 0.6 Re_p^{0.5} Sc^{0.33}. \quad (12.56)$$

12.6.2 Spherical Particles in a Packed Bed

k_S in a packed bed depends on the liquid velocity around the particles. For the range $10 < Re_p < 10^4$, Sherwood number in packed beds has been correlated by the equation [23]:

$$Sh = 0.95 Re_p^{0.5} Sc^{0.33}. \quad (12.57)$$

12.7 Experimental Aspects

Applying diffusion-reaction theory to real biocatalysts requires prior measurement of several parameters. This section considers some experimental aspects of heterogeneous reactions.

12.7.1 Observed Reaction Rate

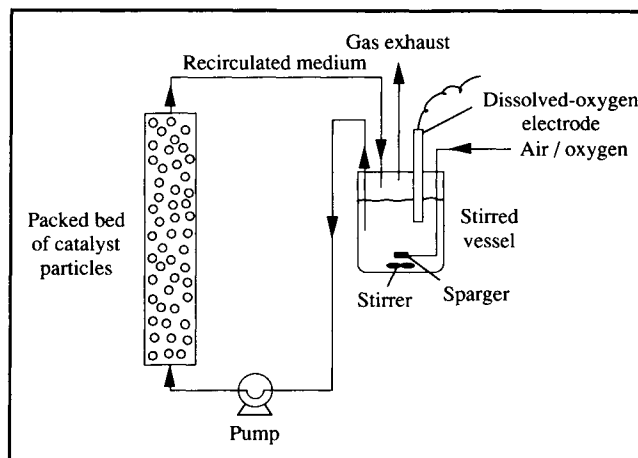
Calculation of Weisz's modulus Φ and the external modulus Ω requires knowledge of $r_{A,obs}$, the observed rate of reaction per volume of catalyst. This information can be obtained from batch concentration data using the methods described in Section 11.2.

When substrate levels change relatively slowly during reaction, batch rate data can be obtained by removing samples of the reaction mixture at various times and analysing for substrate concentration. However if substrate is consumed rapidly or if oxygen uptake rates are required, continuous *in situ* monitoring is necessary. The equipment shown in Figure 12.13 is configured for measurement of oxygen consumption by immobilised cells or enzymes. Catalyst particles are packed into a column and liquid medium recirculated through the packed bed. Oxygen is sparged into a stirred vessel connected to the column; dissolved-oxygen tension is measured using an electrode. The catalyst is allowed to reach steady state at a particular oxygen tension; the gas supply is then switched off and the initial rate of oxygen uptake recorded. Observed rates must be measured at steady state so that the results do not reflect rapidly-changing transient conditions. Unless sufficient time is allowed for the system to establish steady state, the experimental results will be unreliable.

The system of Figure 12.13 is assumed to be well mixed so that the oxygen tension measured by the electrode is the same everywhere in the bulk liquid. Actually, oxygen concentrations entering and leaving the packed bed will be different because oxygen is consumed by the catalyst. However, if only 1–2% of the oxygen entering the column is consumed during each pass through the bed, from a practical point of view the system can be regarded as perfectly mixed. For this to occur, the recirculation flow rate must be sufficiently high [24]. In oxygen uptake experiments, temperature control is also very important as the solubility and mass-transfer properties of oxygen are temperature dependent (see Sections 9.6.4 and 9.8.2).

Observed reaction rates for catalytic reactions depend on the cell or enzyme loading per volume of catalyst. Therefore, experimental measurements apply only to the cell or enzyme density tested. Increasing the amount of cells or enzyme in the catalyst will increase $r_{A,obs}$ and the likelihood of mass-transfer restrictions.

Figure 12.13 Batch recirculation reactor for measuring oxygen uptake by immobilised cells or enzymes.



12.7.2 Effective Diffusivity

The value of the effective diffusivity $\mathcal{D}_{Ac}$ reflects the ease with which compound A diffuses within the catalyst matrix, and depends strongly on the pore structure of the solid. Effective diffusivities are normally lower than corresponding molecular diffusivities in water because porous solids offer more resistance to diffusion. Some experimental $\mathcal{D}_{Ac}$ values are given in Table 12.8.

As cells can pose a significant barrier to diffusion, $\mathcal{D}_{Ac}$ for immobilised-cell preparations must be determined with the biomass present. During the measurement, external mass-transfer limitations must be overcome with high liquid flow rates around the catalyst. If experimental values of $\mathcal{D}_{Ac}$ are higher than the molecular diffusion coefficient, this may indicate the presence of convective mass transfer in the catalyst [3–6]. Techniques for measuring effective diffusivity are described in several papers [28–33]; however, accurate measurement of $\mathcal{D}_{Ac}$ is difficult in most systems.

12.8 Minimising Mass-Transfer Effects

To improve overall rates of reaction in bioprocesses, mass-transfer restrictions must be minimised or eliminated. In this section we consider practical ways of achieving this objective based on the equations presented in Tables 12.4 and 12.6.

12.8.1 Internal Mass Transfer

Internal mass-transfer effects are eliminated when the internal effectiveness factor is equal to 1; η_i approaches unity as the observable Thiele modulus Φ is reduced. From Table 12.4, Φ is decreased by:

Table 12.8 Effective diffusivity values

Substance	Catalyst	Temperature (°C)	$\mathcal{D}_{Ac}$ ($m^2 s^{-1}$)	$\frac{\mathcal{D}_{Ac}^*}{\mathcal{D}_{Aw}}$	Ref.
Oxygen	Agar (2% w/v) containing <i>Candida lipolytica</i> cells	30°C	1.94×10^{-9}	0.70	[25]
	Microbial aggregates from domestic waste- treatment plant	20°C	1.37×10^{-9}	0.62	[26]
	Trickling-filter slime	25°C	0.82×10^{-9}	—	[27]
Glucose	Microbial aggregates from domestic waste- treatment plant	20°C	0.25×10^{-9}	0.37	[26]
	Glass-fibre discs containing <i>Saccharomyces uvarum</i> cells	30°C	0.30×10^{-9}	0.43	[3]
	Calcium alginate (3 wt%)	30°C	0.62×10^{-9}	0.87	[28]
	Calcium alginate (2.4– 2.8 wt%) containing 50 wt% bakers' yeast	30°C	0.26×10^{-9}	0.37	[28]
Sucrose	Calcium alginate (2% w/v)	25°C	0.48×10^{-9}	0.86	[29]
	Calcium alginate (2% w/v) containing 12.5% (v/v) <i>Catharanthus roseus</i> cells	25°C	0.14×10^{-9}	0.25	[29]
L-tryptophan	Calcium alginate (2%)	30°C	0.67×10^{-9}	1.0	[30]
	κ -Carrageenan (4%)	30°C	0.58×10^{-9}	0.88	[31]
Lactate	Agar (1%) containing 1% Ehrlich ascites tumour cells	37°C	1.4×10^{-9}	0.97	[32]
	Agar (1%) containing 6% Ehrlich ascites tumour cells	37°C	0.7×10^{-9}	0.48	[32]
Ethanol	κ -Carrageenan (4%)	30°C	1.01×10^{-9}	0.92	[31]
	Calcium alginate (1.4– 3.8 wt%)	30°C	1.25×10^{-9}	0.92	[28]
	Calcium alginate (2.4– 2.8 wt%) containing 50 wt% bakers' yeast	30°C	0.45×10^{-9}	0.33	[28]
Nitrate	Compressed film of nitrifying organisms	—	1.4×10^{-9}	0.90	[26]
Ammonia	Compressed film of nitrifying organisms	—	1.3×10^{-9}	0.80	[26]

* $\mathcal{D}_{Aw}$ is the molecular diffusivity in water at the temperature of measurement.

- (i) reducing the observed reaction rate $r_{A,obs}$;
- (ii) reducing the size of the catalyst;
- (iii) increasing the effective diffusivity $\mathcal{D}_{Ae}$; and
- (iv) increasing the surface concentration of substrate C_{As} .

All these changes impact directly on the effectiveness of mass transfer in supplying substrate to the particle.

Paradoxically, reducing the reaction rate $r_{A,obs}$ improves the effectiveness of mass transfer aimed at increasing the reaction rate. When the catalyst is very active with a high demand for substrate, mass transfer is likely to be slow relative to reaction so that steep concentration gradients are produced. However, limiting the reaction rate by operating at sub-optimum conditions or using an organism or enzyme with low intrinsic activity does not achieve the overall goal of higher conversion rates. Because $r_{A,obs}$ is the reaction rate per volume of catalyst, another way of reducing $r_{A,obs}$ is to reduce the cell or enzyme loading in the solid. This reduces the demand for substrate per particle so that mass transfer has a better chance of supplying it at a sufficient rate. Therefore, if the same mass of cells or enzyme is distributed between more particles, the rate of conversion will increase. However, using more particles may mean that a larger reactor is required.

Because Φ is proportional to the square of catalyst size (R^2 for spheres or b^2 for flat plates), reducing the catalyst size has a more dramatic effect on Φ than changes in any other variable. It is therefore a good way to improve the reaction rate. In principle, mass-transfer limitations can be completely overcome if the particle size is decreased sufficiently. However, it is often extremely difficult in practice to reduce particle dimensions to this extent [34]. Even if mass-transfer effects were eliminated, operation of large-scale reactors with tiny, highly compressible gel particles raises new problems. Some degree of internal mass-transfer restriction must usually be tolerated.

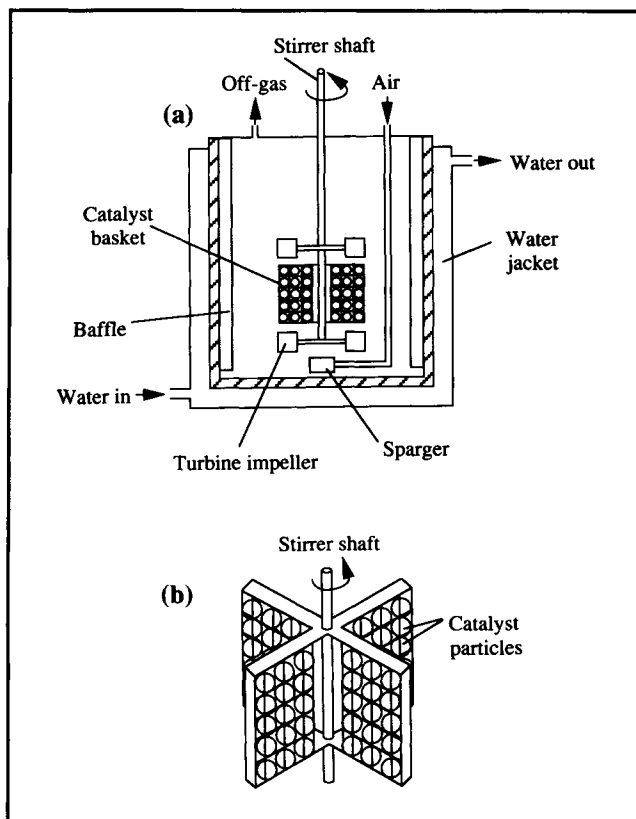
12.8.2 External Mass Transfer

External mass-transfer effects decrease as the observable modulus Ω is reduced. From the equations in Table 12.6, this can be achieved by:

- (i) reducing the observed reaction rate $r_{A,obs}$;
- (ii) reducing the size of the catalyst;
- (iii) increasing the mass-transfer coefficient k_S ; and
- (iv) increasing the bulk concentration of substrate C_{Ab} .

Decreasing the catalyst size and increasing the mass-transfer coefficient reduce the thickness of the boundary layer and facilitate external mass transfer. k_S is increased most readily by raising the liquid velocity outside the catalyst. k_S is a function of other system properties such as liquid viscosity, density, and substrate

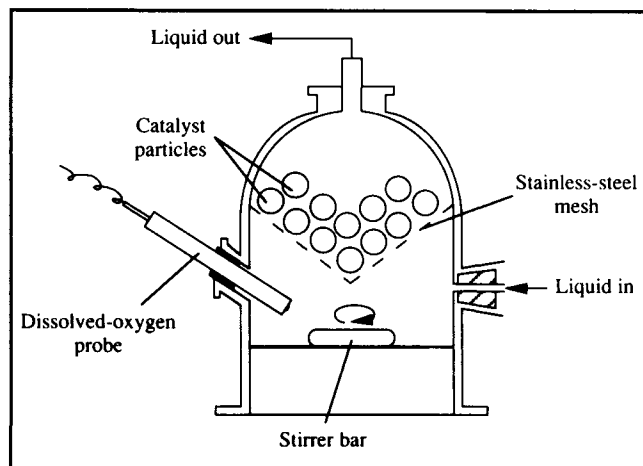
Figure 12.14 Spinning-basket reactor for minimising external mass-transfer effects: (a) reactor configuration; (b) detail of spinning basket. (From D.G. Tajbl, J.B. Simons and J.J. Carberry, 1966, *Heterogeneous catalysis in a continuous stirred tank reactor*. *Ind. Eng. Chem. Fund.* 5, 171–175.)



diffusivity; changes in these variables can also reduce boundary-layer effects to some extent. External mass transfer is more rapid at high bulk substrate concentrations; the higher the concentration, the greater is the driving force for mass transfer across the boundary layer. By reducing the demand for substrate transfer across the boundary layer, decreasing $r_{A,obs}$ as described in the previous section also reduces external limitations.

In large-scale reactors, external mass-transfer problems may be unavoidable if sufficiently high liquid velocities cannot be achieved. However, when evaluating biocatalyst kinetics in the laboratory, it is advisable to eliminate fluid boundary layers to simplify analysis of the data. Several laboratory reactor configurations allow almost complete elimination of interparticle and interphase concentration gradients [35]. Recycle reactors such as that shown in Figure 12.13 have been employed extensively for study of immobilised cell and enzyme reactions; operation with high liquid velocity through the bed reduces

Figure 12.15 Stirred laboratory reactor for minimising external mass-transfer effects. (From K. Sato and K. Toda, 1983, Oxygen uptake rate of immobilized growing *Candida lipolytica*. *J. Ferment. Technol.* 61, 239–245.)



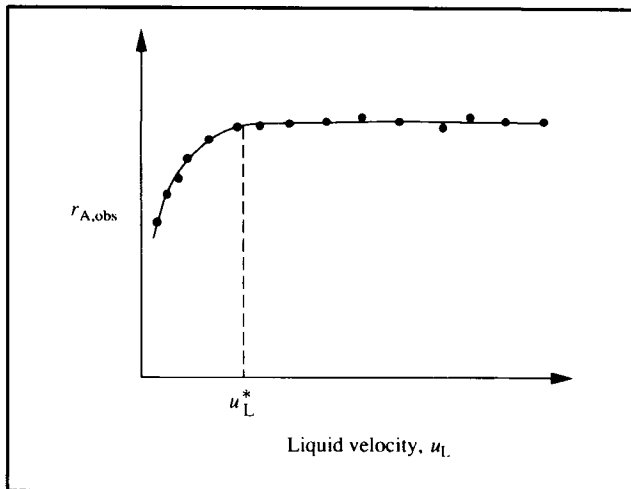
boundary-layer effects. Another suitable configuration is the *spinning-basket reactor* [36, 37] illustrated in Figure 12.14. By rotating the baskets at high speed, high relative velocity is achieved between the particles and the surrounding fluid. Slip velocities obtained using this apparatus are significantly greater than with freely-suspended particles. Another laboratory design aimed at increasing the slip velocity is the stirred vessel shown in Figure 12.15. In this vessel, catalyst particles are held relatively stationary in a wire-mesh cage while liquid is agitated at high speed.

Elimination of external mass-transfer effects can be verified by calculating the observable parameter Ω as described in Section 12.5. However, if the mass-transfer coefficient is not known accurately, an experimental test may be used instead [24]. Consider again the apparatus of Figure 12.13. If boundary layers around the particles affect the reaction, increased liquid velocity through the bed will improve conversion rates by reducing the film thickness and bringing C_{As} closer to C_{Ab} . As illustrated in Figure 12.16, at sufficiently high liquid velocity, external mass-transfer effects can be removed using this procedure; further increases in pump speed do not change the overall reaction rate. Therefore, if we can identify a liquid velocity u_L^* at which reaction rate becomes independent of liquid velocity, operation at $u_L > u_L^*$ will ensure that $\eta_e = 1$. For stirred reactors, a similar relationship holds between $r_{A,obs}$ and agitation speed.

12.9 Evaluating True Kinetic Parameters

The intrinsic kinetics of zero-order, first-order and Michaelis–Menten reactions are represented by the param-

Figure 12.16 Relationship between observed reaction rate and external liquid velocity.

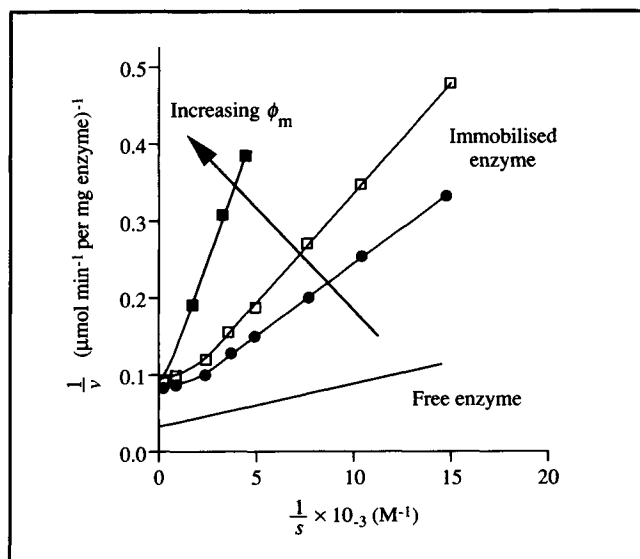


eters k_0 , k_1 , v_{max} and K_m . In general, it cannot be assumed that the values of these parameters will be the same before and after cell or enzyme immobilisation; significant changes can be wrought during the immobilisation process. As an example, Figure 12.17 shows Lineweaver–Burk plots for free and immobilised β -galactosidase enzyme. According to Eq. (11.37), the slopes and intercepts of the lines in Figure 12.17 indicate values of K_m/v_{max} and $1/v_{max}$, respectively. The steeper slopes and higher intercepts obtained for the immobilised enzyme indicate that immobilisation reduces v_{max} ; this is a commonly observed result. The value of K_m can also be affected [9].

As described in Sections 11.3 and 11.4, kinetic parameters for homogeneous reactions can be determined directly from experimental rate data. However, evaluating the true kinetic parameters of immobilised cells and enzymes is somewhat more difficult. The observed rate of reaction is not the true rate at all points in the catalyst; mass-transfer processes effectively ‘mask’ true kinetic behaviour. For example, v_{max} and K_m for immobilised catalysts cannot be estimated using the classical plots described in Section 11.4; under the influence of mass transfer, these plots no longer give straight lines over the entire range of substrate concentration [38, 39].

As illustrated in Figure 12.17 for β -galactosidase, Lineweaver–Burk plots for immobilised enzymes are non-linear; the shape and slope of the curves depend on the magnitude of the Thiele modulus. Deviation from linearity in such plots is easily obscured by the scatter in real experimental data and the distortion of errors due to the Lineweaver–Burk linearisation. At low substrate concentrations, i.e. large values

Figure 12.17 Lineweaver–Burk plots for free and immobilised β -galactosidase. Enzyme concentrations within the gel are: 0.10 mg ml^{-1} ($\bullet$); 0.17 mg ml^{-1} ($\square$); and 0.50 mg ml^{-1} ($\blacksquare$). (Data from P.S. Bunting and K.J. Laidler, 1972, Kinetic studies on solid-supported β -galactosidase. *Biochemistry* 11, 4477–4483.)



of $1/v$, the Lineweaver–Burk plot appears linear as first-order kinetics are approached. Even if we mistakenly interpret one of the immobilised-enzyme curves in Figure 12.17 as a straight line and evaluate apparent values of $v_{\max}$ and K_m , we can check whether or not we have found the true kinetic parameters by changing the particle radius and substrate concentration over a wide range of values. True kinetic parameters do not vary with these conditions which affect mass transfer into the catalyst; if the slope and/or intercept change with Thiele modulus, it is soon evident that the system is subject to diffusional limitations. Thus, apparent linearity of Lineweaver–Burk or similar plots is insufficient demonstration of the absence of mass-transfer restrictions. Studies have shown that the effect of diffusion is more pronounced in Eadie–Hofstee plots than Lineweaver–Burk or Langmuir plots [39]; however, all three can be approximated by straight lines over certain intervals.

Although the Lineweaver–Burk plot for immobilised β -galactosidase is non-linear, we should not conclude that immobilised enzymes fail to obey Michaelis–Menten kinetics. The kinetic form of reactions is generally maintained upon immobilisation of cells and enzymes [9]; non-linearity of the Lineweaver–Burk plot is due to the effect of mass transfer on $r_{A,\text{obs}}$.

Several methods have been proposed for determining $v_{\max}$ and K_m in heterogeneous catalysts [38–41]. The most straightforward approach is experimental; it involves reducing the particle size and catalyst loading and increasing the external liquid velocity to eliminate all mass-transfer resistances. Measured rate data can then be analysed for kinetic parameters as if the reaction were homogeneous. However, because it is usually very difficult or impossible to completely remove intraparticle effects, procedures involving a series of experiments coupled with theoretical analysis have also been proposed. In these methods, rate data are collected at high and low substrate concentrations using different particle sizes. At high substrate levels, it is assumed that the reaction is zero order with $\eta_1 = 1$; at low concentrations, first-order kinetics are assumed. These assumptions simplify the analysis, but may not always be valid.

When adequate computing facilities are available, true values of $v_{\max}$ and K_m can be extracted from diffusion-limited data using iterative calculations based on numerical integration and non-linear regression [7, 9]. Many iteration loops may be required before convergence to the final parameter values.

12.10 General Comments on Heterogeneous Reactions in Bioprocessing

Before concluding this chapter, some general observations and rules of thumb for heterogeneous reactions are outlined.

- (i) *Importance of oxygen mass-transfer limitations.* In aerobic reactions, mass transfer of oxygen is more likely to limit reaction rate than mass transfer of most other substrates. The reason is the poor solubility of oxygen in aqueous solutions. Whereas the sugar concentration in a typical fermentation broth is around $20\text{--}50 \text{ kg m}^{-3}$, oxygen concentration under 1 atm air pressure is limited to about $8 \times 10^{-3} \text{ kg m}^{-3}$. Therefore, because C_{A_s} is so low for oxygen, the observable Thiele modulus Φ can be several orders of magnitude greater than for other substrates. In anaerobic systems, the rate-limiting substrate is more difficult to identify.
- (ii) *Relationship between internal effectiveness factor and concentration gradient.* Depending on the reaction kinetics, the severity of intraparticle concentration gradients can be inferred from the value of the internal effectiveness factor. For first-order kinetics, reaction rate is directly proportional to substrate concentration; therefore $\eta_{11} = 1$ implies that concentration gradients do not exist in the catalyst. Conversely, if $\eta_{11} < 1$ we can conclude that concentration

gradients are present. In contrast, for zero-order reactions $\eta_{i0} = 1$ does not imply that gradients are absent; as long as $C_A > 0$, the reaction rate is unaffected. Concentration gradients can be so steep that C_A is reduced to almost zero within the catalyst, but η_{i0} remains equal to 1. On the other hand, $\eta_{i0} < 1$ implies that the concentration gradient is very severe and that some fraction of the particle volume is starved of substrate.

- (iii) *Relative importance of internal and external mass-transfer limitations.* For porous catalysts, it has been demonstrated with realistic values of mass transfer and diffusion parameters that external mass-transfer limitations do not exist unless internal limitations are also present [42]. Concentration differences between the bulk liquid and external catalyst surface are never observed without larger internal gradients developing within the particle. On the other hand, if internal limitations are known to be present, external limitations may or may not be important depending on conditions. Significant external mass-transfer effects can occur when reaction does not take place inside the catalyst, for example, if cells or enzymes are attached only to the exterior surface.
- (iv) *Operation of catalytic reactors.* Certain solid-phase properties are desirable for operation of immobilised-cell and -enzyme reactors. For example, in packed-bed reactors, large, rigid and uniformly-shaped particles promote well-distributed and stable liquid flow. Solids in packed columns should also have sufficient mechanical strength to withstand their own weight. These requirements are in direct conflict with those needed for rapid intraparticle mass transfer; diffusion is facilitated in particles that are small, soft and porous. Because blockages and large pressure drops through the bed must be avoided, mass-transfer rates are usually compromised. In stirred reactors, soft, porous gels are readily destroyed at the agitation speeds needed to eliminate external boundary-layer effects.
- (v) *Product effects.* Products formed by reaction inside catalysts must diffuse out under the influence of a concentration gradient. The concentration profile for product is the reverse of that for substrate; concentration is highest at the centre of the catalyst and lowest in the bulk liquid. If product inhibition affects cell or enzyme activity, high intraparticle concentrations may inhibit progress of the reaction. Immobilised enzymes which produce or consume H^+ ions are often affected; because enzyme reactions are very sensitive to pH, small local variations due to slow diffusion of ions can have a significant influence on reaction rate [43].

12.11 Summary of Chapter 12

At the end of Chapter 12 you should:

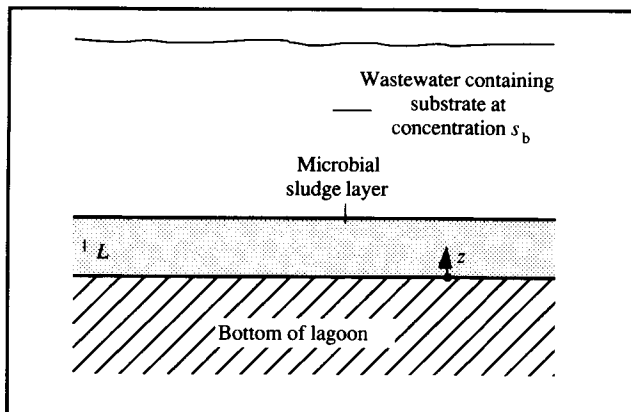
- (i) know what heterogeneous reactions are and when they occur in bioprocessing;
- (ii) understand the difference between *observed* and *true reaction rates*;
- (iii) know how concentration gradients arise in solid-phase catalysts;
- (iv) understand the concept of the *effectiveness factor*;
- (v) be able to apply the *Thiele modulus* and *observable Thiele modulus* to determine the effect of internal mass transfer on reaction rate;
- (vi) be able to quantify external mass-transfer effects from measured data;
- (vii) know how to minimise internal and external mass-transfer restrictions; and
- (viii) understand that it is generally difficult to determine true kinetic parameters for heterogeneous biological reactions.

Problems

12.1 Diffusion and reaction in a waste treatment lagoon

Industrial wastewater is often treated in large shallow lagoons. Consider such a lagoon covering land of area A . Microorganisms form a sludge layer of thickness L at the bottom of the lagoon; this sludge remains essentially undisturbed by movement of the liquid. As indicated in Figure 12P1.1, distance from the bottom of the lagoon is measured by coordinate z . Assume that microorganisms are distributed uniformly in the sludge.

Figure 12P1.1 Lagoon for wastewater treatment.



At steady state, wastewater is fed into the lagoon so that the bulk concentration of digestible substrate remains constant at s_b . Cells consume substrate diffusing into the sludge layer, thereby establishing a concentration gradient across thickness L .

- (a) Set up a shell mass balance on substrate by considering a thin slice of sludge of thickness Δz perpendicular to the direction of diffusion. The rate of microbial reaction per unit volume sludge is:

$$r_s = k_1 s$$

where s is the concentration of substrate in the sludge layer (gmol cm^{-3}) and k_1 is the first-order rate constant (s^{-1}). The effective diffusivity of substrate in sludge is $\mathcal{D}_{\text{Se}}$. Obtain a differential equation relating s and z .

Hint: Area A is constant for flat-plate geometry and can be cancelled from all terms of the mass-balance equation.

- (b) External mass-transfer effects at the liquid–sludge interface are negligible. What are the boundary conditions for this problem?
(c) The differential equation obtained in (a) is solved by making the substitution:

$$s = Ne^{pz}$$

where N and p are constants.

- (i) Substitute this expression for s into the differential equation derived in (a) to obtain an equation for p . (Remember that $\sqrt{p^2} = p$ or $-p$.)
(ii) Because there are two possible values of p , let:

$$s = Ne^{pz} + Me^{-pz}.$$

Apply the boundary condition at $z=0$ to this expression and obtain a relationship between N and M .

- (iii) Use the boundary condition at $z=L$ to find N and M explicitly. Obtain an expression for s as a function of z .
(iv) Use the definition of $\cosh x$:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

to prove that:

$$\frac{s}{s_b} = \frac{\cosh(z\sqrt{k_1/\mathcal{D}_{\text{Se}}})}{\cosh(L\sqrt{k_1/\mathcal{D}_{\text{Se}}})}.$$

- (d) At steady state, rate of substrate consumption must be equal to the rate at which substrate enters the sludge. As substrate enters the sludge by diffusion, the overall rate of reaction can be evaluated using Fick's law:

$$r_{A,\text{obs}} = \mathcal{D}_{\text{Se}} A \left. \frac{ds}{dz} \right|_{z=L}$$

where

$$\left. \frac{ds}{dz} \right|_{z=L} \text{ means } \frac{ds}{dz} \text{ evaluated at } z=L.$$

Use the equation for s from (c) to derive an equation for $r_{A,\text{obs}}$.

Hint: The derivative of $\cosh ax = a \sinh ax$ where a is a constant and:

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

- e) Show from the result of (d) that the internal effectiveness factor is given by the expression:

$$\eta_{\text{il}} = \frac{\tanh \phi_1}{\phi_1}$$

where

$$\phi_1 = L \sqrt{\frac{k_1}{\mathcal{D}_{\text{Se}}}}$$

and:

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

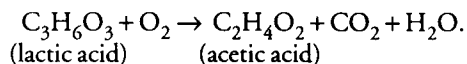
- (f) Plot the concentration profiles through a sludge layer of thickness 2 cm for the following sets of conditions:

	(1)	(2)	(3)
$k_1 (\text{s}^{-1})$	4.7×10^{-8}	2.0×10^{-7}	1.5×10^{-4}
$\mathcal{D}_{\text{Se}} (\text{cm}^2 \text{s}^{-1})$	7.5×10^{-7}	2.0×10^{-7}	6.0×10^{-6}

Take s_b to be $10^{-5} \text{ gmol cm}^{-3}$. Label the profiles with corresponding values of ϕ_1 and η_{il} . Comment on the general relationship between ϕ_1 , the shape of the concentration profile, and the value of η_{il} .

12.2 Oxygen profile in immobilised-enzyme catalyst

L-Lactate 2-monoxygenase from *Mycobacterium smegmatis* is immobilised in spherical agarose beads. The enzyme catalyses the reaction:



Beads 4 mm in diameter are immersed in a well-mixed solution containing 0.5 mM oxygen. A high lactic acid concentration is provided so that oxygen is the rate-limiting substrate. The effective diffusivity of oxygen in agarose is $2.1 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. K_m for the immobilised enzyme is 0.015 mM; $v_{\max}$ is 0.12 mol s^{-1} per kg enzyme. The beads contain $0.012 \text{ kg enzyme m}^{-3}$ gel. External mass-transfer effects are negligible.

- Plot the oxygen concentration profile inside the beads.
- What fraction of the catalyst volume is active?
- Determine the largest bead size that allows the maximum conversion rate?

12.3 Effect of oxygen transfer on recombinant cells

Recombinant *E. coli* cells contain a plasmid derived from pBR322 incorporating genes for the enzymes β -lactamase and catechol 2,3-dioxygenase from *Pseudomonas putida*. To produce the desired enzymes the organism requires aerobic conditions. The cells are immobilised in spherical beads of carrageenan gel. The effective diffusivity of oxygen is $1.4 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. Uptake of oxygen is zero-order with intrinsic rate constant $10^{-3} \text{ mol s}^{-1} \text{ m}^{-3}$ particle. The concentration of oxygen at the surface of the catalyst is $8 \times 10^{-3} \text{ kg m}^{-3}$. Cell growth is negligible.

- What is the maximum particle diameter for aerobic conditions throughout the catalyst?
- For particles half the diameter calculated in (a), what is the minimum oxygen concentration in the beads?
- The density of cells in the gel is reduced by a factor of five. If specific activity is independent of cell loading, what is the maximum particle size for aerobic conditions?

12.4 Ammonia oxidation by immobilised cells

Thiosphaera pantotropha is being investigated for aerobic oxidation of ammonia to nitrite in wastewater treatment. The organism is immobilised in spherical agarose particles of diameter 3 mm. The effective diffusivity of oxygen in the particles is $1.9 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. The immobilised cells are placed in a flow

chamber for measurement of oxygen uptake rate. Using published correlations, the liquid–solid mass-transfer coefficient for oxygen is calculated as $6 \times 10^{-5} \text{ m s}^{-1}$. When the bulk oxygen concentration is $6 \times 10^{-3} \text{ kg m}^{-3}$, the observed rate of oxygen consumption is $2.2 \times 10^{-5} \text{ kg s}^{-1} \text{ m}^{-3}$ catalyst.

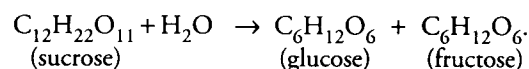
- What effect does external mass-transfer have on respiration rate?
- What is the effectiveness factor?
- For optimal activity of *T. pantotropha*, oxygen levels must be kept above the critical level, $1.2 \times 10^{-3} \text{ kg m}^{-3}$. Is this condition satisfied?

12.5 Microcarrier culture and external mass transfer

Mammalian cells form a monolayer on the surface of microcarrier beads of diameter 120 μm and density $1.2 \times 10^3 \text{ kg m}^{-3}$. The culture is maintained in spinner flasks in serum-free medium of viscosity $10^{-3} \text{ N s m}^{-2}$ and density 10^3 kg m^{-3} . The diffusivity of oxygen in the medium is $2.3 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. The observed rate of oxygen uptake is $0.015 \text{ mol s}^{-1} \text{ m}^{-3}$ at a bulk oxygen concentration of 0.2 mol m^{-3} . What is the effect of external mass transfer on reaction rate?

12.6 Immobilised-enzyme reaction kinetics

Invertase catalyses the reaction:



Invertase from *Aspergillus oryzae* is immobilised in porous resin particles of diameter 1.6 mm at a density of $0.1 \text{ } \mu\text{mol enzyme g}^{-1}$. The effective diffusivity of sucrose in the resin is $1.3 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$. The resin is placed in a spinning-basket reactor operated so that external mass-transfer effects are eliminated. At a sucrose concentration of 0.85 kg m^{-3} , the observed rate of conversion is $1.25 \times 10^{-3} \text{ kg s}^{-1} \text{ m}^{-3}$ resin. K_m for the immobilised enzyme is 3.5 kg m^{-3} .

- Calculate the effectiveness factor.
- Determine the true first-order reaction constant for immobilised invertase.
- Assume that specific enzyme activity is not affected by steric hindrance or conformational changes as enzyme loading increases. This means that k_1 should be directly proportional to enzyme concentration in the resin. Plot changes in effectiveness factor and reaction rate as a function of enzyme loading from $0.01 \text{ } \mu\text{mol g}^{-1}$ to

$2.0 \mu\text{mol g}^{-1}$. Comment on the relative benefit of increasing the concentration of enzyme in the resin.

12.7 Mass-transfer effects in plant cell culture

Suspended *Catharanthus roseus* cells form spherical clumps approximately 1.5 mm in diameter. Oxygen uptake is measured using the apparatus of Figure 12.13; medium is recirculated with a superficial liquid velocity of 0.83 cm s^{-1} . At a bulk concentration of 8 mg l^{-1} , oxygen is consumed at a rate of 0.28 mg per g wet weight per hour. Assume that the density and viscosity of the medium are similar to water, the specific gravity of wet cells is 1, and oxygen uptake is zero order. The effective diffusivity of oxygen in the clumps is $9 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, or half that in the medium.

- Does external mass transfer affect the oxygen-uptake rate?
- To what extent does internal mass transfer affect oxygen uptake?
- Roughly, what would you expect the profile of oxygen concentration to be within the aggregates?

12.8 Respiration in mycelial pellets

Aspergillus niger cells are observed to form aggregates of average diameter 5 mm. The effective diffusivity of oxygen in the aggregates is $1.75 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. In a fixed-bed reactor, the oxygen-consumption rate at a bulk oxygen concentration of $8 \times 10^{-3} \text{ kg m}^{-3}$ is $8.7 \times 10^{-5} \text{ kg s}^{-1} \text{ m}^{-3}$ biomass. The liquid–solid mass-transfer coefficient is $3.8 \times 10^{-5} \text{ m s}^{-1}$.

- Is oxygen uptake affected by external mass transfer?
- What is the external effectiveness factor?
- What reaction rate would be observed if both internal and external mass-transfer resistances were eliminated?
- If only external mass-transfer effects were removed, what would be the reaction rate?

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Suggestions for Further Reading

Immobilised Cells and Enzymes

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Engineering Analysis of Mass Transfer and Reaction (see also refs 1, 7, 14 and 38–41)

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Reactor Engineering

The reactor is the heart of any fermentation or enzyme conversion process. Design of bioreactors is a complex task, relying on scientific and engineering principles and many rules of thumb. Specifying aspects of the reactor and its operation involves several critical decisions.

- (i) *Reactor configuration.* For example, should the reactor be a stirred tank or an air-driven vessel without mechanical agitation?
- (ii) *Reactor size.* What size reactor is required to achieve the desired rate of production?
- (iii) *Processing conditions inside the reactor.* What reaction conditions such as temperature, pH and dissolved-oxygen tension should be maintained in the vessel, and how will these parameters be controlled? How will contamination be avoided?
- (iv) *Mode of operation.* Will the reactor be operated batch-wise or as a continuous-flow process? Should substrate be fed intermittently? Should the reactor be operated alone or in series with others?

Decisions made in reactor design have a significant impact on overall process performance, yet there are no simple or standard design procedures available which specify all aspects of the vessel and its operation. Reactor engineering brings together much of the material already covered in Chapters 7–12 of this book. Knowledge of reaction kinetics is essential for understanding how biological reactors work. Other areas of bioprocess engineering such as mass and energy balances, mixing, mass transfer and heat transfer are also required.

13.1 Reactor Engineering in Perspective

Before starting to design a reactor, some objectives have to be defined. Simple aims like 'Produce 1 g of monoclonal antibody per day', or 'Produce 10 000 tonnes of amino acid per year', provide the starting point. Other objectives are also relevant; in industrial processes the product should be made at the lowest possible cost to maximise the company's commercial advantage. In some cases, economic objectives are overridden by safety concerns, the need for high-product purity or regula-

tory considerations. The final reactor design will be a reflection of all these process requirements and, in most cases, represents a compromise solution to conflicting demands.

In this section, we will consider the various contributions to bioprocessing costs for different types of product, and examine the importance of reactor engineering in improving overall process performance. As shown in Figure 13.1, the value of products made by bioprocessing covers a wide range. Typically, products with the highest value are those from

Figure 13.1 Range of value of fermentation products. (From P.N. Royce, 1993, A discussion of recent developments in fermentation monitoring and control from a practical perspective. *Crit. Rev. Biotechnol.* 13, 117–149.)

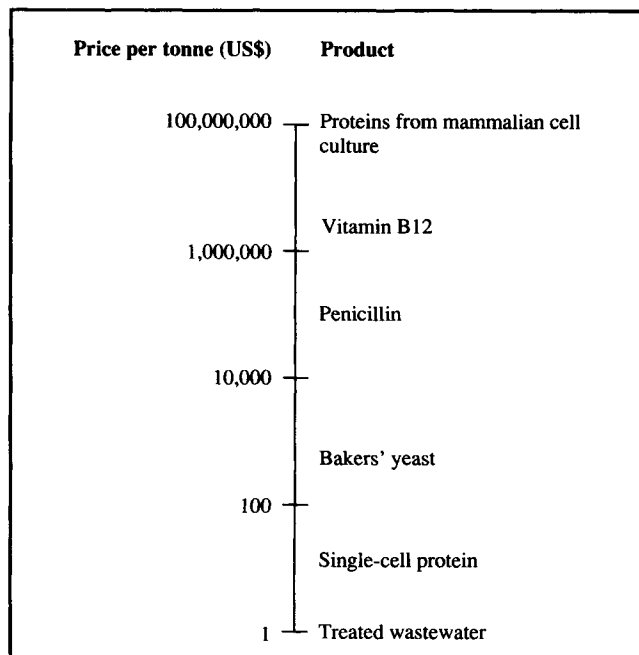
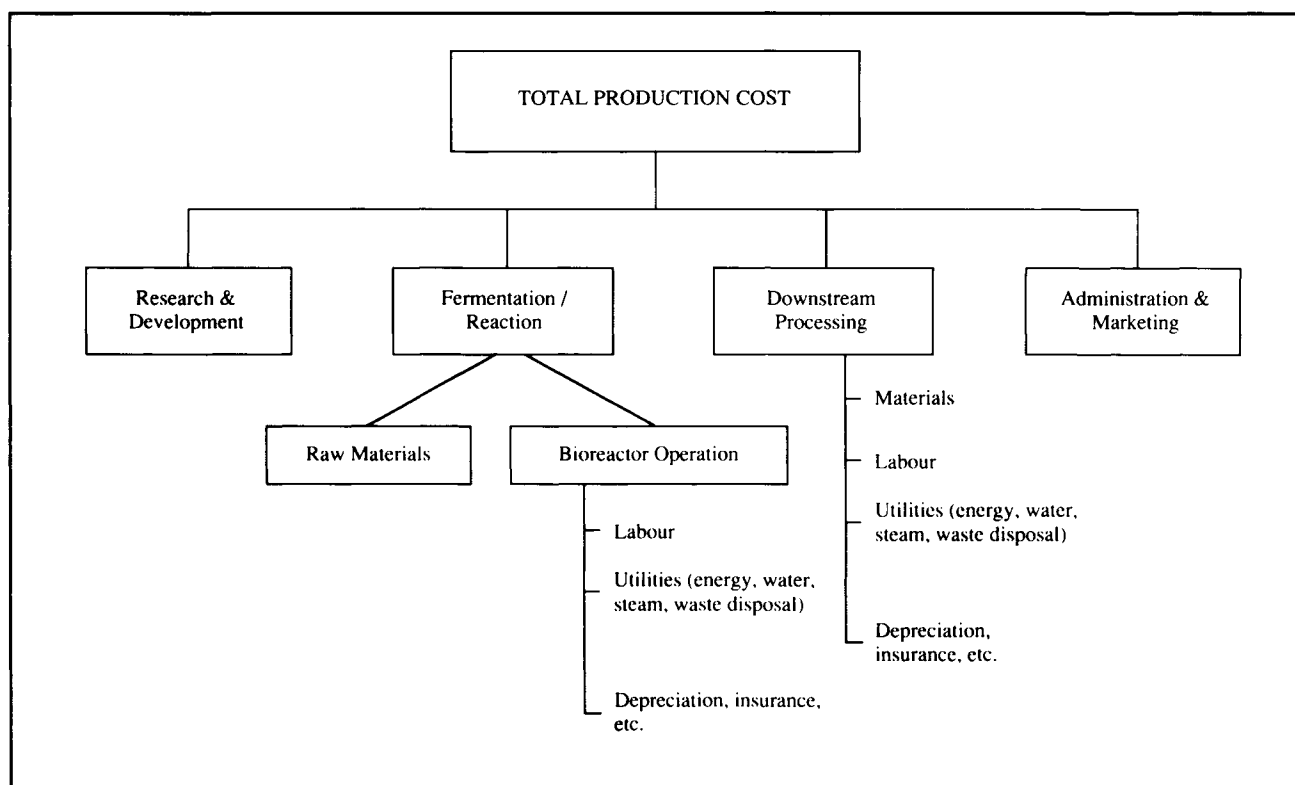


Figure 13.2 Contributions to total production cost in bioprocessing.

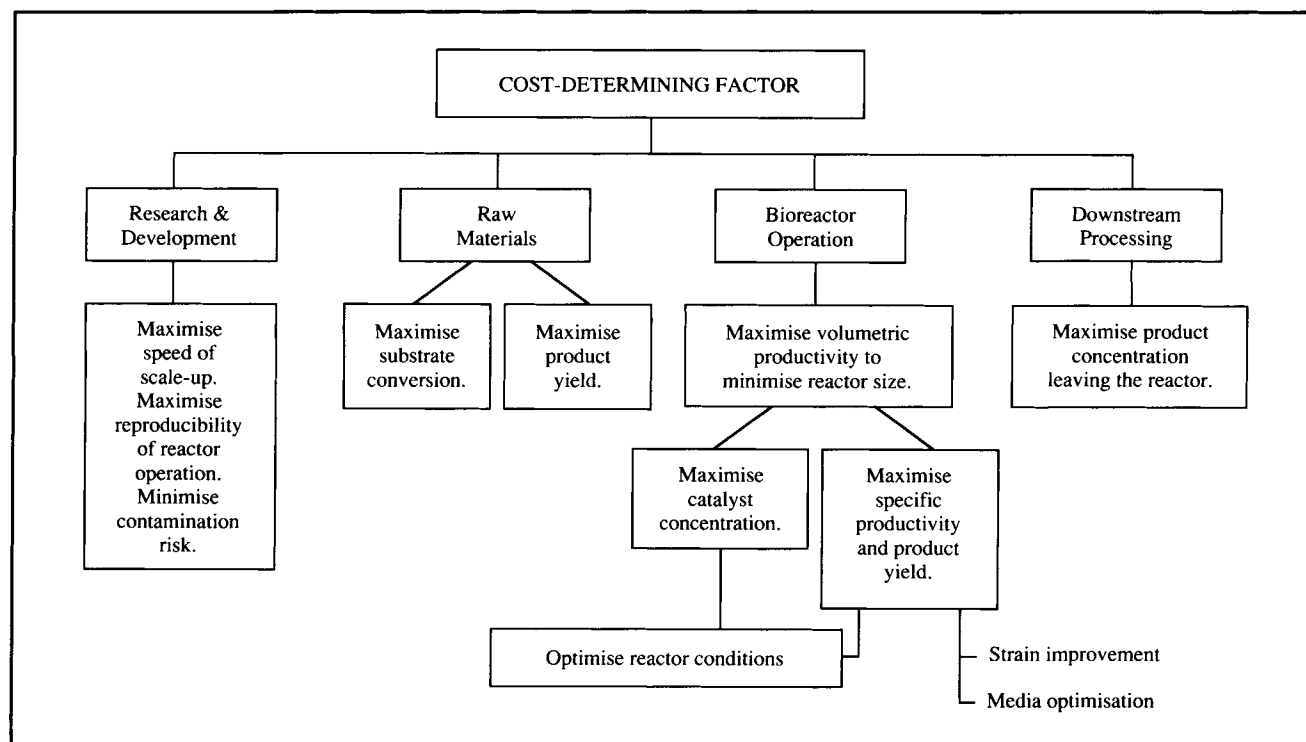


mammalian cell culture, such as therapeutic proteins and monoclonal antibodies. At the opposite end of the scale is treatment of waste, where the overriding objective is minimal financial outlay for the desired level of purity. To reduce the cost of any bioprocess, it is first necessary to identify which aspect of it is cost-determining. Break-down of production costs varies from process to process; however, a general scheme is shown in Figure 13.2. The following components are important: (i) research and development; (ii) the fermentation or reaction step; (iii) downstream processing; and (iv) administration and marketing. In most bioprocesses, the cost of administration and marketing is relatively small. Products for which the cost of reaction dominates include biomass such as bakers' yeast and single-cell protein, catabolic metabolites such as ethanol and lactic acid, and bioconversion products such as high-fructose corn syrup and 6-aminopenicillanic acid. Intracellular products such as proteins have high downstream-processing costs compared with reaction; other examples in this category are antibiotics, vitamins and amino acids. For new, high-value biotechnology products such as recombinant proteins and antibodies, actual processing costs are only a small part of the total because of the enormous

investment required for research and development and regulatory approval. Getting the product into the marketplace quickly is the most important cost-saving measure in these cases; any savings made by improving the efficiency of the reactor are generally trivial in comparison. However, for the majority of fermentation products outside this high-value category, bioprocessing costs make a significant contribution to the final price.

If the reaction step dominates the cost structure, this may be because of the high cost of the raw materials required or the high cost of reactor operation. The relative contributions of these factors depends on the process. As an example, to produce high-value antibiotics, the cost of 100 m³ media is US\$25 000–100 000 [1]. In contrast, the cost of energy, i.e. electricity, to operate a 100 m³ stirred-tank fermenter including agitation, air compression and cooling water for a 6-d antibiotic fermentation is about US\$8000 [1, 2]. Clearly then, energy costs for reactor operation are much less important than raw-material costs for this fermentation process. For high-value, low-yield products such as antibiotics, vitamins, enzymes and pigments, media represents 60–90% of the fermentation costs [1]. For low-cost, high-yield metabolites such

Figure 13.3 Strategies for bioreactor design as a function of the cost-determining factors in the process.



as ethanol, citric acid, biomass and lactic acid, raw material costs range from 40% of fermentation costs for citric acid to about 70% for ethanol produced from molasses [1, 3]. The remainder of the operating cost of bioreactors consists mainly of labour and utilities costs.

As indicated in Figure 13.3, identifying the cost structure of bioprocesses assists in defining the objectives for reactor design. Even if the reaction itself is not cost-determining, aspects of reactor design may still be important. If the cost of research and development is dominating, design of the reactor is directed towards the need for rapid scale-up; this is more important than maximising conversion or minimising operating costs. For new biotechnology products intended for therapeutic use, regulatory guidelines require that the entire production scheme be validated and process control guaranteed for consistent quality and safety; reproducibility of reactor operation is therefore critical. When the cost of raw materials is significant, maximising substrate conversion and product yield in the reactor have high priority. If downstream processing is expensive, the reactor is designed and operated to maximise the product concentration leaving the vessel; this avoids the expense of recovering product from dilute solutions. When reaction costs are significant, the reactor should

be as small as possible to reduce both operating and capital costs. To achieve the desired total production rate using a small vessel, the volumetric productivity of the reactor must be sufficiently high (see Section 11.1.3).

As indicated in Figure 13.3, volumetric productivity depends on the concentration of catalyst and its specific rate of production. To achieve high volumetric rates, the reactor must therefore allow maximum catalyst activity at the highest practical catalyst concentration. For tightly packed cells or cell organelles, the physical limit on concentration is of the order $200 \text{ kg dry weight m}^{-3}$; for enzymes in solution, the maximum concentration depends on the solubility of enzyme in the reaction mixture. The extent to which these limiting concentrations can be approached depends on the functioning of the reactor. For example, if mixing or mass transfer is inadequate, oxygen or nutrient starvation will occur and the maximum cell density achieved will be low. Alternatively, if shear levels in the reactor are too high, cells will be disrupted and enzymes inactivated so that the effective concentration of catalyst is reduced.

Maximum specific productivity is obtained when the catalyst is capable of high levels of production and conditions in the reactor allow the best possible catalytic function. For

simple metabolites such as ethanol, butanol and acetic acid which are linked to energy production in the cell, the maximum theoretical yield is limited by the thermodynamic and stoichiometric principles outlined in Section 4.6. Accordingly, there is little scope for increasing production titres of these materials; reduced production costs and commercial advantage rely mostly on improvements in reactor operation which allow the system to achieve close to the maximum theoretical yield. In contrast, it is not unusual for strain improvement and media optimisation programmes to improve yields of antibiotics and enzymes by over 100-fold, particularly in the early stages of process development. Therefore, for these products, identification of high-producing strains and optimal environmental conditions is initially more rewarding than improving the reactor design and operation.

13.2 Bioreactor Configurations

The cylindrical tank, either stirred or unstirred, is the most common reactor in bioprocessing. Yet, a vast array of fermenter configurations is in use in different bioprocess industries. Novel bioreactors are constantly being developed for special applications and new forms of biocatalyst such as plant and animal tissue and immobilised cells and enzymes.

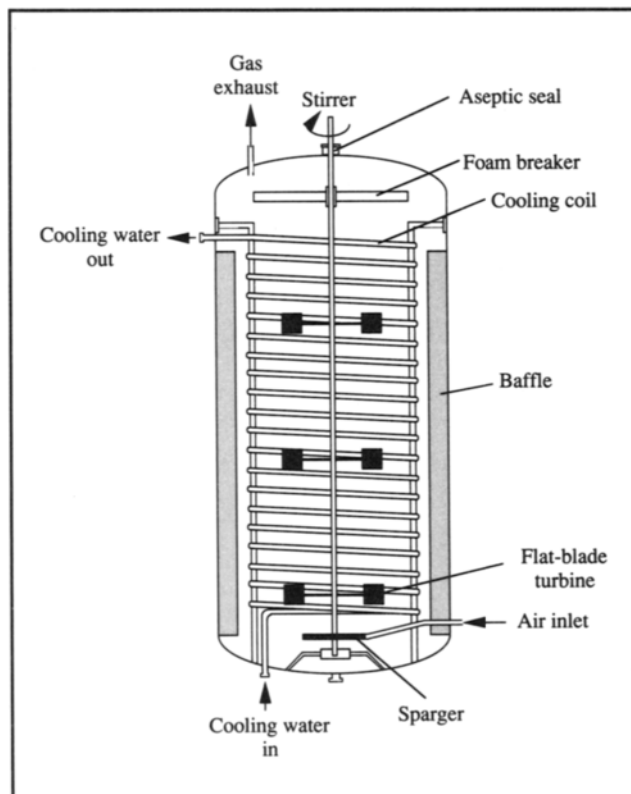
Much of the challenge in reactor design lies in the provision of adequate mixing and aeration for the large proportion of fermentations requiring oxygen; reactors for anaerobic culture are usually very simple in construction without sparging or agitation. In the following discussion of bioreactor configurations, aerobic operation will be assumed.

13.2.1 Stirred Tank

A conventional stirred, aerated bioreactor is shown schematically in Figure 13.4. Mixing and bubble dispersion are achieved by mechanical agitation; this requires a relatively high input of energy per unit volume. Baffles are used in stirred reactors to reduce vortexing. A wide variety of impeller sizes and shapes is available to produce different flow patterns inside the vessel; in tall fermenters, installation of multiple impellers improves mixing. The mixing and mass-transfer functions of stirred reactors are described in detail in Chapters 7 and 9.

Typically, only 70–80% of the volume of stirred reactors is filled with liquid; this allows adequate headspace for disengagement of droplets from the exhaust gas and to accommodate any foam which may develop. If foaming is a problem, a supplementary impeller called a *foam breaker* may

Figure 13.4 Typical stirred-tank fermenter for aerobic culture.



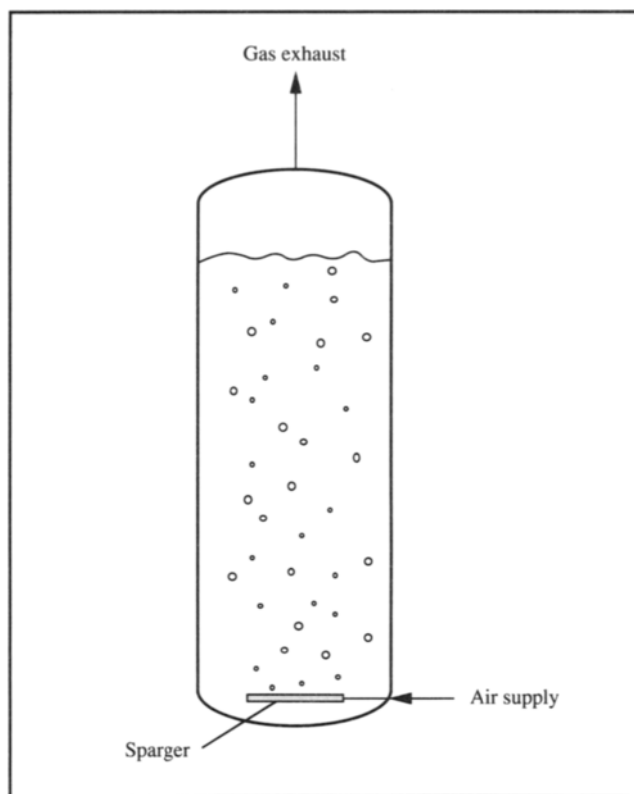
be installed as shown in Figure 13.4. Alternatively, chemical antifoam agents are added to the broth; because antifoams reduce the rate of oxygen transfer (see Section 9.6.3), mechanical foam dispersal is generally preferred.

The aspect ratio of stirred vessels, i.e. the ratio of height to diameter, can be varied over a wide range. The least expensive shape to build has an aspect ratio of about 1; this shape has the smallest surface area and therefore requires the least material to construct for a given volume. However, when aeration is required, the aspect ratio is usually increased. This provides for longer contact times between the rising bubbles and liquid and produces a greater hydrostatic pressure at the bottom of the vessel.

As shown in Figure 13.4, temperature control and heat transfer in stirred vessels can be accomplished using internal cooling coils. Alternative cooling equipment for bioreactors is illustrated in Figure 8.1 (p. 165). The relative advantages and disadvantages of different heat-exchange systems are discussed in Section 8.1.1.

Stirred fermenters are used for free- and immobilised-

Figure 13.5 Bubble-column bioreactor.



enzyme reactions, and for culture of suspended and immobilised cells. Care is required with particulate catalysts which may be damaged or destroyed by the impeller at high speeds. As discussed in Section 7.14, high levels of shear can also damage sensitive cells, particularly in plant and animal cell culture.

13.2.2 Bubble Column

Alternatives to the stirred reactor include vessels with no mechanical agitation. In bubble-column reactors, aeration and mixing are achieved by gas sparging; this requires less energy than mechanical stirring. Bubble columns are applied industrially for production of bakers' yeast, beer and vinegar, and for treatment of wastewater.

Bubble columns are structurally very simple. As shown in Figure 13.5, they are generally cylindrical vessels with height greater than twice the diameter. Other than a sparger for entry of compressed air, bubble columns typically have no internal structures. A height-to-diameter ratio of about 3:1 is common in bakers' yeast production; for other applications, towers with height-to-diameter ratios of 6:1 have been used. Perforated

horizontal plates are sometimes installed in tall bubble columns to break up and redistribute coalesced bubbles. Advantages of bubble columns include low capital cost, lack of moving parts, and satisfactory heat- and mass-transfer performance. As in stirred vessels, foaming can be a problem requiring mechanical dispersal or addition of antifoam to the medium.

Bubble-column hydrodynamics and mass-transfer characteristics depend entirely on the behaviour of the bubbles released from the sparger. Different flow regimes occur depending on the gas flow rate, sparger design, column diameter and medium properties such as viscosity. *Homogeneous flow* occurs only at low gas flow rates and when bubbles leaving the sparger are evenly distributed across the column cross-section. In homogeneous flow, all bubbles rise with the same upward velocity and there is no backmixing of the gas phase. Liquid mixing in this flow regime is also limited, arising solely from entrainment in the wakes of the bubbles. Under normal operating conditions at higher gas velocities, large chaotic circulatory flow cells develop and *heterogeneous flow* occurs as illustrated in Figure 13.6. In this regime, bubbles and liquid tend to rise up the centre of the column while a corresponding downflow of liquid occurs near the walls. Liquid circulation entrains bubbles so that some backmixing of gas occurs.

Liquid mixing time in bubble columns depends on the flow regime. For heterogeneous flow, the following equation has been proposed [4] for the upward liquid velocity at the centre of the column for $0.1 < D < 7.5$ m and $0 < u_G < 0.4$ m s⁻¹:

$$u_L = 0.9(g D u_G)^{0.33} \quad (13.1)$$

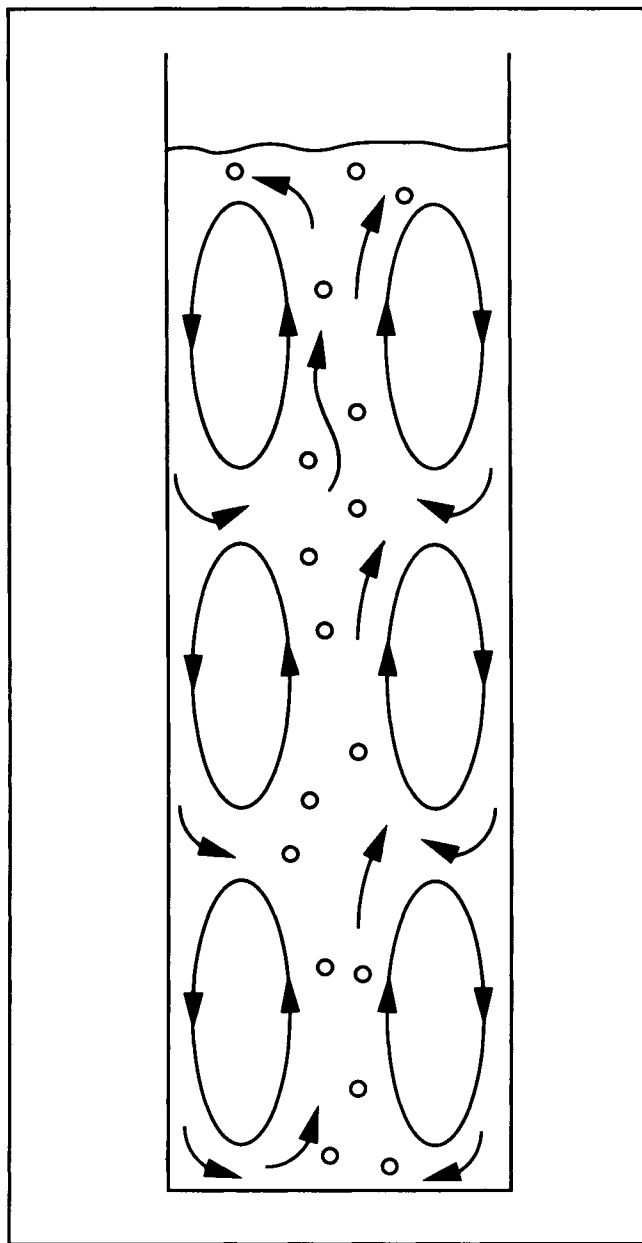
where u_L is linear liquid velocity, g is gravitational acceleration, D is column diameter, and u_G is gas superficial velocity. u_G is equal to the volumetric gas flow rate at atmospheric pressure divided by the reactor cross-sectional area. From this equation, an expression for the mixing time t_m (see Section 7.9.4) can be obtained [5]:

$$t_m = 11 \frac{H}{D} (g u_G D^{-2})^{-0.33} \quad (13.2)$$

where H is the height of the bubble column.

As discussed in Section 9.6.1, values for gas-liquid mass-transfer coefficients in reactors depend largely on bubble diameter and gas hold-up. In bubble columns containing non-viscous liquids, these variables depend solely on the gas flow rate. However, as exact bubble sizes and liquid circulation

Figure 13.6 Heterogeneous flow in a bubble column.



patterns are impossible to predict in bubble columns, accurate estimation of the mass-transfer coefficient is difficult. The following correlation has been proposed for non-viscous media in heterogeneous flow [4, 5]:

$$k_L a \approx 0.32 u_G^{0.7} \quad (13.3)$$

where $k_L a$ is the combined volumetric mass-transfer coefficient and u_G is the gas superficial velocity. Eq. (13.3) is valid for bubbles with mean diameter about 6 mm, $0.08 \text{ m} < D < 11.6 \text{ m}$, $0.3 \text{ m} < H < 21 \text{ m}$, and $0 < u_G < 0.3 \text{ m s}^{-1}$. If smaller bubbles are produced at the sparger and the medium is non-coalescing, $k_L a$ will be larger than the value calculated using Eq. (13.3), especially at low values of u_G less than about 10^{-2} m s^{-1} [4].

13.2.3 Airlift Reactor

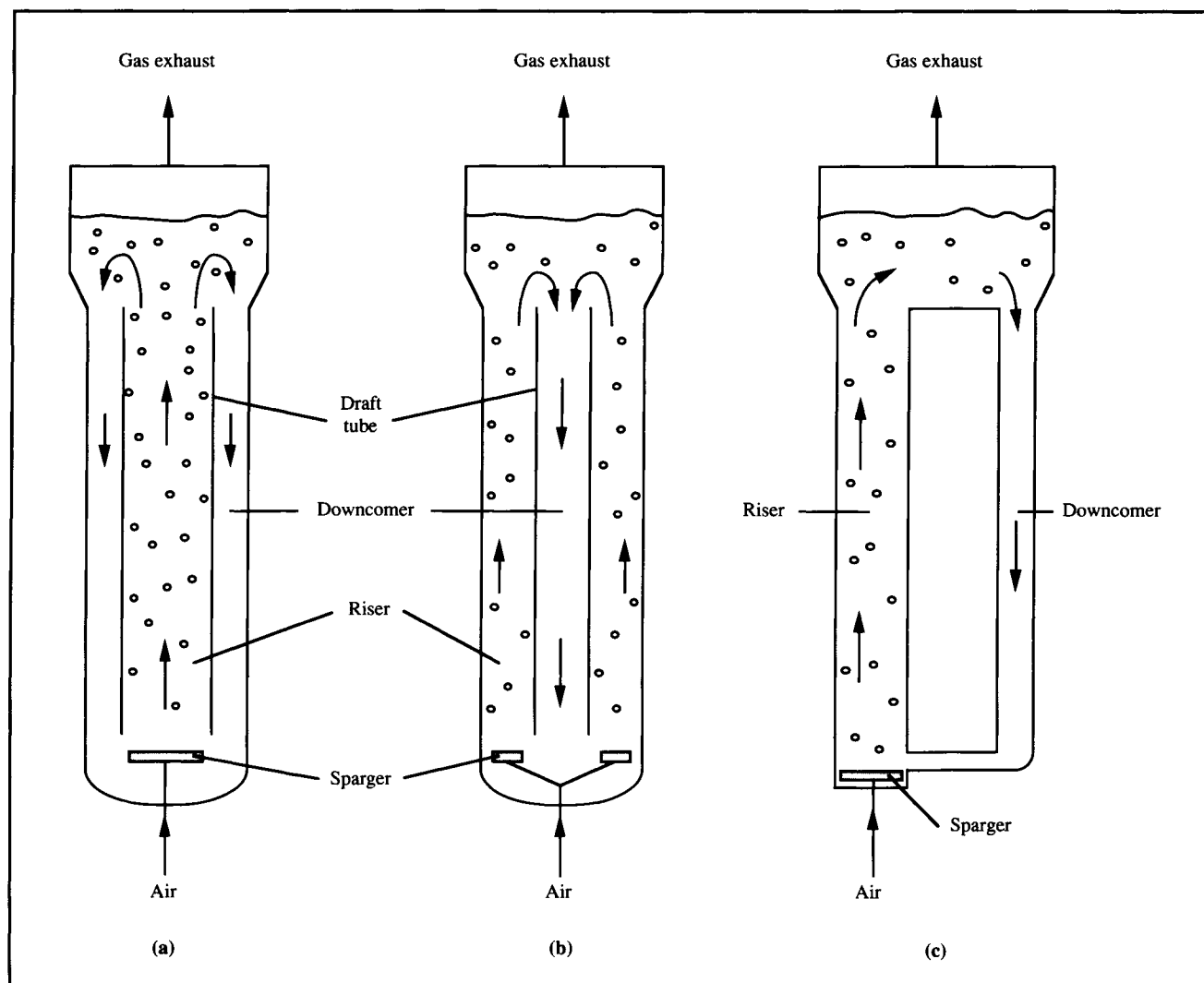
As in bubble columns, mixing in airlift reactors is accomplished without mechanical agitation. Airlift reactors are often chosen for culture of plant and animal cells and immobilised catalysts because shear levels are significantly lower than in stirred vessels.

Several types of airlift reactor are in use. Their distinguishing feature compared with the bubble column is that patterns of liquid flow are more defined owing to the physical separation of up-flowing and down-flowing streams. As shown in Figure 13.7, gas is sparged into only part of the vessel cross-section called the *riser*. Gas hold-up and decreased fluid density cause liquid in the riser to move upwards. Gas disengages at the top of the vessel leaving heavier bubble-free liquid to recirculate through the *downcomer*. Liquid circulates in airlift reactors as a result of the density difference between riser and downcomer.

Figure 13.7 illustrates the most common airlift configurations. In the *internal-loop vessels* of Figures 13.7(a) and 13.7(b), the riser and downcomer are separated by an internal baffle or *draft tube*; air may be sparged into either the draft tube or the annulus. In the *external-loop or outer-loop* airlift of Figure 13.7(c), separate vertical tubes are connected by short horizontal sections at the top and bottom. Because the riser and downcomer are further apart in external-loop vessels, gas disengagement is more effective than in internal-loop devices. Fewer bubbles are carried into the downcomer, the density difference between fluids in the riser and downcomer is greater, and circulation of liquid in the vessel is faster. Accordingly, mixing is usually better in external-loop than internal-loop reactors.

Airlift reactors generally provide better mixing than bubble columns except at low liquid velocities when circulatory flow-patterns similar to those shown in Figure 13.6 develop. The airlift configuration confers a degree of stability to liquid flow compared with bubble columns; therefore, higher gas flow rates can be used without incurring operating problems such as slug flow or spray formation. Several empirical correlations have been developed for liquid velocity, circulation time and

Figure 13.7 Airlift reactor configurations.



mixing time in airlift reactors; however there is considerable discrepancy between the results [6]. Equations derived from hydrodynamic models are also available [6, 7]; these are usually relatively complex and, because liquid velocity and gas hold-up are not independent, require iterative numerical solution.

Gas hold-up and gas-liquid mass-transfer rates in internal-loop airlifts are similar to those in bubble columns [6]. However, in external-loop devices, near-complete gas disengagement increases the liquid velocity and decreases the air hold-up [8, 9] so that mass-transfer rates at identical gas velocities are lower than in bubble columns [6]. Therefore, by comparison with Eq. (13.3) for bubble columns, for external-loop airlifts:

$$k_L a < 0.32 u_G^{0.7}. \quad (13.4)$$

Several other empirical mass-transfer correlations have been developed for Newtonian and non-Newtonian fluids in airlift reactors [6].

Performance of airlift devices is influenced significantly by the details of vessel construction [6, 10, 11]. For example, in internal-loop airlifts, changing the distance between the lower edge of the draft tube and the base of the reactor alters the pressure drop in this region and affects liquid velocity and gas hold-up. The depth of draft-tube submersion from the top of the liquid also influences mixing and mass-transfer characteristics.

Airlift reactors have been applied in production of single-cell protein from methanol and gas oil; they are also used for plant and animal cell culture and in municipal and industrial waste treatment. Large airlift reactors with capacities of thousands of cubic metres have been constructed. Tall internal-loop airlifts built underground are known as *deep-shaft reactors*; very high hydrostatic pressure at the bottom of these vessels considerably improves gas–liquid mass-transfer. The height of airlift reactors is typically about 10 times the diameter; for deep-shaft systems the height-to-diameter ratio may be increased up to 100.

13.2.4 Stirred and Air-Driven Reactors: Comparison of Operating Characteristics

For low-viscosity fluids, adequate mixing and mass transfer can be achieved in stirred tanks, bubble columns and airlift vessels. When a large fermenter ($50\text{--}500\text{ m}^3$) is required for low-viscosity culture, a bubble column is an attractive choice because it is simple and cheap to install and operate. Mechanically-agitated reactors are impractical at volumes greater than about 500 m^3 as the power required to achieve adequate mixing becomes extremely high (see Section 7.11).

If the culture has high viscosity, sufficient mixing and mass transfer cannot be provided by air-driven reactors. Stirred vessels are more suitable for viscous liquids because greater power can be input by mechanical agitation. Nevertheless, mass-transfer rates decline rapidly in stirred vessels at viscosities greater than $50\text{--}100\text{ cP}$ [5].

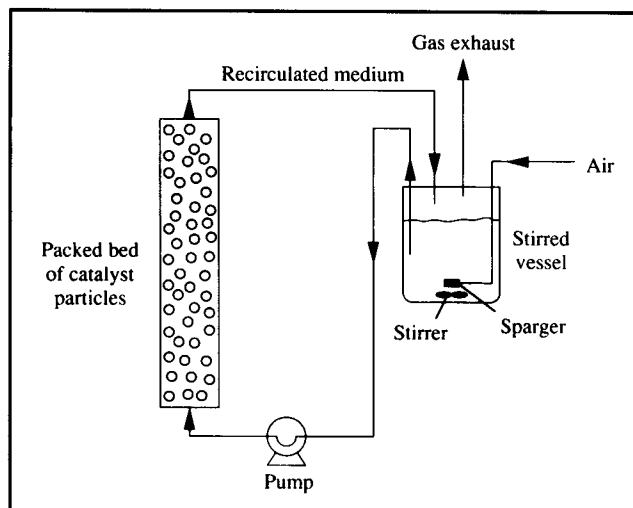
Heat transfer can be an important consideration in the choice between air-driven and stirred reactors. Mechanical agitation generates much more heat than sparging of compressed gas. When the heat of reaction is high, such as in production of single-cell protein from methanol, removal of frictional stirrer heat can be a problem so that air-driven reactors may be preferred.

Stirred-tank and air-driven vessels account for the vast majority of bioreactor configurations used for aerobic culture. However, other reactor configurations may be used in particular processes.

13.2.5 Packed Bed

Packed-bed reactors are used with immobilised or particulate biocatalysts. The reactor consists of a tube, usually vertical, packed with catalyst particles. Medium can be fed either at the top or bottom of the column and forms a continuous liquid phase between the particles. Damage due to particle attrition is minimal in packed beds compared with stirred reactors. Packed-bed reactors have been used commercially with

Figure 13.8 Packed-bed reactor with medium recycle.



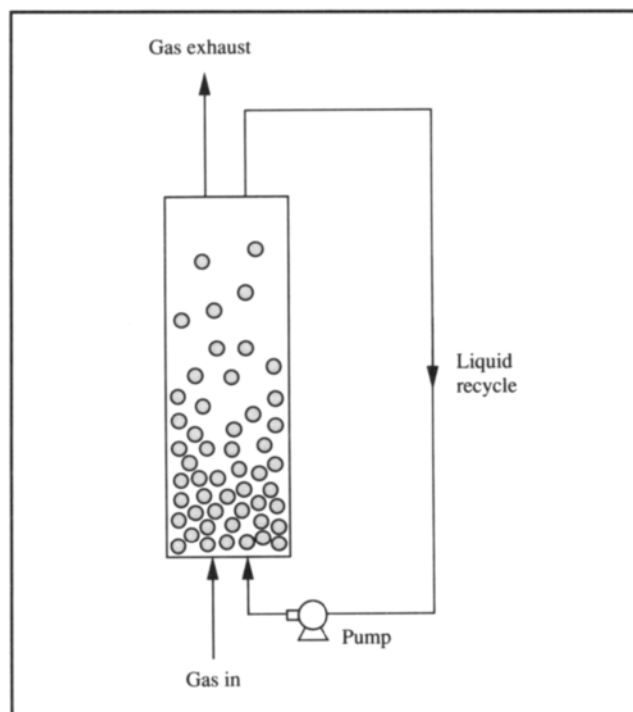
immobilised cells and enzymes for production of aspartate and fumarate, conversion of penicillin to 6-aminopenicillanic acid, and resolution of amino acid isomers.

Mass transfer between the liquid medium and solid catalyst is facilitated at high liquid flow rates through the bed; to achieve this, packed beds are often operated with liquid recycle as shown in Figure 13.8. The catalyst is prevented from leaving the column by screens at the liquid exit. The particles should be relatively incompressible and able to withstand their own weight in the column without deforming and occluding liquid flow. Recirculating medium must also be clean and free of debris to avoid clogging the bed. Aeration is generally accomplished in a separate vessel; if air is sparged directly into the bed, bubble coalescence produces gas pockets and flow *channelling* or maldistribution. Packed beds are unsuitable for processes which produce large quantities of carbon dioxide or other gases which can become trapped in the packing.

13.2.6 Fluidised Bed

When packed beds are operated in upflow mode with catalyst beads of appropriate size and density, the bed expands at high liquid flow rates due to upward motion of the particles. This is the basis for operation of fluidised-bed reactors as illustrated in Figure 13.9. Because particles in fluidised beds are in constant motion, channelling and clogging of the bed are avoided and air can be introduced directly into the column. Fluidised-bed reactors are used in waste treatment with sand or similar material supporting mixed microbial populations. They are also used with flocculating organisms in brewing and for production of vinegar.

Figure 13.9 Fluidised-bed reactor.



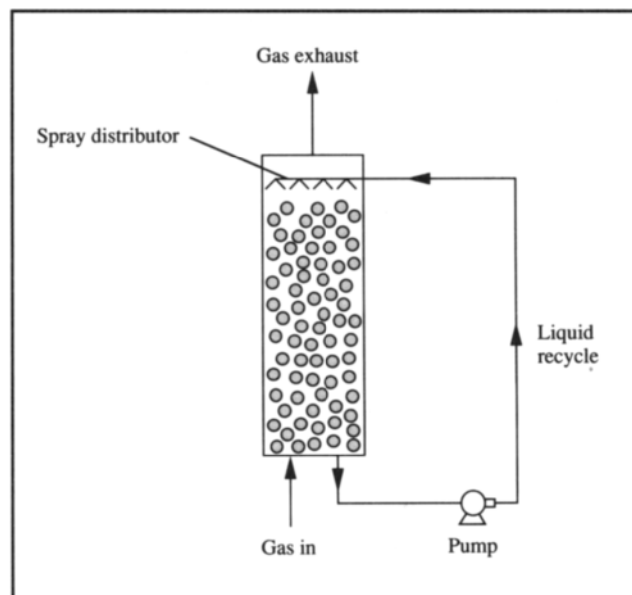
13.2.7 Trickle Bed

The trickle-bed reactor is another variation of the packed bed. As illustrated in Figure 13.10, liquid is sprayed onto the top of the packing and trickles down through the bed in small rivulets. Air may be introduced at the base; because the liquid phase is not continuous throughout the column, air and other gases move with relative ease around the packing. Trickle-bed reactors are used widely for aerobic wastewater treatment.

13.3 Practical Considerations For Bioreactor Construction

Industrial bioreactors for sterile operation are usually designed as steel pressure vessels capable of withstanding full vacuum up to about 3 atm positive pressure at 150–180°C. A hole is provided on large vessels to allow workers entry into the tank for cleaning and maintenance; on smaller vessels the top is removable. Flat headplates are commonly used with laboratory-scale fermenters; for larger vessels a domed construction is less expensive. Large fermenters are equipped with a lighted vertical sight-glass for inspecting the contents of the reactor. Nozzles for medium, antifoam, acid and alkali addition, air-exhaust pipes, pressure gauge, and a rupture disc for

Figure 13.10 Trickle-bed reactor.

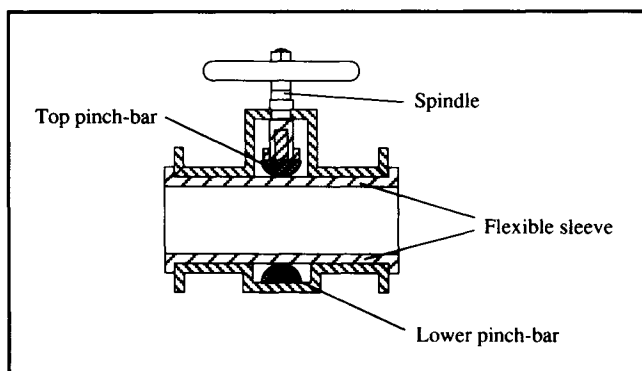


emergency pressure release, are normally located on the head-plate. Side ports for pH, temperature and dissolved-oxygen sensors are a minimum requirement; a steam-sterilisable sample outlet should also be provided. The vessel must be fully draining via a harvest nozzle located at the lowest point of the reactor. If the vessel is mechanically agitated, either a top- or bottom-entering stirrer is installed.

13.3.1 Aseptic Operation

Most fermentations outside of the food and beverage industry are carried out using pure cultures and aseptic conditions. Keeping the reactor free of unwanted organisms is especially important for slow-growing cultures which can be quickly over-run by contamination. Fermenters must be capable of operating aseptically for a number of days, sometimes months.

Typically, 3–5% of fermentations in an industrial plant are lost due to failure of sterilisation procedures. However, the frequency and causes of contamination vary considerably from process to process. For example, the nature of the product in antibiotic fermentations affords some protection from contamination; fewer than 2% of production-scale antibiotic fermentations are lost through contamination by microorganisms or phage [12]. In contrast, a contamination rate of 17% has been reported for industrial-scale production of β -interferon from human fibroblasts cultured in 50-litre bioreactors [13].

Figure 13.11 Pinch valve.

Industrial fermenters are designed for *in situ* steam sterilisation under pressure. The vessel should have a minimum number of internal structures, ports, nozzles, connections and other attachments to ensure that steam reaches all parts of the equipment. For effective sterilisation, all air in the vessel and pipe connections must be displaced by steam. The reactor should be free of crevices and stagnant areas where liquid or solids can accumulate; polished welded joints are used in preference to other coupling methods. Small cracks or gaps in joints and fine fissures in welds are a haven for microbial contaminants and are avoided in fermenter construction whenever possible. After sterilisation, all nutrient medium and air entering the fermenter must be sterile. As soon as flow of steam into the fermenter is stopped, sterile air is introduced to maintain a slight positive pressure in the vessel and discourage

entry of air-borne contaminants. Filters preventing passage of microorganisms are fitted to exhaust-gas lines; this serves to contain the culture inside the fermenter and insures against contamination should there be a drop in operating pressure.

Flow of liquids to and from the fermenter is controlled using valves. Because valves are a potential entry point for contaminants, their construction must be suitable for aseptic operation. Common designs such as simple gate and globe valves have a tendency to leak around the valve stem and accumulate broth solids in the closing mechanism. Although used in the fermentation industry, they are unsuitable if a high level of sterility is required. Pinch and diaphragm valves such as those shown in Figures 13.11 and 13.12 are recommended for fermenter construction. These designs make use of flexible sleeves or diaphragms so that the closing mechanism is isolated from the contents of the pipe and there are no dead spaces in the valve structure. Rubber or neoprene capable of withstanding repeated sterilisation cycles is used to fashion the valve closure; the main drawback is that these components must be checked regularly for wear to avoid valve failure. To minimise costs, ball and plug valves are also used in fermenter construction.

With stirred reactors, another potential entry point for contamination is where the stirrer shaft enters the vessel. The gap between the rotating stirrer shaft and the fermenter body must be sealed; if the fermenter is operated for long periods, wear at the seal opens the way for air-borne contaminants. Several types of stirrer seal have been developed to prevent contamination. On large fermenters, mechanical seals are commonly used [14].

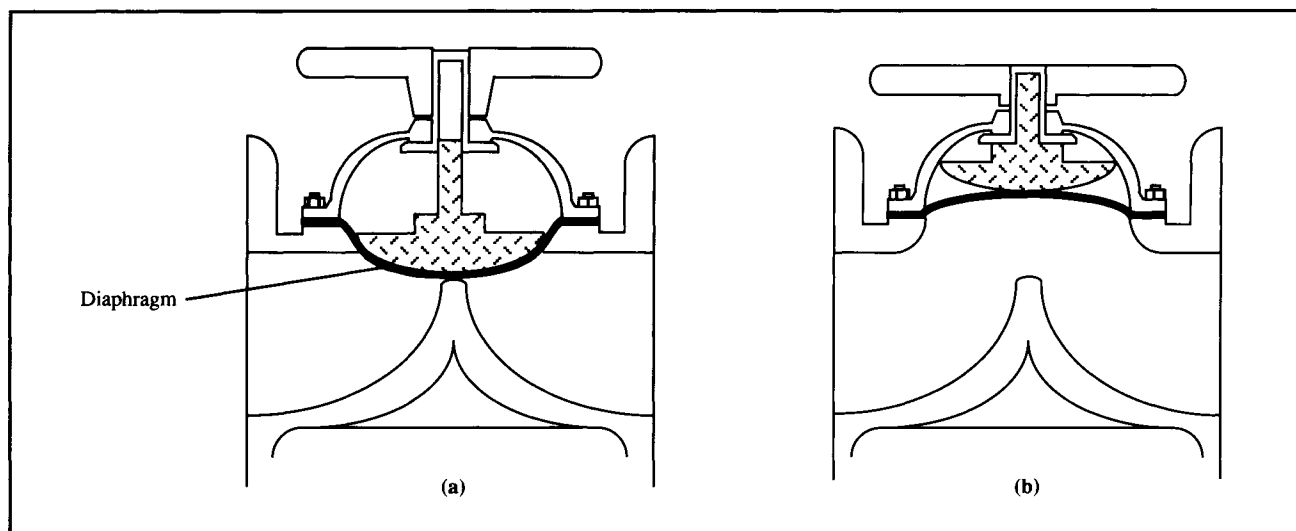
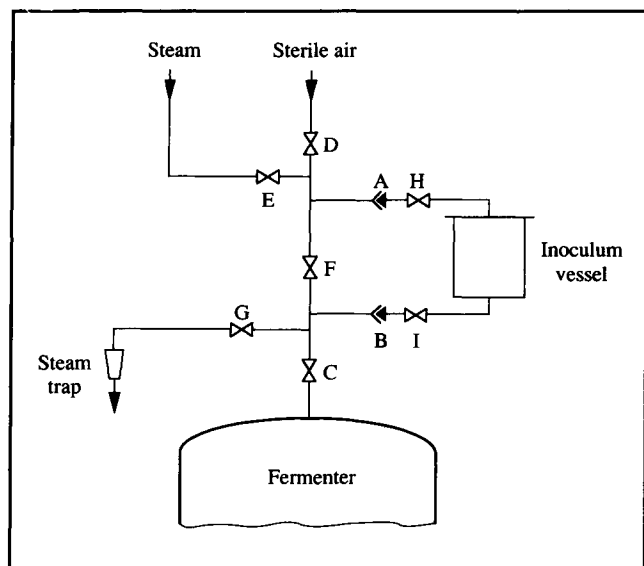
Figure 13.12 Weir-type diaphragm valve in (a) closed and (b) open positions.

Figure 13.13 Pipe and valve connections for aseptic transfer of inoculum to a large-scale fermenter. (From A. Parker, 1958, *Sterilization of equipment, air and media*. In: R. Steel, Ed, *Biochemical Engineering*, pp. 97–121, Heywood, London.)



In these devices, one part of the assembly is stationary while the other rotates on the shaft; the precision-machined surfaces of the two components are pressed together by springs or expanding bellows and cooled and lubricated with water. Mechanical seals with running surfaces of silicon carbide paired with tungsten carbide are often specified for fermenter application. Stirrer seals are especially critical if the reactor is designed with a bottom-entering stirrer; double mechanical seals may be installed to prevent fluid leakage. On smaller vessels, magnetic drives can be used to couple the stirrer shaft with the motor; with these devices, the shaft does not pierce the fermenter body. A magnet in a housing on the outside of the fermenter is driven by the stirrer motor; inside, another magnet is attached to the end of the stirrer shaft and held in place by bearings. Sufficient power can be transmitted using magnetic drives to agitate vessels up to at least 800 litres in size [15]. However, the suitability of magnetic drives for viscous broths, especially when high oxygen-transfer rates are required, is limited.

13.3.2 Fermenter Inoculation and Sampling

Consideration must be given in design of fermenters for aseptic inoculation and sample removal. Inocula for large-scale fermentations are transferred from smaller reactors; to prevent contamination during this operation, both vessels are maintained under positive air pressure. The simplest aseptic

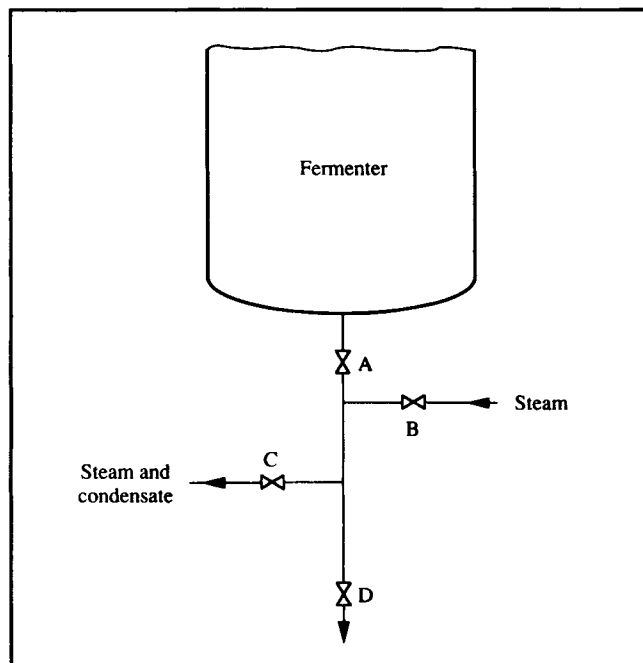
transfer method is to pressurise the inoculum vessel using sterile air; culture is then effectively blown into the larger fermenter. An example of the pipe and valve connections required for this type of transfer is shown in Figure 13.13. The fermenter and its piping and the inoculum tank and its piping including valves H and I are sterilised separately before culture is added to the inoculum tank. With valves H and I closed, the small vessel is joined to the fermenter at connections A and B. Because these connectors were open prior to being joined, they must be sterilised before the inoculum tank is opened. With valves D, H, I and C closed and A and B slightly open, steam flows through E, F and G and bleeds slowly from A and B. After about 20 minutes' steam sterilisation, valves E and G and connectors A and B are closed; the route from inoculum tank to fermenter is now sterile. Valves D and C are opened for flow of sterile air into the fermenter to cool the line under positive pressure. Valve F is then closed, valves H and I are opened and sterile air is used to force the contents of the inoculum tank into the fermenter. The line between the vessels is emptied of most residual liquid by blowing through with sterile air. Valves D, C, H and I are then closed to isolate both the fermenter and the empty inoculum tank which can now be disconnected at A and B.

Sampling ports are fitted to fermenters to allow removal of broth for analysis. An arrangement for sampling which preserves aseptic operation is shown in Figure 13.14. Initially, valves A and D are closed; valves B and C are open to maintain a steam barrier between the reactor and the outside environment. Valve C is then closed, valve B partially closed and valve D partially opened to allow steam and condensate to bleed from the sampling port D. For sampling, A is opened briefly to cool the pipe and carry away any condensate that would dilute the sample; this broth is discarded. Valve B is then closed and a sample collected through D. When sampling is complete, valve A is closed and B opened for re-sterilisation of the sample line; this prevents any contaminants which entered while D was open from travelling up to the fermenter. Valve D is then closed and valve C re-opened.

13.3.3 Materials of Construction

Fermenters are constructed from materials that can withstand repeated steam sterilisation and cleaning cycles. Materials contacting the fermentation medium and broth should also be non-reactive and non-absorptive. Glass is used to construct fermenters up to about 30 litres capacity. The advantages of glass are that it is smooth, non-toxic, corrosion-proof and transparent for easy inspection of the vessel contents. Because entry ports are required for medium, inoculum, air and instruments such as pH

Figure 13.14 Pipe and valve connections for a simple fermenter sampling port.



and temperature sensors, glass fermenters are usually equipped with stainless steel headplates containing many screw fittings.

Most pilot- and large-scale fermenters are made of corrosion-resistant stainless steel, although mild steel with stainless steel cladding has also been used. Cheaper grades of stainless steel may be used for the jacket and other surfaces isolated from the broth. Copper and copper-containing materials must be avoided in all parts of the fermenter contacting the culture because of its toxic effect on cells. Interior steel surfaces are polished to a bright 'mirror' finish to facilitate cleaning and sterilisation of the reactor; welds on the interior of the vessel are ground flush before polishing. Electropolishing is preferred over mechanical polishing which leaves tiny ridges and grooves in the metal to accumulate dirt and microorganisms.

13.3.4 Sparger Design

The sparger, impeller and baffles determine the effectiveness of mixing and oxygen mass transfer in stirred bioreactors. Design of impellers and baffles is discussed in Section 7.9. Three types of sparger are commonly used in bioreactors: porous, orifice and nozzle. *Porous spargers* of sintered metal, glass or ceramic are used mainly in small-scale applications; gas throughput is limited because the sparger poses a high resistance to flow. Cells growing through the fine holes and

blocking the sparger can also be a problem. *Orifice spargers*, also known as *perforated pipes*, are constructed by making small holes in piping which is then fashioned into a ring or cross and placed at the base of the reactor; individual holes must be large enough to minimise blockages. Orifice spargers have been used to a limited extent for production of yeast and single-cell protein and in waste treatment. *Nozzle spargers* are used in many agitated fermenters from laboratory to production scale. These spargers consist of a single open or partially-closed pipe providing a stream of air bubbles; advantages compared with other sparger designs include low resistance to gas flow and small risk of blockage. Other sparger designs have also been developed. In *two-phase ejector-injectors*, gas and liquid are pumped concurrently through a nozzle to produce tiny bubbles; in *combined sparger-agitator* designs for smaller fermenters, a hollow stirrer-shaft is used for delivery of air. Irrespective of sparger design, provision should be made for in-place cleaning of the interior of the pipe.

13.3.5 Evaporation Control

Aerobic cultures are continuously sparged with air; however, most components of air are inert and leave directly through the exhaust gas line. If air entering the fermenter is dry, water is continually stripped from the medium and leaves the system as vapour. Over a period of days, evaporative water loss can be significant. This problem is more pronounced in air-driven reactors because the gas flow rates required for good mixing and mass transfer are generally higher than in stirred reactors.

To combat evaporation problems, air sparged into fermenters may be pre-humidified by bubbling through columns of water outside the fermenter; humid air entering the fermenter has less capacity for evaporation than dry air. Fermenters are also equipped with water-cooled condensers to return to the broth any vapours carried by the exit gas. Evaporation can be a particular problem when products or substrates are more volatile than water. For example, *Acetobacter* species are used to produce acetic acid from ethanol in a highly aerobic process requiring large quantities of air. It has been reported for stirred-tank reactors operated at air flow rates between 0.5 and 1.0 vvm (vvm means volume of gas per volume of liquid per minute) that from a starting alcohol concentration of 5%, 30–50% of the substrate is lost within 48 hours due to evaporation [16].

13.4 Monitoring and Control of Bioreactors

The environment inside bioreactors should allow optimal catalytic activity. Parameters such as temperature, pH,

dissolved-oxygen concentration, medium flow rate, stirrer speed and sparging rate have a significant effect on the outcome of fermentation and enzyme reactions. To provide the desired environment, system properties must be monitored and control action taken to rectify any deviations from the desired values. Fermentation monitoring and control is an active area of research aimed at improving the performance of bioprocesses and achieving uniform and reliable fermenter operation.

Various levels of process control exist in the fermentation industry. Manual control is the simplest, requiring a human operator to manipulate devices such as pumps, motors and valves. Automatic feedback control is used to maintain parameters at specified values. With increasing application of computers in the fermentation industry, there is also scope for implementing advanced control and optimisation strategies based on fermentation models.

13.4.1 Fermentation Monitoring

Any attempt to understand or control the state of a fermentation depends on knowledge of critical variables which affect the process. These parameters can be grouped into three categories: physical, chemical and biological. Examples of process variables in each group are given in Table 13.1. Many of the physical measurements listed are well established in the fermentation industry; others are currently being developed or are the focus of research into new instrumentation.

Despite the importance of fermentation monitoring, industrially-reliable instruments and sensors capable of rapid, accurate and direct measurements are not available for many process variables. For effective control of fermentations based on measured data, the time taken to complete the measure-

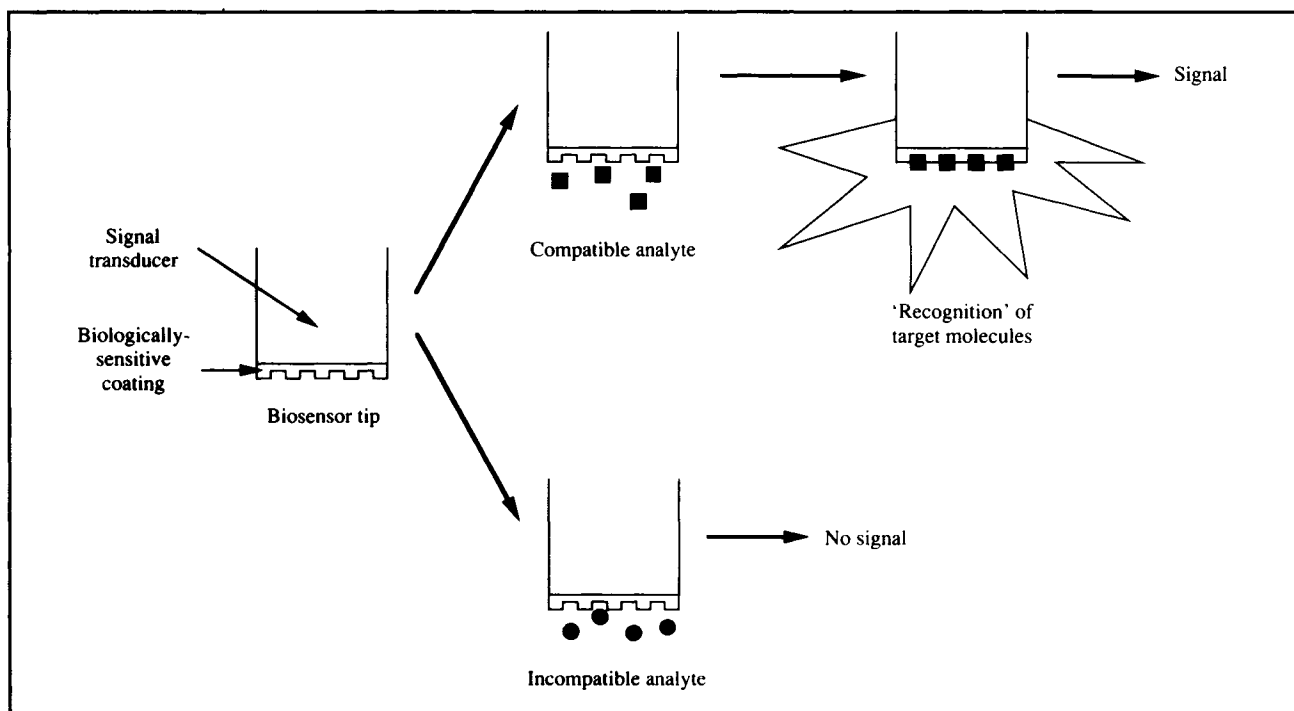
ment should be compatible with the rate of change of the variable being monitored. For example, in a typical fermentation, the time scale for change in pH and dissolved-oxygen tension is several minutes, while the time scale for change in culture fluorescence is less than 1 second. For other variables such as biomass concentration, an hour or more may pass before measurable changes occur. The frequency and speed of measurement must be consistent with these time scales. Ideally, measurements should be made *in situ* and on-line, i.e. in or near the reactor during operation, so that the result is available for timely control action. Many important variables such as biomass concentration and broth composition cannot currently be measured on-line because of the lack of appropriate instruments. Instead, samples must be removed from the reactor and taken to the laboratory for off-line analysis. Because fermentation conditions can change during laboratory analysis, control action based on the measurement is not as effective. Off-line measurements are used in industrial fermentations for analysis of biomass, carbohydrate, protein, phosphate and lipid concentrations, enzyme activity and broth rheology. Samples are usually taken manually every 4–8 h; the results are available 2–24 h later.

Examples of measurements which can be made on-line in industry are temperature, pressure, pH, dissolved-oxygen tension, flow rate, stirrer speed, power consumption, foam level, broth weight and gas composition. Instruments for taking these measurements are relatively commonplace; detailed descriptions can be found elsewhere [17–19]. The availability of an on-line measurement or its use in the laboratory does not necessarily mean it is applied in commercial-scale processing. Owing to the cost of installation and the financial consequences of instrument failure during fermentation, measurement devices used in industry must meet stringent performance

Table 13.1 Parameters measured or controlled in bioreactors

<i>Physical</i>	<i>Chemical</i>	<i>Biological</i>
Temperature	pH	Biomass concentration
Pressure	Dissolved O ₂	Enzyme concentration
Reactor weight	Dissolved CO ₂	Biomass composition
Liquid level	Redox potential	(such as DNA, RNA,
Foam level	Exit gas composition	protein, ATP/ADP/AMP,
Agitator speed	Conductivity	NAD/NADH levels)
Power consumption	Broth composition	Viability
Gas flow rate	(substrate, product, ion	Morphology
Medium flow rate	concentrations, etc.)	
Culture viscosity		
Gas hold-up		

Figure 13.15 Operating principle for biosensors. (From E.A.H. Hall, 1991, *Biosensors*, Prentice-Hall, New Jersey.)



criteria. These include: accuracy to within 1–2% of full scale, reliable operation for at least 80% of the time, low maintenance needs, steam sterilisability, simple and fast calibration, and maximum drift of less than 1–2% of full scale [1]. Sterilisable pH probes have been proven reliable if properly grounded and are used widely in industrial fermentations; their low failure rate is assisted by replacement every four or five fermentations. However, a variety of problems is associated in the industrial environment with electrodes for dissolved carbon dioxide, redox potential and specific ions; these instruments are therefore confined mostly to laboratory applications. Drift and interference from air bubbles are well-known limitations with dissolved-oxygen probes in industrial fermenters; deposits and microbial fouling of the membrane also reduce measurement reliability. Even if electrodes are steam sterilisable, probe life can be significantly reduced by repeated sterilisation cycles.

Development of new on-line measurements for chemical and biological variables is a challenging area of bioprocess engineering research. One of the areas targeted is *biosensors* [20, 21]. There are many different biosensor designs; however, in general, biosensors incorporate a biological sensing element in close proximity or integrated with a signal transducer. As illustrated schematically in Figure 13.15, the binding specificity of particular biological molecules immobilised on the

probe is used to 'sense' or recognise target species in the fermentation broth; ultimately, an electrical signal is generated after the interaction between sensing molecules and analyte is detected by the transducer. Biosensors which use immobilised cells to recognise broth components are also being developed. Biosensors with enzyme, antibody and cell sensing-elements have been tested for a wide range of substrates and products including glucose, sucrose, ethanol, acetic acid, ammonia, penicillin, urea, riboflavin, cholesterol and amino acids. However, although several biosensors are available commercially, they are not used for routine monitoring of large-scale fermentations because of intractable problems with robustness, sterilisation, limited life-span and long-term stability. At this stage, because of the sensitivity of their biological components, biosensors appear to be more suitable for medical diagnostics and environmental analyses than *in situ* bioprocess applications. Nevertheless, biosensors may be used for rapid analysis of broth samples removed from fermenters.

Another approach to on-line chemical and biological measurements involves use of automatic sampling devices linked to analytical equipment for high-performance liquid chromatography (HPLC), image analysis, NMR, flow cytometry or fluorometry. For these techniques to work, methods must be available for on-line aseptic sampling and analysis without blockage or interference from gas bubbles and cell solids.

Development in this area has centred around *flow injection analysis (FLA)*, a sample-handling technique for removing cell-free medium from the fermenter and delivering a pulse of analyte to *ex situ* measurement devices. To date, routine on-line measurement of the most fundamental fermentation parameter, biomass concentration, has not been achieved. Several procedures based on culture turbidity, light scattering, fluorescence, calorimetry, piezoelectric membranes, radio-frequency dielectric permittivity and acoustics have been developed; however, they have not proved sufficiently accurate and reliable under industrial conditions. Air bubbles and background particles readily interfere with optical readings, while the relationship between biomass density and other culture properties such as fluorescence is affected by pH, dissolved-oxygen tension and substrate levels [19]. Other difficulties have been encountered in development of automatic sampling devices; the most important of these is the high risk of contamination and blockage. Further details about on-line measurement techniques can be found elsewhere [22, 23].

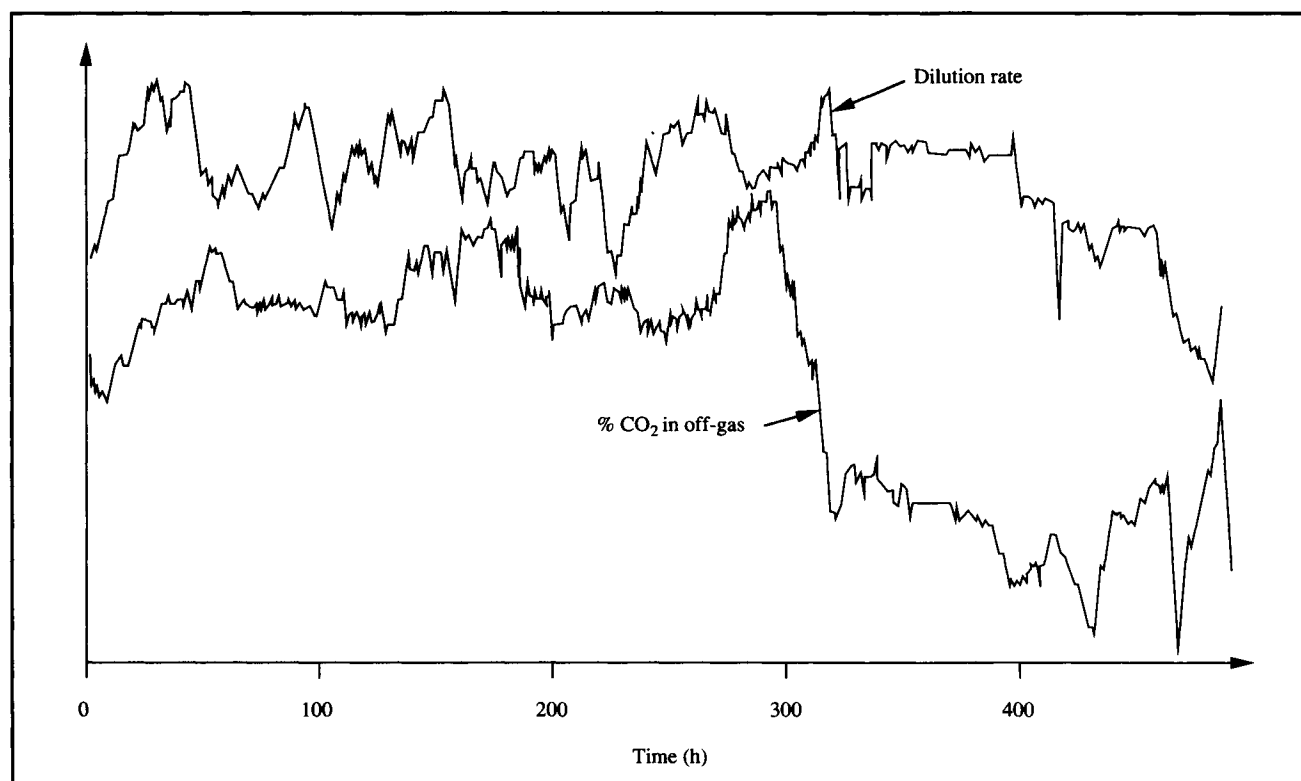
In a large fermentation factory, thousands of different vari-

ables may be monitored at any given time. The traditional device for recording on-line process information is the chart recorder; however, with increasing application of computers in the fermentation industry, digital data logging is widespread. Because of the enormous quantity of information obtained from continuous monitoring of bioprocesses, an increasing problem in industry is the management and effective utilisation of gigabytes of data.

13.4.2 Measurement Analysis

Any attempt to analyse or apply the results of fermentation monitoring must consider the errors and spurious or transient results incorporated into the data. Noise and variability are particular problems with certain fermentation measurements; for example, probes used for pH and dissolved-oxygen measurements are exposed to rapid fluctuations and heterogeneities in the broth so that noise can seriously affect the accuracy of point readings. Figure 13.16 shows typical results from on-line measurement of dilution rate and carbon dioxide

Figure 13.16 On-line measurements of dilution rate and off-gas carbon dioxide during an industrial mycelial fermentation. (From G.A. Montague, A.J. Morris and A.C. Ward, 1989, Fermentation monitoring and control: a perspective. *Biotechnol. Genet. Eng. Rev.* 7, 147–188.)



evolution in a production-scale fermenter; in many cases, *signal conditioning* or *smoothing* must be carried out to reduce the noise in these data before they can be applied for process control or modelling. Most modern data acquisition and logging systems contain facilities for signal conditioning. Unwanted pseudo-random noise can be filtered out using analogue filter circuits or by averaging values over successive measurements. Alternatively, unfiltered signals can be digitised and filtering algorithms applied using computer software. Measurement drift cannot be corrected using electronic circuitry; instruments must be periodically recalibrated during long fermentations to avoid loss of accuracy.

Raw data are sometimes used to calculate *derived variables* which characterise the performance of the fermenter. The most common derived variables are oxygen-uptake rate, rate of carbon dioxide evolution, the respiratory quotient RQ , and the mass-transfer coefficient $k_L a$. As oxygen-uptake rate is usually calculated from the difference between two quantities of similar magnitude (gas inlet and outlet oxygen levels; see Section 9.10.1), noise in this variable can be significant [24] and affect the quality of other dependent variables such as $k_L a$ and RQ .

13.4.3 Fault Analysis

Faults in reactor operation affect 15–20% of fermentations [25]. Fermentation measurements can be used to detect faults; for example, signals from a flow sensor could be used to detect blockages in a pipe and trigger an alarm in the factory control room. Normally however, the sensors themselves are the most likely components to fail; rates of failure of some fermentation instruments are listed in Table 13.2. Failure of a sensor might

be detected as an unexpected change or rate of change in its signal, or a change in the noise characteristics. Several approaches can be used to reduce the impact of faults on large-scale fermentations; these include comparison of current measurements with historical values, cross-checking between independent measurements, using multiple and back-up sensors, and building hardware redundancy into computer systems and power supplies.

13.4.4 Process Modelling

In modern approaches to fermentation control, a reasonably accurate mathematical model of the reaction and reactor environment is required. Using process models, we can progress beyond environmental control of bioreactors into the realm of direct biological control. Development of fermentation models is aided by information from measurements taken during process operation.

Models are mathematical relationships between variables. Traditionally, models are based on a combination of 'theoretical' relationships which provide the structure of the model, and experimental observations which provide the numerical values of coefficients. For biological processes, specifying the model structure can be difficult because of the complexity of cellular processes and the large number of environmental factors which affect cell culture. Usually, bioprocess models are much-simplified approximate representations deduced from observation rather than from theoretical laws of science. As an example, a frequently-used mathematical model for batch fermentation consists of the Monod equation for growth and an expression for rate of substrate consumption as a function of biomass concentration:

Table 13.2 Reliability of fermentation equipment

(From S.W. Carleysmith, 1987, *Monitoring of bioprocessing*, In: J.R. Leigh, Ed, *Modelling and Control of Fermentation Processes*, pp. 97–117, Peter Peregrinus, London; and P.N. Royce, 1993, *A discussion of recent developments in fermentation monitoring and control from a practical perspective*, Crit. Rev. Biotechnol. 13, 117–149)

Equipment	Reliability	Mean time between failures (weeks)
Temperature probe	—	150–200
pH probe	98%	9–48
Dissolved-oxygen probe	50–80%	9–20
Mass spectrometer	—	10–50
Paramagnetic O ₂ analyser	—	24
Infrared CO ₂ analyser	—	52
Computer system	—	4

$$\frac{dx}{dt} = \mu x = \frac{\mu_{\max} s x}{K_S + s} \quad (13.5)$$

$$\frac{-ds}{dt} = \frac{\mu x}{Y_{XS}} + m_S x. \quad (13.6)$$

This model represents a combination of Eqs (11.52), (11.60) and (11.72). The form of the equations was determined from experimental observation of a large number of different culture systems. In principle, once values of the parameters $\mu_{\max}$, K_S , Y_{XS} and m_S have been determined, we can use the model to estimate cell concentration x and substrate concentration s as a function of time. A common problem with fermentation models is that the model parameters can be difficult to measure or change with time.

We have already encountered several model equations in this book; examples include the kinetic, yield and maintenance relationships introduced in Chapter 11 and the stoichiometric equations of Chapter 4 relating masses of substrates, biomass and products during reaction. Other models were derived in Chapter 12 for heterogeneous reactions; dependence of culture parameters such as cell concentration on physical conditions in the bioreactor are represented in simple form by the equations of Sections 8.6.1 and 9.5.2. Process models vary in form but have the unifying feature that they predict outputs from a set of inputs. When models are used for fermentation control, they are usually based on mass- and energy-balance equations for the system.

Development of a comprehensive model covering all key aspects of a particular bioprocess and able to predict the effects of a wide range of culture variables is a demanding exercise. Accurate models applicable to a range of process conditions are rare. As many aspects of fermentation are poorly understood, it is difficult to devise mathematical models covering these areas. For example, the response of cells to spatial variations in dissolved-oxygen and substrate levels in fermenters, or the effect of impeller design on microbial growth and productivity, is not generally incorporated into models because the subject has been inadequately studied. Evidence that all important fermentation variables have not yet been identified is the significant batch-to-batch variation that occurs in the fermentation industry.

13.4.5 State Estimation

As described in Section 13.4.1, it is not possible to measure on-line all the key variables or states of a fermentation process.

Often, considerable delays are involved in off-line measurement of important variables such as biomass, substrate and product concentrations. Such delays make effective control of the reactor difficult if control action is dependent on the value of these parameters, but must be undertaken more quickly than off-line analysis allows. One approach to this problem is to use available on-line measurements in conjunction with mathematical models of the process to estimate unknown variables. The computer programs and numerical procedures developed to achieve this are called *software sensors*, *estimators* or *observers*. The *Kalman filter* is a well-known type of observer applicable to linear process equations; non-linear systems can be treated using the *extended Kalman filter* [26]. The success of Kalman filters and other observers depends largely on the accuracy and robustness of the process model used.

Several techniques for state estimation are available. As a simple example, on-line measurements of carbon dioxide in fermenter off-gas can be applied with an appropriate process model to estimate biomass concentration during penicillin fermentation [27]. As shown in Figure 13.17, the results were satisfactory for 100-m³ fed-batch fermentations over a period of more than 8 days. Direct state estimation can achieve reasonable results as long as the process model remains valid and the estimator is able to reduce measurement noise. However, if

Figure 13.17 Measured and estimated biomass concentrations in a large-scale fed-batch penicillin fermentation. (From G.A. Montague, A.J. Morris and J.R. Bush, 1988, Considerations in control scheme development for fermentation process control. *IEEE Contr. Sys. Mag.* 8, April, 44–48.)

